Supplify Complete Handbook

Restaurant & F&B supplier marketplace — product, onboarding, and technical reference

Version 1.0
Generated 2026-06-17
Repository commit `ab5695e195079adde17df8b8082f193551daf2d8`

This handbook reflects implementation status in the repository at generation time. Where documentation and code disagree, the code is authoritative. See Part XVI — Implementation Status.

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Title	Supplify Complete Handbook
Version	1.0
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Repository	supplify_erp

Single-volume onboarding, product, operations, and technical reference assembled from `docs/onboarding/01` – `20`.

Disclaimer

This handbook describes Supplify **as implemented in the repository at the commit above**, not as a marketing promise. Capabilities marked **Partial**, **UI-only**, or **disabled by default** in the feature inventory and acceptance criteria may be incomplete, environment-gated, or absent from demo seeds.

Implementation status (honest summary):

- **Core B2B order flow** (cart → supplier fulfill → receive → invoice) is production-intent and well tested.
- **Admin platform**, **monetization/tiers**, and **hospitality add-ons** (reservations, B2C, staff) are shipped with varying demo polish.
- **Known gaps** include: supplier Settings Delivery Zones/Contacts tabs (UI-only, hidden); restaurant finance opening balance hardcoded `0`; delivery rollover cron disabled unless `DELIVERY_ROLLOVER_ENABLED=true`; driver Keycloak users not in `seed:full`; quote prices informational at checkout.
- **554 API routes** and **127 feature inventory rows** are code-verified; always re-run route discovery and tests after major releases.

For claim-level traceability, see Part XX — Source Evidence Index. For pass/fail definitions, see Part XIII — Acceptance Criteria. Part XVI — Implementation Status expands the honest assessment.

Audience: Parts VII–XI and XV are marked *(Internal Technical Reference)* for engineers, DevOps, and implementation partners.

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Feature Inventory Summary

> **Generated:** 2026-06-17 · **Commit:** `ab5695e195079adde17df8b8082f193551daf2d8`
> **Full inventory:** 00-feature-inventory.md (127 rows — not duplicated here)

Metric	Count
Inventory rows	**127**
Domains covered	**22**
API routes (discovered)	**554**
Frontend route entries	**80**
Permission keys	**52**
Subscription tiers	**4** × 2 tenant types

Status legend

Status	Meaning
Full	End-to-end implemented and tested in code
Partial	Works with known gaps or doc/code drift
UI-only	Frontend without backend persistence
Backend-only	API without complete UI
Deprecated	Legacy path still present
Unverified	Could not confirm without running environment

Domains in the full inventory include: Authentication & Tenancy; Ordering, Fulfillment & Delivery; Catalog, Inventory & Reorder; Finance, Deals & Onboarding; Staff, Reservations, Platform & Integrations. See the linked file for per-feature status, limitations, and evidence paths.

Part I — Executive Overview

Audience: Executives, product leaders, solution architects, and onboarding partners who need a concise but accurate picture of what Supplify is, who it serves, and how the platform is built.

Source of truth: Application code, database migrations, and route inventories in this repository (metrics verified 2026-06-17).

What Supplify Is

Supplify is a **restaurant–supplier marketplace and operations platform**. It connects food-service buyers (restaurants) with distributors and producers (suppliers) in a single, role-based system where ordering, catalog management, fulfillment, receiving, finance, reservations, staff operations, and platform administration share one data model and one API.

The product is not a lightweight ordering widget. It is an end-to-end B2B supply chain workspace with:

- **Unified ordering** — Restaurants browse supplier catalogs, build carts, place orders, schedule re-orders via quick lists, and track status through delivery and invoicing.
- **Supplier operations** — Suppliers manage products, warehouses, fulfillment boards, driver dispatch, GPS tracking, promotions, and receivables.
- **Restaurant operations** — Restaurants receive goods, record quality, manage on-hand inventory, run front-of-house reservations, and reconcile invoices.
- **Monetization** — Tiered subscriptions (Free Trial, Silver, Gold, Platinum) gate features and usage meters for both tenant types.
- **Growth and discovery** — Supplier customer import, referral programs, public mini-store catalogs, quote requests, and consumer B2C ordering extend reach beyond logged-in B2B users.
- **Platform control** — Admins manage tenants, plans, feature flags, limit overrides, deal approvals, impersonation, and observability.

The canonical product description in code aligns with `docs/product/overview.md`: "*Supplify is a restaurant–supplier marketplace: ordering, receiving, fulfillment, finance, reservations, and platform admin.*"

User Ecosystem

Supplify serves multiple personas across authenticated workspaces, public portals, and mobile-adjacent experiences.

Core B2B tenants

Persona	Keycloak / app role	Primary workspace
Restaurant operator	`RESTAURANT`	Orders, receiving, inventory, reservations, finance
Supplier operator	`SUPPLIER`	Catalog, fulfillment, warehouses, promotions, growth
Platform administrator	`ADMIN`	Tenant management, billing, feature toggles, audit

Each restaurant and supplier tenant is a **workspace** with its own subscription, branding, team members, and tenant-scoped RBAC roles (see `apps/api/src/lib/role-matrix.js`).

Extended personas

Persona	Access pattern	Purpose
Driver	Supplier workspace role `Driver` with `DRIVER_DELIVERIES_*` permissions	Assigned deliveries, status updates, proof of delivery
Restaurant staff (FOH)	Role `FOH Staff` or `Receiving Staff`	Reservations, receiving, limited order visibility
Accountant	Role `Accountant` on either side	Invoices, payments, subscription view
Staff portal user	Keycloak `staff_portal` → `STAFF_PORTAL` app role	PTO, shift swaps, self-service dashboard at `/staff`
Reservation guest	Public, unauthenticated	Book, confirm, cancel, waitlist at `/reserve`
Consumer (B2C)	Public storefront at `/order/:restaurantSlug`	Menu browse, checkout, order tracking, loyalty
Prospective tenant	`PENDING` until registration completes	`/register/complete` → `/app/activate`

Team members inside a tenant are not separate Keycloak platform roles; they are **users assigned tenant roles** (`tenant_user_roles`) with granular permissions from `permission-keys.js`.

Value Propositions by Persona

Restaurants

Restaurants gain a **single pane of glass** for procurement and back-of-house coordination:

- **Less friction in ordering** — Browse linked supplier catalogs, save quick lists, schedule recurring orders, and chat in context next to products and orders.
- **Visibility through delivery** — Track in-flight orders, view supplier GPS when enabled, and record receiving with optional quality photos (plan-gated `receiving_quality`).
- **Financial clarity** — Invoices tied to orders; payment recording; supplier account statements (with known limitations documented in the product guide).
- **Operational breadth** — Inventory and waste tracking, disputes, deals redemptions, contract pricing, quote requests, and FOH reservations on one platform.
- **Scalable controls** — Multi-branch inventory, advanced roles, and smart reorder unlock as plans upgrade from Silver through Platinum.

Suppliers

Suppliers reduce missed orders and manual coordination:

- **Centralized order intake** — All restaurant orders in one fulfillment workflow with decline reasons, amendments, and calendar views.
- **Catalog and pricing power** — Product CRUD, bulk CSV import, image ZIP import, contract pricing, and optional public mini-store at `/supplier/:slug`.
- **Logistics** — Warehouse management, multi-warehouse routing (Gold+), delivery board, driver assignment, route planning, and GPS tracking.
- **Revenue tools** — Invoicing from delivered orders, promotions/deals with admin approval, growth program for customer acquisition, and command-center analytics.
- **Relationship management** — Chat, reviews, disputes resolution, and restaurant connection requests.

Drivers

Drivers interact through a focused **delivery-only surface** (`/app/driver-deliveries`) with permissions limited to viewing and updating assigned deliveries. They do not access catalog, billing, or team administration — reducing training burden and security exposure.

Platform administrators

Admins operate the **control plane**:

- Tenant lifecycle: plans, locks, Free Trial extension, sponsorship limits
- Feature flags: global and per-tenant overrides
- Limit overrides: temporary meter caps (orders/day, chats/day, etc.)
- Deal moderation: approve/reject supplier promotions
- Impersonation: enter a tenant workspace for support
- Health and audit: cron job failures, audit logs, growth settings

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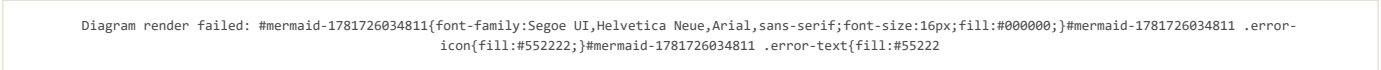
Platform at a Glance

Architecture



Application layout

Layer	Location	Responsibility
API	`apps/api`	554 HTTP routes, RBAC, subscriptions, business logic, cron
Web	`apps/web`	80 frontend routes (`apps/web/src/App.tsx`), React SPA
Database	`apps/api/db/migrations`	175 SQL migrations, schema evolution since `0000`
Tests	`apps/api/**/*.test.js` , `apps/web/**/*.test.{ts,tsx}` , `tests/e2e`	213 API test files, 309 web test files (per bootstrap metrics)

Technology stack

The monorepo uses **pnpm workspaces** with a Node.js/Express API and a Vite-powered React frontend. PostgreSQL is the system of record; Redis is optional but recommended for Socket.IO fan-out and order-calendar caching in multi-replica deployments (Railway). Keycloak provides OIDC identity; MinIO (or S3-compatible storage) handles uploads for chat attachments, product images, and bulk import archives. A sibling **supplify-mobile** repository consumes the same API with parity expectations documented in `docs/mobile/MOBILE_FEATURE_PARITY.md` . Local development is orchestrated via `pnpm dev` with Docker profiles for Keycloak, Postgres, and optional full-stack nginx.

Subscription tiers

Four **active self-serve tiers** apply to both restaurants and suppliers (separate `subscription_plan` rows per tenant type):

Code	Name	Monthly (USD)	Positioning
`free`	Free Trial	$0	Time-limited evaluation (default 30 days, admin 7–90); read-only after expiry
`silver`	Silver	$49	First paid tier; single-location core
`gold`	Gold	$149	Daily operations; multi-branch, analytics, smart reorder
`platinum`	Platinum	$349	High limits; full feature catalog

An `enterprise` plan code exists in the database but is **inactive for self-serve** (`requires_admin_assignment`). Bronze was removed in migration `0116` ; legacy `bronze` input maps to `silver` .

Feature gates and meter limits are enforced server-side via `requireFeature()` middleware and `plan-enforcement.js` , keyed from `feature-keys.js` .

Domain coverage (high level)

Domain	Restaurant	Supplier	Admin	Public
Auth & registration	✓	✓	✓	—
Catalog & pricing	browse	manage	—	mini-store
Ordering & quick lists	✓	fulfill	—	B2C menu
Fulfillment & GPS	track	✓	—	B2C track
Receiving & disputes	✓	view	—	—
Finance & invoices	✓	✓	—	—
Deals & promotions	redeem	manage	approve	—
Reservations	✓	—	—	guest portal
Staff & labour	✓	team	—	self-service
Chat & notifications	✓	✓	—	—
Growth & referrals	benefit	✓	config	register ref
Quote requests	✓	respond	—	—
Reports	✓	✓	metrics	—
Subscriptions	✓	✓	manage	—

Key Platform Metrics

These figures are generated from repository artifacts and should be re-verified after major releases.

Metric	Value	How to verify
API routes	554	`docs/audits/route-inventory.json` (`count` field)
Frontend routes	80	`apps/web/src/App.tsx` route definitions
SQL migrations	175	`apps/api/db/migrations/*.sql`
API test files	213	`apps/api/**/*.test.js`
Web test files	309	`apps/web/**/*.test.{ts,tsx,jsx}`
Plan tiers (active)	4	`free` , `silver` , `gold` , `platinum`
Workspace system roles	16	7 restaurant + 9 supplier in `role-matrix.js`
Mermaid diagrams (valid)	50	`scripts/check-mermaid-diagrams.mjs`

Ecosystem Diagram

The following diagram shows how value flows between participants on the platform.

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Public consumers and reservation guests connect at the edges: `/order/:slug` for B2C and `/reserve` for table booking, both backed by the same API and tenant data.

Strategic Outcomes

For the business

Supplify monetizes through **subscription tiers** aligned to operational maturity, with upgrade paths triggered by limit proximity (orders/day, chats/day, branches, products) and feature gates. Conversion events (`VIEW_PLANS` , `OPEN_UPGRADE`) are recorded for funnel analysis (`POST /api/subscriptions/conversion-event`).

The **supplier growth program** (migration `0169`) turns suppliers into acquisition channels: CSV import, connection requests, referral tokens, sponsorship, and rewards on first paid conversion.

For operations

A **single PostgreSQL schema** with 175 migrations means auditability and consistent reporting across tenants. In-process cron jobs handle scheduled orders, billing, trial expiry, promotions expiry, reorder forecasts, and optional delivery rollover (disabled by default).

For engineering

RBAC is **permission-first**: route guards resolve `tenant_user_roles` → permissions, with system roles seeded from `role-matrix.js` per tenant. Admin impersonation uses a separate token path without weakening tenant isolation for normal users.

Known Limitations (Executive Summary)

The following items are **implemented partially** or **disabled by default**. Full detail appears in the Complete Product Guide.

Area	Status
Supplier Settings → Delivery Zones tab	UI exists; tab hidden (`DELIVERY_ZONES_ENABLED = false`); warehouse zone API is separate
Restaurant finance opening balance	Hardcoded `0` in account statement summary (`TODO` in API)
Delivery rollover cron	Registered but no-op unless `DELIVERY_ROLLOVER_ENABLED=true`

These are product honesty markers, not blockers for core B2B ordering flows.

Implementation Evidence

Topic	Primary paths
Product definition	`docs/product/overview.md`, `docs/sales/02_solution.md`, `docs/sales/03_platform_overview.md`
Route inventory	`docs/audits/route-inventory.json`
Bootstrap metrics	`docs/onboarding/_artifacts/bootstrap-metrics.md`
Frontend routing	`apps/web/src/App.tsx`
RBAC & roles	`apps/api/src/lib/role-matrix.js`, `apps/api/src/lib/tenant-roles.js`, `docs/architecture/rbac-permission-matrix.md`
Subscriptions & tiers	`docs/product/tier-matrix.md`, `docs/product/plans.md`, `apps/api/db/migrations/0116 – 0120`
Auth & registration	`apps/api/src/lib/rbac.js`, `docs/features/tenant-registration.md`
Feature catalog	`docs/product/features.md`, `docs/product/feature-catalog-technical.md`
Cron jobs	`docs/operations/cron-jobs.md`, `apps/api/src/lib/register-cron-jobs.js`
Demo readiness audit	`docs/audits/SUPPLIFY_DEMO_READINESS_AUDIT.md`

Next Steps for Readers

- **Business stakeholders:** Continue to 02-complete-product-guide.md for domain-by-domain behavior, order lifecycle, and tenant model.
- **Technical onboarding:** Run `pnpm setup && pnpm dev`, then use `docs/product/features.md` verification tables and `docs/qa/regression-checklist.md`.
- **Sales narrative:** See `docs/sales/` (01_problem through enterprise_checklist).

Part II — Complete Product Guide

Audience: Product managers, business analysts, implementation consultants, and engineers who need end-to-end knowledge of every Supplify domain.

Source of truth: Application code, migrations, route inventories, and existing product documentation in this repository.

Introduction

This guide documents **all major product domains** in Supplify: what each does, who can access it, how it connects to adjacent domains, and where to find implementation evidence in the codebase. It is written for readers who will configure tenants, demo the product, or extend features — not as a marketing overview.

Supplify's data model centers on **tenants** (`RESTAURANT`, `SUPPLIER`, `ADMIN`) with per-tenant subscriptions, team members, and RBAC. B2B orders flow from restaurant cart through supplier fulfillment to receiving, invoicing, and optional disputes. Parallel tracks cover reservations (FOH), consumer B2C ordering, quote requests, growth programs, and platform administration.

User Types and Roles

Platform roles (Keycloak → `app_user.role`)

Role	Set by	Workspace access
`PENDING`	First login before org setup	`/register/complete` only
`RESTAURANT`	Registration or Keycloak `restaurant` role	Restaurant sidebar and APIs
`SUPPLIER`	Registration or Keycloak `supplier` role	Supplier sidebar and APIs
`ADMIN`	Keycloak `admin` role or seeded admin emails	`/app/admin` control plane
`STAFF_PORTAL`	Keycloak `staff_portal` / `staff_portal_user`	`/staff/login`, `/staff/dashboard` only

Resolution logic lives in `apps/api/src/lib/rbac.js` (`upsertUser`). Platform roles take precedence over staff portal for dual-role Keycloak users.

Restaurant workspace system roles (`role-matrix.js`)

#	Role	Purpose
1	**Owner**	Full access; immutable main admin
2	**Restaurant Manager**	Daily ops: orders, receiving, catalog view, reservations; no billing/team admin
3	**Purchaser**	Browse catalog, create and track orders, chat
4	**Receiving Staff**	Receive deliveries, open receiving disputes; cannot create orders
5	**Accountant**	Invoices, payments, subscriptions view
6	**Viewer**	Read-only across workspace views
7	**FOH Staff**	Reservations create/edit/view only

Supplier workspace system roles (`role-matrix.js`)

#	Role	Purpose
8	**Owner**	Full supplier access
9	**Supplier Manager**	Orders (accept/decline/fulfill), catalog, fulfillment, growth view, customer import
10	**Warehouse Manager**	Warehouses, fulfillment board, delivery ops
11	**Order Fulfillment Staff**	Fulfillment and delivery board; no billing or catalog import
12	**Driver**	Assigned deliveries only (`DRIVER_DELIVERIES_VIEW` , `DRIVER_DELIVERIES_MANAGE`)
13	**Catalog Manager**	Products, catalog, pricing, import
14	**Promotions Manager**	Deals, promotions, reorder intelligence
15	**Accountant**	Finance and receivables only
16	**Viewer**	Read-only supplier workspace

Additional personas (17+)

#	Persona	Notes
17	**Platform admin** (`ADMIN`)	Separate from tenant Owner; uses admin dashboard
18	**Staff portal user**	Labour self-service; provisioned from restaurant Team tab
19	**Reservation guest**	Unauthenticated public booking
20	**Consumer (B2C)**	Guest or light account on `/order/:restaurantSlug`

Custom tenant roles can be created with subsets of permissions; system role names are protected (`RESERVED_SYSTEM_ROLE_NAMES` in `tenant-roles.js`).

Implementation evidence: `apps/api/src/lib/role-matrix.js` , `apps/api/src/lib/permission-keys.js` , `docs/architecture/rbac-permission-matrix.md` , `apps/web/src/components/Sidebar.tsx` , `apps/web/src/components/RequirePermission.tsx` .

Tenant Model

Each commercial organization is a **tenant** with type `RESTAURANT` or `SUPPLIER` . Users belong to tenants through `tenant_user` membership and hold one or more `tenant_user_roles` .

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Registration flow: `POST /api/register/complete` creates tenant, org, default system roles, supplier catalog/warehouse (supplier path), and subscription with `lock_reason = pending_activation` . Activation clears lock via Free checkout or paid billing (`docs/features/tenant-registration.md`).

Multi-branch: Restaurants use `branch` records; suppliers use org branches (`/app/org`) and warehouses. Plan feature `multi_branch` gates expansion.

Implementation evidence: `apps/api/src/routes/register.routes.js` , `apps/api/src/lib/billing/` , `apps/api/db/migrations/0041_rbac_tenant_roles.sql` , `apps/web/src/pages/RegisterCompletePage.tsx` , `apps/web/src/pages/AccountActivationPage.tsx` .

Authentication and Session Management

Login

- **OIDC authorization code flow** via Keycloak (`/login` , `/auth/login`).
- Session cookies are HTTP-only; API validates JWT on each request.
- Expired sessions redirect to `/login?expired=true` .

Guards

- `AuthGuard` (web) blocks unauthenticated access to `/app/*` .
- `requireAuth` , `requireRole` , `requirePermission` (API) enforce server-side access.
- `billingAccess` middleware blocks writes when subscription is locked (pending activation, Free Trial expired, suspended).

Invitations

- Team invites: `/invite` → `POST` acceptance APIs.
- Branch invites: `/invite/branch` .
- Restaurant invitations from suppliers: migration `0087_restaurant_invitations.sql` .

Legal compliance

- `legal_acceptances` (migration `0129`); `/legal/reaccept` for policy updates.

Implementation evidence: `apps/api/src/routes/auth.routes.js`, `apps/web/src/components/AuthGuard.tsx`, `apps/web/src/lib/authRedirect.ts`, `apps/api/src/middlewares/billingAccess.js`, `tests/e2e/suites/critical_e2e/auth.spec.ts`.

Supplier and Restaurant Onboarding

Supplier onboarding

After activation, suppliers configure:

- **Profile & business** — `SupplierSettingsPage` tabs: profile, business, notifications, plan, team, drivers, branches, warehouses, activity.
- **Catalog seed** — Created at registration; products added via Products page, CSV bulk upload, or ZIP image import.
- **Warehouse** — Default warehouse created at registration; additional warehouses plan-gated (`warehouses` feature).
- **Fulfillment** — Enable driver management, fulfillment board (`/app/fulfillment`), command center (`/app/command-center`) on eligible plans.

Restaurant onboarding

- **Wizard** — `/app/onboarding` → `restaurant-onboarding` API for setup steps.
- **Supplier linking** — Browse `/app/suppliers`, follow/connect, block list support.
- **Branch setup** — Settings and branch invitations for multi-location (Gold+).
- **Push notifications** — Optional PWA opt-in during onboarding.

Implementation evidence: `apps/api/src/routes/restaurant-onboarding.routes.js`, `apps/web/src/pages/RestaurantOnboardingPage.tsx`, `apps/web/src/pages/SupplierSettingsPage.tsx`, `apps/api/scripts/seed-demo-users.js`.

Catalog, Products, and Pricing

Supplier catalog management

Capability	Web	API
Product CRUD	`/app/products`, `/app/products/:id`	`/api/products`
Bulk CSV upload	Products page	`/api/products/import`
Image ZIP import	ProductImageImportDialog	`/api/products/import-images`
Contract pricing	`/app/contract-pricing`	`/api/prices`, contract pricing routes
Public mini-store	Settings catalog link card	`GET /api/public/suppliers/:idOrSlug`

Restaurants see supplier catalogs through authenticated product APIs with **server-side price resolution** (`resolveProductPricesBatch`). Contract prices override list prices per restaurant relationship.

Restaurant pricing views

- **My contract prices** — `/app/my-prices` for negotiated rates.
- **Restaurant internal pricing** — `/api/restaurant-pricing` for cost/menu pricing (separate from supplier catalog).

Warehouses and inventory (supplier)

- **Warehouses tab** — Full CRUD wired to `/api/warehouses`.
- **Per-warehouse zones** — `GET/POST /api/warehouses/:id/zones` for warehouse-scoped delivery zones (distinct from Settings Delivery Zones tab).
- **Stock levels** — `/app/inventory` → `/api/inventory`.

> **Partial feature — Supplier Settings Delivery Zones:** The Delivery Zones and Contacts tabs in Supplier Settings are **UI-only**. `DELIVERY_ZONES_ENABLED` and `CONTACTS_TAB_ENABLED` are `false` in `supplierSettingsShared.tsx`. Warehouse zone APIs exist separately; the settings tab was never wired. See `docs/audits/SUPPLIFY_DEMO_READINESS_AUDIT.md`.

Implementation evidence: `apps/api/src/routes/products.routes.js`, `apps/api/src/routes/warehouses.routes.js`, `apps/api/src/services/public-supplier-catalog.service.js`, `apps/web/src/pages/PublicSupplierCatalogPage.tsx`, `apps/web/src/components/supplier/settings/supplierSettingsShared.tsx`.

Ordering Lifecycle

Cart and placement

1. Restaurant adds items to cart (`/app/cart`) from supplier catalogs.
2. Checkout validates plan limits (orders/day), supplier connection, and minimums.
3. Order created with status `PLACED` (or `DRAFT` if saved).
4. Notifications fan out to supplier team via `notifyTenantUsers`.

Status workflow

PostgreSQL enum `order_status` (evolved across migrations `0021`, `0028`, `0069`, `0110`):

Status	Meaning
`DRAFT`	Saved, not submitted
`PLACED`	Submitted by restaurant
`PENDING_APPROVAL`	Awaiting internal approval (legacy path; `approvals_budgets` feature removed)
`ACKNOWLEDGED`	Supplier accepted
`PROCESSING`	Being picked/packed
`SHIPPED / IN_TRANSIT`	Out for delivery
`DELIVERED`	Supplier marked delivered
`RECEIVED_PARTIAL`	Restaurant received partial shipment
`RECEIVED_FULL`	Restaurant received complete shipment
`RECEIVED_WITH_DISPUTE`	Received but dispute open
`INVOICED`	Invoice generated
`COMPLETED`	Terminal success
`CANCELLED`	Cancelled or **declined** (`cancelled_by = SUPPLIER` shows "Declined by supplier")

Adjacent ordering features

Feature	Description
Quick lists	Saved templates; scheduled re-order via cron (`scheduled-orders.service.js`)
Order calendar	Redis-backed delivery calendar (`/api/orders/calendar`)
Order amendments	Post-place line changes; plan `order_amendments`
Supplier decline	Required `decline_reason`; migration `0108` cancellation metadata
Smart reorder	Forecast job `reorder-forecast.job.js`; plan `smart_reorder`

Implementation evidence: `apps/api/db/migrations/0021_update_order_status_enum.sql`, `apps/api/src/routes/orders.routes.js`, `apps/web/src/lib/orderStatusDisplay.ts`, `apps/api/src/lib/order-statuses.js`, `docs/features/order-decline.md`, `tests/e2e/suites/critical_e2e/orders.spec.ts`.

Fulfillment, Drivers, and GPS

Supplier fulfillment board

- **Route:** `/app/fulfillment`
- **APIs:** `/api/fulfillment/*`, assignment rollover endpoint, route planning (migration `0127`)
- **Features:** Assign drivers, plan routes, update assignment status, proof of delivery

Driver experience

- **Route:** `/app/driver-deliveries` (Driver role)
- **Permissions:** `DRIVER_DELIVERIES_VIEW`, `DRIVER_DELIVERIES_MANAGE` only
- **GPS:** Driver location pings; restaurant tracking panel on order detail (`RestaurantOrderTrackingPanel`)

Delivery rollover

Incomplete deliveries can roll to the next calendar day based on supplier timezone cutoff. The cron job is **registered but disabled by default**:

- `DELIVERY_ROLLOVER_ENABLED` defaults to `false` (`apps/api/src/config/env.js`, `register-cron-jobs.test.js`)
- Manual run: `node apps/api/scripts/run-delivery-rollover.mjs --force`
- Per-assignment API: `POST /api/fulfillment/assignments/:id/rollover-to-tomorrow`

> **Partial feature — Delivery rollover cron:** Operational only when env `DELIVERY_ROLLOVER_ENABLED=true`. Otherwise the hourly tick is a no-op. See `docs/operations/cron-jobs.md`.

Implementation evidence: `apps/api/src/routes/fulfillment/routes.js`, `apps/api/src/services/delivery-rollover.service.js`, `apps/web/src/pages/FulfillmentPage.tsx`, `apps/web/src/pages/DriverDeliveriesPage.tsx`, `docs/features/drivers-and-gps-tracking.md`.

Receiving and Quality

Restaurants record goods-in against orders at `/app/receiving`:

- Match delivered lines to ordered quantities
- Photo capture when plan includes `receiving_quality`
- Status transitions to `RECEIVED_PARTIAL` or `RECEIVED_FULL`
- Opens path to invoicing and inventory updates

Receiving staff role can manage receiving without order creation rights.

Implementation evidence: `apps/api/src/routes/receiving.routes.js`, `apps/web/src/pages/ReceivingPage.tsx`, `tests/api/receiving-delivered.spec.ts`.

Disputes and Returns

Disputes bridge receiving and finance:

- Open from received orders; sets status `RECEIVED_WITH_DISPUTE` (migration `0110`)
- Tracks resolution types: replacement orders, credit notes
- Plan feature `disputes_returns` gates access
- Free tier includes limited supplier free disputes (migration `0109`)

Web: `/app/disputes`, `/app/disputes/:id`
API: `/api/disputes`

Implementation evidence: `apps/api/src/services/disputes.service.js`, `apps/api/db/migrations/0072_disputes.sql`, `apps/web/src/pages/disputes/`.

Inventory

Restaurant inventory

- **Route:** `/app/restaurant-inventory`
- **Capabilities:** On-hand tracking, par levels, expiry reminders, waste analytics tab (`waste_tracking` plan feature)
- **API:** `/api/restaurant-inventory`

Supplier inventory

- **Route:** `/app/inventory`
- **Capabilities:** Stock per warehouse, reservations against orders
- **API:** `/api/inventory`

Multi-branch restaurant inventory requires `multi_branch` + `inventory_management` on appropriate tiers.

Implementation evidence: `apps/api/db/migrations/0004_restaurant_inventory.sql`, `apps/api/db/migrations/0014_restaurant_inventory_enhancements.sql`, `docs/features/waste-tracking.md`.

Finance, Invoices, and Payments

Invoice lifecycle

`DRAFT → ISSUED → PARTIALLY_PAID → PAID → VOID` with auto-invoicing from delivered orders, tax handling, and credit notes for disputes/returns.

Restaurant and supplier views

- **Shared UI:** `/app/invoices`
- **APIs:** `/api/invoices`, `/api/payments`, `/api/restaurant-finance`

Account statements

Restaurant finance APIs provide per-supplier statement views with aging analysis.

> **Partial feature — Restaurant finance opening balance:** The account statement summary sets `openingBalance: 0` with an explicit `TODO: Calculate from previous period` in `restaurant-finance.routes.js`. Charges, payments, and closing balance within the selected period are calculated; opening balance is not rolled from prior periods.

Implementation evidence: `docs/product/finance-implementation.md`, `apps/api/src/routes/invoices.routes.js`, `apps/api/src/routes/restaurant-finance.routes.js` (line ~795), `apps/api/src/jobs/invoice-overdue.job.js`.

Deals, Promotions, and Loyalty

Supplier promotions

- **Route:** `/app/promotions`
- **Workflow:** Create deal → `pending_approval` → admin approves → active
- **Boosts:** Featured placement pricing (migration `0150`)
- **API:** `/api/promotions`

Restaurant deals feed

- **Route:** `/app/deals`
- Discovery of supplier promotions; redemption limits per plan (`supplier_deals`, `supplier_deals_redeem`)

Loyalty

- Restaurant loyalty programs: `/app/loyalty`
- Consumer rewards: `/order/:slug/rewards`, `/app/consumer-loyalty`
- Migration `0160_loyalty_programs.sql`

Implementation evidence: `apps/api/src/services/deal-promotions.service.js`, `apps/api/db/migrations/0095_deal_promotions_system.sql`, `tests/api/promotions-deals-gates.spec.ts`.

Reservations (Front of House)

Restaurant cockpit

- **Route:** `/app/reservations`
- Floor plan, booking board, table assignment, waitlist
- Role `FOH Staff` limited to reservation permissions

Public guest portal

Route	Purpose
`/reserve`	Booking entry
`/reserve/:restaurantIdOrSlug`	Tenant-specific portal
`/reserve/manage/:token`	Guest cancel/reschedule
`/reserve/waitlist/:token/accept`	Waitlist offer acceptance

Waitlist auto-promotion is plan-gated (`waitlist_auto_promo`).

Implementation evidence: `apps/api/db/migrations/0033_reservations_system.sql`, `apps/api/src/routes/reservations.routes.js`, `docs/features/reservations-foh.md`.

Staff and Labour

Restaurant staff directory

- **Route:** `/app/staff`
- Roster, shifts, role assignment, staff portal account provisioning

Staff self-service portal

- **Routes:** `/staff/login`, `/staff/dashboard`
- Keycloak `STAFF_PORTAL` role; PTO and shift swaps
- Migration `0108_staff_portal_accounts.sql`
- Staff users receive **403** on main `/app` APIs

Implementation evidence: `apps/api/src/routes/staff.routes.js`, `apps/api/src/lib/staff-portal-auth.js`, `apps/web/src/components/StaffPortalGuard.tsx`.

Notifications and Realtime

Channels

Channel	Scope
In-app	Bell + toasts; `useNotificationAlerts` in Layout
Email	SMTP (Resend/Mailpit); digest job
WhatsApp	Guest notifications; settings toggle
Web Push	PWA; `/api/push`, service worker

Realtime transport

- **Socket.IO** on API origin; Redis adapter when `REDIS_URL` set
- Chat and notification alerts use shared socket connection

Chat

- **Route:** `/app/chat`
- **API:** `/api/chat`
- Daily message limits by plan (`chats_per_day` meter)
- Attachments via MinIO `/api/files`

Implementation evidence: `docs/product/notifications-summary.md` , `apps/api/src/services/notification.service.js` , `apps/api/src/routes/chat.routes.js` , `apps/web/src/hooks/useChatRealtime.ts` .

Admin Platform

Admin dashboard

- **Routes:** `/app/admin` , `/app/admin/:tab` , `/app/admin/restaurants` , `/app/admin/suppliers`
- **API:** `/api/admin-dashboard/*`

Key admin capabilities

Capability	API / UI
Tenant management	Admin → Restaurants / Suppliers tabs
Plan changes	Subscription admin actions
Feature toggles	`/api/admin-dashboard/feature-flags`
Limit overrides	`/api/admin-dashboard/limit-overrides`
Free Trial length	Platform settings (7–90 days, default 30)
Growth program config	`/api/admin-dashboard/growth-settings`
Deal approvals	Admin → Deals tab
Impersonation	`/api/admin-dashboard/impersonate`
Audit logs	`/api/admin-dashboard/audit-logs`
Health	`/api/admin-dashboard/health` (cron failures)

Admin RBAC uses `ADMIN` tenant scope with `SUPER_ADMIN` role when `ALLOW_AUTO_SUPER_ADMIN` is enabled.

Implementation evidence: `docs/sales/06_admin_and_operations.md` , `apps/api/src/routes/admin-dashboard/` , `tests/api/admin-rbac.spec.ts` , `tests/api/admin-impersonation.spec.ts` .

Subscriptions, Billing, and Entitlements

Plans

Free Trial (`free`), Silver ($49), Gold ($149), Platinum ($349) — separate plan rows per `RESTAURANT` and `SUPPLIER` tenant type. Enterprise catalog entry exists but is admin-assignment only.

Enforcement

- **Feature keys** — `apps/api/src/lib/feature-keys.js` ; checked via `requireFeature()`
- **Meters** — orders/day, chats/day, branches, products, warehouses, etc.
- **Locks** — `pending_activation` , `free_sandbox_expired` , `SUSPENDED`

Self-serve flows

- View entitlements: `GET /api/subscriptions/current` , `GET /api/subscriptions/usage/:meterType`
- Upgrade: billing checkout, `UpgradeModal` , conversion events
- Admin override: extend trial, unlock, change plan

Implementation evidence: `docs/product/subscriptions.md` , `docs/product/tier-matrix.md` , `apps/api/src/routes/subscriptions.routes.js` , `apps/api/src/lib/plan-enforcement.js` , `tests/e2e/suites/critical_e2e/subscription-limits.spec.ts` .

Consumer B2C Ordering

Restaurants can operate a **public storefront** for end consumers (distinct from B2B supplier ordering):

Route	Purpose
`/order/:restaurantSlug`	Storefront landing
`/order/:slug/menu`	Menu browse
`/order/:slug/checkout`	Guest checkout
`/order/:slug/track`	Order tracking
`/order/:slug/receipt/:token`	Receipt
`/order/:slug/account`	Light consumer account
`/order/:slug/rewards`	Loyalty rewards

Restaurant admin for B2C

- **Menu admin:** `/app/consumer-menu` → categories, items, modifiers
- **Consumer orders:** `/app/consumer-orders`
- **Fulfillment config:** Per-branch delivery/takeaway/dine-in; delivery zones on branches (API-backed, used at checkout via `deliveryZones.ts`)

Migrations `0161_consumer_ordering.sql`, `0163_consumer_b2c_complete.sql`, `0164_consumer_ordering_hours.sql`.

Implementation evidence: `apps/api/src/routes/consumer/`, `apps/api/src/services/consumer-order.service.js`, `apps/web/src/pages/consumer/`, `apps/web/src/lib/deliveryZones.ts`.

Supplier Growth Program

Suppliers acquire restaurants through:

- **CSV import** — Match existing tenants or mark import-only prospects
- **Connection requests** — Existing Supplify restaurants must accept
- **Invites** — Email, WhatsApp link, copy link for non-users
- **Sponsorship** — Pay for prospect's first month (plan limits per year)
- **Referral tokens** — `/register?ref=` → 30-day trial + first-paid discount

Route: `/app/customer-growth` (requires `GROWTH_VIEW`; import needs `CUSTOMERS_IMPORT`)

Implementation evidence: `docs/features/supplier-customer-growth.md`, `apps/api/db/migrations/0169_supplier_growth_program.sql`, `apps/api/src/jobs/sponsorship-expiry.job.js`.

Quote Requests and Supplier Mini-Store

Quote requests (RFQ)

Restaurants send multi-supplier quote requests; suppliers respond per line item; restaurants compare and optionally add winning lines to cart (manual checkout — quoted prices are informational at order create).

Route	Actor
`/app/quote-requests`	Restaurant list
`/app/quote-requests/new`	Create RFQ
`/app/quote-requests/:id`	Compare responses
`/app/quote-requests/supplier`	Supplier inbox
`/app/quote-requests/supplier/:id`	Supplier response form

API: `/api/quote-requests/*`
Notifications: `quote_request_received`, `quote_response_received`

Public mini-store

- **Route:** `/supplier/:idOrSlug` (no prices for anonymous; priced endpoint for authenticated restaurants)
- **Toggle:** `supplier.public_catalog_enabled` in settings

Implementation evidence: `docs/product/QUOTE_REQUESTS_AND_SUPPLIER_MINISTORE.md`, `apps/api/db/migrations/0153_quote_requests_and_public_catalog.sql`, `apps/api/src/services/quote-requests.service.js`.

Reports and Analytics

- **Route:** `/app/reports`
- **API:** `/api/reports`
- **Plan gate:** `reports` (basic KPIs on Silver; advanced on Gold/Platinum)
- Restaurant: spend charts, usage dashboards
- Supplier: revenue, order analytics, promotion performance

Indexes added in migration `0071_reports_analytics_indexes.sql`.

Implementation evidence: `apps/api/src/routes/reports.routes.js`, `apps/web/src/pages/reports/ReportsPage.tsx`.

Partial and Disabled Features — Summary

Feature	State	Evidence
Supplier Settings → Delivery Zones	UI tab hidden; not wired to API	`DELIVERY_ZONES_ENABLED = false` in `supplierSettingsShared.tsx`; audit in `SUPPLIFY_DEMO_READINESS_AUDIT.md`
Supplier Settings → Contacts	Same as above	`CONTACTS_TAB_ENABLED = false`
Restaurant finance opening balance	Hardcoded `0`	`restaurant-finance.routes.js` ~line 795
Delivery rollover cron	Disabled unless env flag	`DELIVERY_ROLLOVER_ENABLED` default `false`; `docs/operations/cron-jobs.md`
Quote price at checkout	Informational only	`QUOTE_REQUESTS_AND_SUPPLIER_MINISTORE.md` §5
Approvals & budgets	Removed	Migration `0114`; feature key removed

Warehouse-level delivery zones (`/api/warehouses/:id/zones`) and consumer B2C branch delivery zones **are** API-backed — only the Supplier Settings aggregate tab remains unwired.

Cron and Background Jobs

All jobs run in-process in the API server (`registerCronJobs`). Notable jobs:

Job	Interval	Notes
Scheduled quick lists	5 min dev / 1 h prod	Places scheduled orders
Subscription billing	1 h	Charges and renewals
Free sandbox expiry	1 h	Locks expired trials
Reorder forecast	24 h	Smart reorder dirty queue
Delivery rollover	1 h	**No-op unless enabled**
Growth sponsorship expiry	1 h	Referral infrastructure

Disable all: `CRONS_ENABLED=false`.

Implementation evidence: `docs/operations/cron-jobs.md`, `apps/api/src/lib/register-cron-jobs.js`, `apps/api/scripts/jobs-registry.mjs`.

Verification and Testing

Layer	Command / location
API unit tests	`pnpm --filter @supplify/api test:run`
Web unit tests	`pnpm --filter @supplify/web test:run`
E2E Playwright	`pnpm e2e:playwright`
Route inventory	`docs/audits/route-inventory.json` (554 routes)
Manual QA	`docs/qa/regression-checklist.md`
RBAC matrix tests	`apps/api/src/lib/rbac-full-app.test.js`

Master Implementation Evidence Index

Domain	Primary paths
Auth	`apps/api/src/lib/rbac.js`, `apps/api/src/routes/auth.routes.js`, `docs/features/tenant-registration.md`
Roles	`apps/api/src/lib/role-matrix.js`, `docs/architecture/rbac-permission-matrix.md`
Catalog	`apps/api/src/routes/products.routes.js`, `apps/web/src/pages/ProductsPage.tsx`
Ordering	`apps/api/src/routes/orders.routes.js`, `apps/web/src/pages/OrdersPage.tsx`
Fulfillment	`apps/api/src/routes/fulfillment/`, `apps/api/src/services/delivery-rollover.service.js`
Receiving	`apps/api/src/routes/receiving.routes.js`
Disputes	`apps/api/src/services/disputes.service.js`
Inventory	`apps/api/src/routes/restaurant-inventory.routes.js`, `apps/api/src/routes/inventory.routes.js`
Finance	`apps/api/src/routes/invoices.routes.js`, `apps/api/src/routes/restaurant-finance.routes.js`
Deals	`apps/api/src/services/deal-promotions.service.js`
Reservations	`apps/api/src/routes/reservations.routes.js`, `apps/api/src/routes/public.routes.js`
Staff	`apps/api/src/routes/staff.routes.js`, `apps/api/src/lib/staff-portal-auth.js`
Notifications	`apps/api/src/services/notification.service.js`
Admin	`apps/api/src/routes/admin-dashboard/`
Subscriptions	`apps/api/src/routes/subscriptions.routes.js`, `docs/product/tier-matrix.md`
Consumer B2C	`apps/api/src/routes/consumer/`, `apps/api/db/migrations/0161_consumer_ordering.sql`
Growth	`apps/api/db/migrations/0169_supplier_growth_program.sql`
Quote requests	`apps/api/src/services/quote-requests.service.js`
Chat	`apps/api/src/routes/chat.routes.js`
Reports	`apps/api/src/routes/reports.routes.js`
Feature catalog	`docs/product/features.md`, `docs/product/feature-catalog-full.md`
Frontend routes	`apps/web/src/App.tsx` (80 routes)
Migrations	`apps/api/db/migrations/` (175 files)

Part III — Supplier Onboarding Guide

End-to-end onboarding for a **supplier** tenant: from first login through catalog, fulfillment, billing, and day-two operations. Routes and APIs match `apps/web/src/App.tsx` and the live API surface as of the current codebase.

Primary persona: Supplier owner or ops manager with `Org Owner` / `SETTINGS_MANAGE` permissions unless noted.

Step 1 — Create your Supplify login (Keycloak)

Field	Detail
Goal	Obtain a Supplify identity before any tenant exists.
Who	Future supplier owner (no tenant yet).
Navigation path	`/login` → **Register** (redirects to `/auth/register` → Keycloak hosted registration).
Required data	Email, password, name (Keycloak fields).
Expected result	Keycloak account exists; first app login creates `app_user` with `role: PENDING`.
Possible errors	Duplicate email in Keycloak; email verification required (realm policy); network/CORS to auth server.
Validation checklist	[ ] Can open `/login` without console errors. [ ] Register completes in Keycloak. [ ] First login redirects away from `/login`.

API: OAuth/session via Keycloak; app session established through normal auth middleware. `GET /api/auth/me` returns `role: "PENDING"` until registration completes.

Step 2 — Complete supplier organization setup

Field	Detail
Goal	Create supplier tenant, organization, default catalog, warehouse scaffold, and subscription row.
Who	Authenticated user with `PENDING` role.
Navigation path	Auto-redirect to `/register/complete` (also enforced by `AuthGuard` when `needsSetup === true`).
Required data	Account type **Supplier**, business name (required), phone (optional), legal acceptance checkboxes. Optional `referralToken` from `/register?ref=…`.
Expected result	`POST /api/register/complete` with `accountType: "SUPPLIER"` creates `supplier`, `supplier_organizations`, default `catalog`, system roles, and `subscription` with `lock_reason = pending_activation`. User role becomes `SUPPLIER`. Redirect to `/app/activate`.
Possible errors	`409` — email already linked to a tenant; `409` — user already has workspace membership; validation on empty business name.
Validation checklist	[ ] `GET /api/register/status` → `{ needsSetup: false }` after complete. [ ] `GET /api/auth/me` → `role: "SUPPLIER"`, tenant id present. [ ] `/app/activate` loads (other `/app/*` routes blocked until unlocked).

API: `GET /api/register/status`, `POST /api/register/complete` (`accountType`, `businessName`, `phone?`, `referralToken?`, legal payload).

Step 3 — Activate your workspace (billing)

Field	Detail
Goal	Clear `pending_activation` so write APIs and full navigation unlock.
Who	Supplier owner on new tenant.
Navigation path	`/app/activate`
Required data	**Activate free plan** (no card) or paid plan via upgrade modal (`POST /api/billing/checkout` with `planId`; stub card `4242424242424242` when `BILLING_GATEWAY=stub`).
Expected result	`account_locked_at` cleared; `GET /api/billing/status` → `access.pendingActivation: false`, `access.isLocked: false`. Redirect to `/app` (supplier home).
Possible errors	`402` on writes while still locked; checkout validation for paid tiers; plan catalog empty if migrations/seeds not run.
Validation checklist	[ ] Free activation succeeds without payment method. [ ] Sidebar appears with supplier sections. [ ] `POST` to catalog/orders no longer returns activation lock.

API: `GET /api/billing/status`, `POST /api/billing/checkout`, `GET /api/subscriptions/entitlements`.

Step 4 — Configure supplier profile & branding

Field	Detail
Goal	Publish trustworthy business identity for restaurants and public mini-store.
Who	User with `SETTINGS_VIEW` / `SETTINGS_MANAGE`.
Navigation path	Sidebar **Settings** → `/app/settings` (renders `SupplierSettingsPage`) or deep link `/app/supplier-settings?tab=profile`
Required data	Legal name, logo, contact email/phone, address, VAT/tax IDs, public slug (used at `/supplier/:idOrSlug`).
Expected result	`GET /api/suppliers/me` reflects updates; `PATCH /api/suppliers/:id` persists profile fields; public catalog page shows branding.
Possible errors	Duplicate slug; permission denied without `SETTINGS_MANAGE`; image upload failures (storage config).
Validation checklist	[ ] Profile tab saves without error. [ ] `/supplier/{slug}` loads publicly. [ ] Logo and name visible on supplier detail for restaurants.

API: `GET /api/suppliers/me`, `PATCH /api/suppliers/:id`.

Step 5 — Business policies, hours, and terms

Field	Detail
Goal	Set MOQ, payment terms, return policy, business hours, blackout dates restaurants see at order time.
Who	Supplier settings manager.
Navigation path	`/app/settings` → **Business** tab (`?tab=business`)
Required data	Minimum order amount, payment terms text, business hours JSON, holiday/blackout dates (optional).
Expected result	Business rules stored on `supplier` row; enforced or displayed in catalog/checkout flows per product docs.
Possible errors	Invalid JSON for hours; numeric validation on MOQ.
Validation checklist	[ ] Business tab persists after refresh. [ ] MOQ visible on supplier profile or order validation where applicable.

API: `PATCH /api/suppliers/:id` (business fields from onboarding migration `0005_supplier_onboarding.sql`).

Step 6 — Warehouses & fulfillment mode

Field	Detail
Goal	Define ship-from locations and fulfillment behavior (required before inventory-backed dispatch on higher tiers).
Who	User with `WAREHOUSES_VIEW` / warehouse manage permissions.
Navigation path	`/app/settings` → **Warehouses** tab (`?tab=warehouses`)
Required data	Warehouse name, address, active flag; fulfillment toggles via `PATCH /api/suppliers/me/fulfillment` .
Expected result	At least one active warehouse; fulfillment settings align with plan entitlements (`fulfillment` feature).
Possible errors	Plan limit on warehouse count; feature gated on Free tier.
Validation checklist	[ ] Default warehouse exists post-registration. [ ] Can add/edit warehouse on entitled plan. [ ] Fulfillment page respects warehouse selection.

API: Warehouse CRUD under `/api/suppliers/me/warehouses` ; `PATCH /api/suppliers/me/fulfillment` .

Step 7 — Team members, roles, and invitations

Field	Detail
Goal	Delegate catalog, orders, fulfillment, or driver access without sharing owner credentials.
Who	Owner or user with staff/team permissions.
Navigation path	`/app/settings` → **Team & roles** (`?tab=team`)
Required data	Invitee email, role (system or custom), optional name.
Expected result	Invite email sent; invitee accepts at `/invite?token=…&type=…` → `POST /api/invites/accept` ; user bound to org workspace.
Possible errors	Seat/role limits on plan; email mismatch on invite accept; expired token.
Validation checklist	[ ] Invite link opens `/invite` . [ ] New member sees sidebar scoped to permissions. [ ] Driver role only sees **My Deliveries** (`/app/driver-deliveries`).

API: Invite validate/accept routes; org role assignment via tenant RBAC.

Step 8 — Register drivers (Gold+ / `driver_management`)

Field	Detail
Goal	Create driver records linked to users for last-mile delivery.
Who	Supplier with `driver_management` entitlement and settings access.
Navigation path	`/app/settings` → **Drivers** tab (`?tab=drivers`)
Required data	Driver name, phone, linked user account (invite or existing user).
Expected result	Driver appears in fulfillment dispatch board assignee list; linked user gets `DRIVER_DELIVERIES_VIEW` .
Possible errors	Feature not on plan (Silver has fulfillment but not `driver_management`); user already linked to another driver; `403` from `requireFeature('driver_management')` .
Validation checklist	[ ] Driver list loads on Gold+. [ ] Assign driver action visible on `/app/fulfillment` . [ ] Driver can log in and open `/app/driver-deliveries` .

API: Driver CRUD under supplier settings / fulfillment APIs.

Step 9 — Build and publish product catalog

Field	Detail
Goal	List SKUs restaurants can browse, price, and order.
Who	User with `CATALOG_VIEW` / `CATALOG_EDIT` .
Navigation path	Sidebar **Products** → `/app/products` ; detail → `/app/products/:id`
Required data	Product name, SKU, unit, price, category, status (`ACTIVE`), images, stock/availability as applicable.
Expected result	`GET /api/products` lists tenant products; restaurants with relationship see items in `/app/products` (restaurant view).
Possible errors	Plan SKU limits; validation on duplicate SKU; inactive catalog.
Validation checklist	[ ] Create product succeeds. [ ] Product visible to test restaurant account. [ ] Edit from detail page persists.

API: `GET /api/products` , `POST /api/products` , `PATCH /api/products/:id` , `GET /api/products/:id` .

Step 10 — Contract pricing for key accounts

Field	Detail
Goal	Offer negotiated prices per restaurant without changing list price globally.
Who	Catalog/pricing manager.
Navigation path	Sidebar **Contract Pricing** → `/app/contract-pricing`
Required data	Restaurant target, product or category, contract price, effective dates.
Expected result	Restaurant sees overrides on **My Prices** (`/app/my-prices` on their side); order lines use contract price when matched.
Possible errors	Restaurant not linked; overlapping contract windows.
Validation checklist	[ ] Contract row saved. [ ] Restaurant **My Prices** shows entry. [ ] Test order reflects contract line price.

API: Contract pricing endpoints under `/api/contract-pricing` (see feature catalog).

Step 11 — Promotions and supplier deals

Field	Detail
Goal	Run time-boxed promotions visible in marketplace and restaurant **Deals** tab.
Who	User with `PROMOTIONS_VIEW` / `PROMOTIONS_MANAGE` on entitled plan.
Navigation path	Sidebar **Deals** → `/app/promotions`
Required data	Deal type, discount, eligibility, schedule, optional payment for featured placement.
Expected result	Active promotion on `GET /api/suppliers` (`has_store_deal`); restaurants browse `/app/deals` .
Possible errors	Promotion limit per plan; admin approval required for some deal types.
Validation checklist	[ ] Promotion activates within schedule. [ ] Badge on supplier list. [ ] Discount applies at cart/checkout.

API: `/api/promotions/*` supplier routes.

Step 12 — Customer growth & restaurant acquisition

Field	Detail
Goal	Invite restaurants via referral links and track conversion.
Who	User with `GROWTH_VIEW` (plan + permission).
Navigation path	Sidebar **Customer Growth** → `/app/customer-growth`
Required data	Referral link (`/register?ref=…`); optional growth campaign metadata.
Expected result	Restaurant signup with `referralToken` records attribution, auto-follow, and trial/discount eligibility per `supplier-customer-growth` docs.
Possible errors	Growth feature disabled on plan; invalid referral token.
Validation checklist	[ ] Referral URL copies from UI. [ ] Test restaurant registration attributes referral. [ ] Restaurant appears under **Restaurants** (`/app/restaurants`).

API: `POST /api/register/complete` with `referralToken` ; growth analytics on supplier growth page.

Step 13 — Acknowledge and process restaurant orders

Field	Detail
Goal	Move orders from placed → acknowledged → processing → shipped.
Who	User with `ORDERS_VIEW` / `ORDERS_MANAGE` .
Navigation path	Sidebar **Orders** → `/app/orders` ; detail → `/app/orders/:id`
Required data	Order id; status transitions; line adjustments if supported.
Expected result	Order timeline updates; restaurant sees mirrored status; notifications fire per preferences.
Possible errors	Invalid status transition; billing lock (`402`) on expired trial.
Validation checklist	[ ] Pending badge on sidebar decrements when orders handled. [ ] Status change reflected on restaurant `/app/orders/:id` . [ ] Chat thread available if enabled.

API: `GET /api/orders` , `GET /api/orders/:id` , `PATCH /api/orders/:id` (status updates).

Step 14 — Fulfillment dispatch board

Field	Detail
Goal	Assign drivers, pick/pack/ship, and track delivery lifecycle (Silver+ `fulfillment`).
Who	User with `FULFILLMENT_VIEW` / `FULFILLMENT_MANAGE`.
Navigation path	Sidebar **Fulfillment** → `/app/fulfillment` (tabs: Dispatch, Routes, Delivery Tracking, Exceptions)
Required data	Warehouse context, driver assignment, delivery status per order.
Expected result	Board shows Unassigned → Assigned → Picked up → Out for delivery → Delivered/Failed; `PATCH /api/orders/:id/delivery-status` is canonical.
Possible errors	Feature off on Free; no driver on Silver (manual ship only); assignment to unlinked driver.
Validation checklist	[ ] Dispatch tab loads on Silver+. [ ] Assign driver updates card. [ ] Delivery tracking drawer shows GPS when enabled.

API: `GET /api/supplier/deliveries/board`, `PATCH /api/orders/:id/delivery-status`, `POST /api/fulfillment/routes`, fulfillment route stop APIs.

Step 15 — Planned routes and activation

Field	Detail
Goal	Batch orders into routes before dispatch-ready, then activate when orders hit `PROCESSING` / `SHIPPED`.
Who	Fulfillment manager.
Navigation path	`/app/fulfillment` → **Routes** tab; dispatch board **Assign to planned route**
Required data	Driver, order selection, route date; activation on ready stops.
Expected result	`delivery_route.status` moves `PLANNED` → `IN_PROGRESS`; eligible stops sync to live dispatch; GPS begins when assignments reach `picked_up` / `out_for_delivery`.
Possible errors	Order on two active routes; cancelled order auto-removed; activate with zero ready stops (route still may start per docs).
Validation checklist	[ ] Planned route badge on dispatch cards. [ ] Activate ready orders promotes stops. [ ] Restaurant map hidden until dispatch starts (privacy).

API: `POST /api/fulfillment/routes`, `POST /api/fulfillment/routes/:id/stops`, `PATCH /api/fulfillment/routes/:id` `{ status: "IN_PROGRESS" }`.

Step 16 — Invoices and revenue visibility

Field	Detail
Goal	Issue and track invoices where `finance_invoices` entitlement is enabled.
Who	User with `INVOICES_VIEW`.
Navigation path	Sidebar **Invoices** → `/app/invoices`
Required data	Linked orders, invoice lines, tax fields per jurisdiction setup.
Expected result	Invoice list and PDF/export per implementation; restaurant sees matching invoice.
Possible errors	Feature not on plan; order not in invoiceable state.
Validation checklist	[ ] Invoice generates from delivered order. [ ] Restaurant `/app/invoices` shows record. [ ] Totals match order lines.

API: `/api/invoices/*` (tenant-scoped).

Step 17 — Command center, dashboard, and reports

Field	Detail
Goal	Monitor ops KPIs, GPS summary, and analytics exports.
Who	Owner/analyst with analytics permissions.
Navigation path	**Command Center** → `/app/command-center`; **Dashboard** → `/app/dashboard`; **Reports** → `/app/reports` (plan-gated)
Required data	Date filters, report type selection.
Expected result	`GET /api/admin/dashboard` returns supplier-scoped stats when impersonating; command center shows GPS today summary; reports export when entitled.
Possible errors	`reports` feature off; empty data on new tenant.
Validation checklist	[ ] Home `/` or `/app` loads supplier home. [ ] Command center shows fulfillment/GPS widgets on Gold+. [ ] Reports page accessible on entitled plan.

API: `GET /api/admin/dashboard` (supplier tenant context), reports endpoints under `/api/reports`.

Step 18 — Disputes and delivery exceptions

Field	Detail
Goal	Resolve quantity/quality issues and failed deliveries with audit trail.
Who	User with fulfillment/dispute permissions.
Navigation path	Sidebar **Disputes** → `/app/disputes` ; detail → `/app/disputes/:id`
Required data	Dispute reason, evidence, resolution notes.
Expected result	Dispute state machine progresses; linked order/receiving records updated per workflow.
Possible errors	Feature `disputes_returns` disabled; permission denied.
Validation checklist	[ ] Create/respond to dispute from list. [ ] Detail page shows order link. [ ] Sidebar badge clears when resolved.

API: `/api/disputes/*` .

Step 19 — Public mini-store and quote inbox

Field	Detail
Goal	Let prospects browse catalog publicly; respond to restaurant RFQs.
Who	Sales/catalog staff.
Navigation path	Public `/supplier/:idOrSlug` ; **Quote inbox** → `/app/quote-requests/supplier` → `/app/quote-requests/supplier/:quoteRequestSupplierId`
Required data	Published products; quote line pricing, lead times, notes.
Expected result	Guest/restaurant browsing without login (optional auth); quote responses visible to restaurant on `/app/quote-requests/:id` .
Possible errors	Unpublished catalog; quote deadline passed.
Validation checklist	[ ] Public URL works logged out. [ ] Quote appears in inbox. [ ] Response visible to restaurant.

API: `GET /api/suppliers/:id` , quote request supplier endpoints.

Step 20 — Reporting, health checks, and troubleshooting

Field	Detail
Goal	Diagnose common production issues without platform admin access.
Who	Supplier owner or support lead.
Navigation path	`/app/settings` → **Plan & usage** (`?tab=plan`); **Activity** tab if `tenant_audit_log` enabled; `/app/chat` for support threads
Required data	Symptom description, order id, driver id, timestamp, browser/device for driver GPS issues.
Expected result	Entitlements explain missing nav items; billing status explains `402` writes; audit shows recent settings changes.
Possible errors	Trial expired (read-only GET allowed, writes `402`); GPS env disabled (`VITE_GPS_TRACKING_ENABLED=false`); plan limit exceeded.
Validation checklist	[ ] `GET /api/billing/status` matches UI lock state. [ ] `GET /api/subscriptions/entitlements` explains missing **Fulfillment** nav. [ ] Driver GPS: location permission + `POST /api/orders/:id/location` succeeds. [ ] Escalate to admin with tenant id + order id.

Quick troubleshooting reference

Symptom	Likely cause	Check
Stuck on `/register/complete`	`needsSetup` true	`GET /api/register/status`
All writes fail `402`	Trial expired or locked	`GET /api/billing/status`
No Fulfillment nav	Plan or feature flag	Entitlements `fulfillment`
No driver assign	`driver_management` off	Upgrade to Gold+
GPS stale on map	Driver permission / env	`GPS_STALE_AFTER_SECONDS` , driver browser
Restaurant cannot see tracking	Dispatch not started	Assignment must be `picked_up` or `out_for_delivery`

Support escalation payload: tenant id (supplier uuid), `subscription_id` , affected `orderId` , screenshot of `/app/fulfillment` or driver portal, and `requestId` from failed API response.

Part IV — Restaurant Onboarding Guide

End-to-end onboarding for a **restaurant** tenant: registration, profile, procurement, receiving, and ongoing operations. Routes and APIs are sourced from `apps/web/src/App.tsx` and live API handlers.

Primary persona: Restaurant owner or purchasing manager unless noted.

Step 1 — Create login and restaurant tenant

Field	Detail
Goal	Register identity and provision restaurant organization.
Who	New restaurant owner (`PENDING` → `RESTAURANT`).
Navigation path	`/login` → Register (`/auth/register`) → `/register/complete`
Required data	Account type **Restaurant**, business name, phone (optional), legal acceptance. Optional `referralToken` from supplier invite (`/register?ref=…`).
Expected result	`POST /api/register/complete` with `accountType: "RESTAURANT"` creates `restaurant` , org, roles, pending subscription; redirect `/app/activate` .
Possible errors	Duplicate tenant email; user already in workspace; validation errors on business name.
Validation checklist	[ ] `GET /api/auth/me` → `role: "RESTAURANT"` . [ ] `/register/complete` not shown again after success. [ ] Activation gate appears.

API: `GET /api/register/status` , `POST /api/register/complete` .

Step 2 — Activate subscription (free or paid)

Field	Detail
Goal	Unlock ordering, inventory, and settings writes.
Who	Restaurant owner.
Navigation path	`/app/activate`
Required data	Free activation (`POST /api/billing/checkout` without card) or paid checkout (stub: `4242424242424242`).
Expected result	`pending_activation` cleared; full sidebar available per entitlements.
Possible errors	Billing middleware blocks routes until unlocked; checkout failure.
Validation checklist	[ ] Navigate to `/app/orders` without redirect to activate. [ ] `GET /api/billing/status` → not locked.

API: `GET /api/billing/status` , `POST /api/billing/checkout` , `GET /api/subscriptions/entitlements` .

Step 3 — Complete restaurant profile (Settings hub)

Field	Detail
Goal	Set legal identity, branding, and operational metadata suppliers see.
Who	Owner or `SETTINGS_MANAGE` .
Navigation path	Sidebar **Settings** → `/app/settings` (restaurant renders `RestaurantOnboardingPage`) or `/app/onboarding?tab=profile`
Required data	Restaurant name, logo, contact email/phone, business type, address.
Expected result	`GET /api/restaurants/me` returns updated profile; suppliers see name on orders.
Possible errors	Permission denied; invalid phone/email format.
Validation checklist	[ ] Profile tab saves. [ ] Summary KPIs on settings header populate after orders exist. [ ] Refresh retains values.

API: `GET /api/restaurants/me` , `PATCH /api/restaurants/me` .

Step 4 — Delivery location coordinates (GPS / ETA)

Field	Detail
Goal	Enable accurate delivery ETA and map destination for supplier drivers (supplier sees pin; restaurant map is driver-only for privacy).
Who	Owner or branch manager.
Navigation path	`/app/onboarding?tab=profile` → **Delivery location** section (or branch settings)
Required data	Latitude, longitude, label, notes; per-branch via branch detail.
Expected result	`PATCH /api/restaurants/me/delivery-location` or `PATCH /api/restaurants/branches/:branchId/delivery-location` stored; tracking shows `destinationCoordinatesAvailable` and ETA when driver active.
Possible errors	Invalid coordinates; branch permission denied.
Validation checklist	[ ] Coordinates saved on main tenant. [ ] Active delivery shows ETA on order detail tracking panel. [ ] Text address alone does not substitute (coords required).

API: `GET/PATCH /api/restaurants/me/delivery-location` , branch variant on `branches/:branchId` .

Step 5 — Invite team and assign roles

Field	Detail
Goal	Add purchasers, receivers, FOH staff with least-privilege roles.
Who	Owner (`SETTINGS_MANAGE` / team permissions).
Navigation path	`/app/onboarding?tab=team`
Required data	Email, role (e.g. receiver, ordering, admin), optional name.
Expected result	Invite link `/invite?token=…&type=rm` (restaurant member) or branch invite `/invite/branch` ; accept binds workspace.
Possible errors	Email mismatch on accept; `advanced_roles` plan gate; seat limits.
Validation checklist	[ ] Invitee completes `/invite` flow. [ ] Sidebar matches role (e.g. receiver sees **Receiving**). [ ] Owner can revoke/adjust role.

API: Invite validate `GET` , accept `POST /api/invites/accept` .

Step 6 — Branches and operational locations

Field	Detail
Goal	Model multi-site restaurants with branch-scoped orders and inventory.
Who	Owner or org admin.
Navigation path	`/app/onboarding?tab=branches` ; branch detail → `/app/org/branches/:supplierId` (org overview `/app/org`)
Required data	Branch name, code, address, delivery location per branch.
Expected result	Orders and quick lists can scope to branch; branch switcher in header when entitled.
Possible errors	Plan branch limit (e.g. Silver: 1 branch); slug conflicts.
Validation checklist	[ ] Branch appears in list. [ ] Branch invite flow works (`/invite/branch`). [ ] Orders attribute to correct branch.

API: Restaurant branch CRUD under `/api/restaurants/branches/*` .

Step 7 — Discover and follow suppliers

Field	Detail
Goal	Build a supplier portfolio for catalog browsing and ordering.
Who	User with `CATALOG_VIEW` .
Navigation path	Sidebar **Suppliers** → `/app/suppliers` ; detail → `/app/suppliers/:id` ; public mini-store `/supplier/:idOrSlug`
Required data	None to browse; follow action to add relationship.
Expected result	`GET /api/suppliers` lists marketplace; follow creates `restaurant_supplier_follow` ; `is_followed` true on detail.
Possible errors	`suppliers_per_restaurant` plan limit on follow; catalog empty until supplier publishes.
Validation checklist	[ ] Supplier list loads with ratings/deal badges. [ ] Follow toggles state. [ ] Followed supplier products appear in **Products**.

API: `GET /api/suppliers` , `GET /api/suppliers/:id` , follow endpoints on supplier relationships router.

Step 8 — Browse catalog, contract prices, and deals

Field	Detail
Goal	Find SKUs at list or negotiated prices before ordering.
Who	Purchaser (`CATALOG_VIEW` , `ORDERS_CREATE`).
Navigation path	**Products** → `/app/products` ; **My Prices** → `/app/my-prices` ; **Deals** → `/app/deals`
Required data	Product filters; supplier relationship for contracted SKUs.
Expected result	Contract prices override list where configured; active deals apply at cart.
Possible errors	No supplier follow — empty catalog; SKU limit on plan.
Validation checklist	[ ] Product detail `/app/products/:id` shows correct unit price. [ ] **My Prices** lists contract rows. [ ] Deal badge visible on supplier with `has_store_deal` .

API: `GET /api/products` , contract pricing read APIs, `/api/promotions` restaurant-facing routes.

Step 9 — Ordering lists (quick lists) and scheduled reorders

Field	Detail
Goal	Save par lists and recurring order templates (`quick_lists` entitlement).
Who	Purchaser.
Navigation path	**Ordering Lists** → `/app/quick-lists`
Required data	List name, lines (product, qty), optional schedule/branch scope.
Expected result	One-click add to cart from list; scheduled orders notify per preferences.
Possible errors	Feature off on plan; product unavailable from supplier.
Validation checklist	[ ] List creates and opens. [ ] Add all to cart populates `/app/cart` . [ ] Scheduled notification preference enabled in settings.

API: Quick list endpoints under restaurant ordering module.

Step 10 — Cart checkout and place orders

Field	Detail
Goal	Submit purchase orders to suppliers.
Who	User with `ORDERS_CREATE` .
Navigation path	**Cart** → `/app/cart` → submit; confirmation in **Orders**
Required data	Cart lines, delivery branch, requested date, PO notes; supplier MOQ met.
Expected result	`POST` order creation → `customer_order` `PLACED` ; supplier sees `/app/orders` ; notifications sent.
Possible errors	Below supplier MOQ; `402` billing lock; daily order limit on plan; out-of-stock SKU.
Validation checklist	[ ] Order appears in `/app/orders` with pending badge. [ ] Supplier acknowledges order. [ ] Order detail `/app/orders/:id` shows lines and status timeline.

API: Cart and `POST /api/orders` (or checkout flow used by `CartPage`).

Step 11 — Quote requests (RFQ)

Field	Detail
Goal	Request custom pricing when catalog price is insufficient.
Who	Purchaser.
Navigation path	**Quote requests** → `/app/quote-requests` ; new → `/app/quote-requests/new` ; detail → `/app/quote-requests/:id`
Required data	Supplier(s), line items, quantities, needed-by date, notes.
Expected result	Suppliers respond via quote inbox; restaurant compares offers on detail page.
Possible errors	Supplier not accepting quotes; validation on empty lines.
Validation checklist	[ ] RFQ creates with status visible. [ ] Supplier response appears. [ ] Convert to order if workflow supported.

API: Quote request CRUD under `/api/quote-requests` .

Step 12 — Track orders and live delivery

Field	Detail
Goal	Monitor fulfillment and see driver location during active delivery (privacy-safe).
Who	User with `ORDERS_VIEW` .
Navigation path	`/app/orders/:id` → `RestaurantOrderTrackingPanel`
Required data	Order in shipped/dispatch states; delivery location coordinates on file.
Expected result	`GET /api/orders/:id/tracking` returns sanitized payload — driver pin only (no destination coords on restaurant map); ETA when `picked_up` or `out_for_delivery` ; 30s poll in UI.
Possible errors	Tracking hidden until dispatch starts; `GPS_ALLOW_RESTAURANT_LIVE_TRACKING` false; no driver GPS.
Validation checklist	[ ] No map before dispatch. [ ] Driver marker appears after pickup/out for delivery. [ ] Driver phone hidden by default. [ ] **Receive order** links to receiving (no auto-receive from GPS).

API: `GET /api/orders/:id/tracking` (restaurant-sanitized via `restaurant-tracking-payload.js`).

Step 13 — Receiving deliveries

Field	Detail
Goal	Confirm quantities received and close the procurement loop (status → `COMPLETED` post-receiving).
Who	User with `RECEIVING_VIEW` / receiving manage.
Navigation path	**Receiving** → `/app/receiving`
Required data	Order id, received quantities, variances, optional photos.
Expected result	Accepts orders in `DELIVERED` or `COMPLETED`; inventory updated when inventory module linked.
Possible errors	Order not in receivable state; permission denied.
Validation checklist	[ ] Delivered order appears in receiving queue. [ ] Confirm updates order status. [ ] Inventory reflects receipt if enabled.

API: Receiving endpoints under `/api/receiving` (see `receiving.md` feature doc).

Step 14 — Restaurant inventory and expiry

Field	Detail
Goal	Track on-hand stock, lots, and expiry by branch.
Who	User with `INVENTORY_VIEW`.
Navigation path	**Inventory** → `/app/restaurant-inventory`
Required data	SKU quantities, lot dates, branch, reorder thresholds.
Expected result	Stock levels adjust from receiving; low-stock notifications if enabled.
Possible errors	Inventory feature plan-gated; branch scope mismatch.
Validation checklist	[ ] On-hand matches post-receiving. [ ] Expiring lots flagged. [ ] Low stock notification preference works.

API: Restaurant inventory routes under `/api/restaurant-inventory` or equivalent module.

Step 15 — Subscription, plan, and notifications

Field	Detail
Goal	Manage billing tier and alert channels.
Who	Owner.
Navigation path	`/app/onboarding?tab=subscription` and `?tab=notifications`; user-level prefs in `/app/settings` (account section)
Required data	Plan selection for upgrade; notification toggles (email, in-app, WhatsApp).
Expected result	Plan change via checkout; `PATCH` notification preferences persist.
Possible errors	Downgrade blocked by usage over new limits; payment failure.
Validation checklist	[ ] Current plan shown with usage meters. [ ] Upgrade modal opens from plan tab. [ ] Order notification test fires on new order.

API: `GET /api/billing/status`, `POST /api/billing/checkout`, notification preference endpoints.

Step 16 — Invoices and spend tracking

Field	Detail
Goal	Reconcile supplier invoices against orders (`finance_invoices` entitlement).
Who	User with `INVOICES_VIEW`.
Navigation path	**Invoices** → `/app/invoices`
Required data	Invoice id from supplier; payment status notes.
Expected result	Invoice list with totals; settings summary shows `totalSpent` KPI.
Possible errors	Feature disabled on plan; no invoices until supplier issues.
Validation checklist	[ ] Invoice list loads. [ ] Totals align with order history. [ ] Export/open PDF if available.

API: `GET /api/invoices`.

Step 17 — Disputes and returns

Field	Detail
Goal	Open disputes for short ships, damage, or quality issues (`disputes_returns` feature).
Who	User with dispute permissions (`RESTAURANT_DISPUTES_ANY_OF`).
Navigation path	**Disputes** → `/app/disputes` ; detail → `/app/disputes/:id`
Required data	Order/line reference, reason, photos/notes.
Expected result	Dispute visible to supplier on `/app/disputes` ; status updates both sides.
Possible errors	Feature off; order not eligible window.
Validation checklist	[ ] Create dispute from order or list. [ ] Sidebar badge updates. [ ] Resolution recorded.

API: `/api/disputes/*` .

Step 18 — Chat and supplier collaboration

Field	Detail
Goal	Coordinate substitutions, delivery instructions, and account issues in-app.
Who	User with `CHAT_VIEW` .
Navigation path	**Chat** → `/app/chat`
Required data	Supplier thread selection; message body.
Expected result	`GET /api/chat/conversations` lists B2B threads (excludes admin support threads from list per audit); real-time or polled messages.
Possible errors	`chats_per_day` limit on Free; supplier not linked.
Validation checklist	[ ] Open thread with followed supplier. [ ] Message delivers both directions. [ ] Unread badge in header.

API: `/api/chat/conversations` , message POST endpoints.

Step 19 — Reports, reviews, hospitality add-ons, and troubleshooting

Field	Detail
Goal	Use analytics and optional modules; resolve common blockers without admin help.
Who	Owner, GM, or IT contact.
Navigation path	**Reports** → `/app/reports` (entitled); `/app/onboarding?tab=reviews` ; hospitality: `/app/reservations` , `/app/staff` , `/app/consumer-menu` , `/app/consumer-orders` , `/app/consumer-loyalty`
Required data	Report date range; review responses; guest-facing slug for `/order/:restaurantSlug` consumer storefront.
Expected result	Reports export on entitled plans; reviews tab shows supplier ratings you can respond to; reservations and guest ordering modules work when respective permissions enabled.
Possible errors	Reports feature off; reservations/staff plan gates; consumer routes need published menu.
Validation checklist	[ ] Reports page loads or shows upgrade CTA. [ ] Reviews tab lists recent supplier reviews. [ ] Guest menu admin saves (`/app/consumer-menu`). [ ] Public guest order path `/order/{slug}/menu` works.

Troubleshooting reference

Symptom	Likely cause	Action
Cannot place order `402`	Trial expired / locked	`/app/onboarding?tab=subscription` or contact admin
Empty **Products**	No supplier follows	Follow suppliers on `/app/suppliers`
No live tracking map	Dispatch not started	Wait for `picked_up` / `out_for_delivery`
ETA missing	No delivery coordinates	Set delivery location (Step 4)
Missing nav item	Plan entitlement	`GET /api/subscriptions/entitlements`
Stuck at activation	`pending_activation`	`/app/activate` or admin unlock

Escalation: Provide restaurant tenant id, order uuid, browser console errors, and API `requestId` from failed responses. Platform admin can impersonate via `/app/admin` → Tenants → **Impersonate**.

Part V — Driver Onboarding Guide

Operational guide for **supplier-linked drivers** using the web driver portal (PWA-friendly). Covers login, deliveries, routes, GPS, status updates, proof of delivery (POD), failures, privacy, and troubleshooting.

Primary persona: User with supplier **Driver** role (`DRIVER_DELIVERIES_VIEW` / `DRIVER_DELIVERIES_MANAGE`), linked to a `drivers` row.

Home route: `/app/driver-deliveries` (sidebar **My Deliveries** under DELIVERIES section — only nav item for driver role).

Step 1 — Receive credentials and log in

Field	Detail
Goal	Access Supplify with a driver-linked account.
Who	New driver (invited by supplier admin).
Navigation path	`/login` (or invite flow `/invite?token=…` if email invite sent)
Required data	Email and password (Keycloak); accept legal terms on invite if applicable.
Expected result	`GET /api/auth/me` returns supplier tenant context with driver permissions; sidebar shows only **My Deliveries**.
Possible errors	User not linked to `drivers` table — fulfillment APIs return `403`; wrong role shows full supplier nav (not driver).
Validation checklist	[ ] Login succeeds. [ ] Redirect to `/app/driver-deliveries` or home with driver nav only. [ ] No access to `/app/fulfillment` without extra permissions.

API: Standard auth session; driver linkage verified by `requireLinkedDriver` on order/fulfillment mutations.

Step 2 — Open the driver deliveries board

Field	Detail
Goal	See all assignments for today and standalone deliveries.
Who	Active driver.
Navigation path	Sidebar **My Deliveries** → `/app/driver-deliveries`
Required data	None — board loads assigned orders from supplier dispatch.
Expected result	`GET /api/supplier/deliveries/board` returns orders with `deliveryStatus`, restaurant name, delivery area, schedule; active vs completed sections.
Possible errors	`403` if not linked driver; empty board if no assignments; feature `driver_management` off at supplier (no assignments created).
Validation checklist	[ ] Page loads without error. [ ] Assigned orders visible. [ ] Refresh button refetches board and route.

Step 3 — Understand delivery statuses and allowed actions

Field	Detail
Goal	Know which buttons appear for each assignment state.
Who	Driver.
Navigation path	`/app/driver-deliveries` — each `DriverDeliveryCard`
Required data	Current `deliveryStatus` on assignment.
Expected result	Status `assigned` / `pending` → actions: **I'm on the way** (`out_for_delivery`), **Problem** (`failed`), **Reschedule** (`rescheduled`). Status `picked_up` / `out_for_delivery` → **Delivered**, **Problem**, **Reschedule**. Terminal: `delivered`, `failed`, `rescheduled` — no further driver actions.
Possible errors	Invalid transition rejected by API with message toast.
Validation checklist	[ ] Primary action label matches status. [ ] Completed orders move to completed section or route stop marked complete.

API: `PATCH /api/orders/:id/delivery-status` with body `{ status, notes?, failure_reason? }` (canonical). Assignment statuses: `assigned` → `picked_up` → `out_for_delivery` → `delivered` | `failed` | `rescheduled`.

Step 4 — Update status: “I'm on the way” and delivered

Field	Detail
Goal	Progress deliveries so restaurants get notifications and live tracking when entitled.
Who	Driver on assigned order.
Navigation path	`/app/driver-deliveries` → card action buttons or sticky action bar for next stop
Required data	Order id; optional notes in textarea per card.
Expected result	`out_for_delivery` starts restaurant-visible tracking window; `delivered` sets `customer_order.status = DELIVERED` and triggers `notifyOrderStatusChange(DELIVERED)`.
Possible errors	Not assigned to order; supplier does not own order; concurrent update conflict.
Validation checklist	[ ] Toast “Delivery status updated”. [ ] Restaurant order shows updated status. [ ] GPS tracking becomes active on `out_for_delivery` (see Step 7).

API: `PATCH /api/orders/:id/delivery-status` via `useUpdateOrderDeliveryStatusMutation`.

Step 5 — Build a route from multiple standalone deliveries

Field	Detail
Goal	Group 2+ active assignments into one ordered route when supplier did not plan a route.
Who	Driver with 2+ eligible standalone deliveries and no active route today.
Navigation path	`/app/driver-deliveries` — **Build my route** card (shown when `standaloneEligibleCount ≥ 2` and no `activeRoute`)
Required data	Optional `{ date: "YYYY-MM-DD" }` (defaults today).
Expected result	`POST /api/fulfillment/routes/build-from-assignments` creates `IN_PROGRESS` route `{Driver name} – Today's route`; orders move into `DriverRoutePanel`; idempotent on repeat.
Possible errors	Fewer than 2 eligible orders; orders already on another route; API error toast "Could not build route".
Validation checklist	[ ] Route panel appears with ordered stops. [ ] Standalone cards for routed orders hidden from active list. [ ] Supplier sees route on `/app/fulfillment` Routes tab.

Step 6 — Navigate the route panel (stops, next stop, reorder)

Field	Detail
Goal	Follow stop sequence and adjust order manually in the field.
Who	Driver on active route (`IN_PROGRESS` or today's planned route).
Navigation path	`/app/driver-deliveries` → **Today's route** (`DriverRoutePanel`)
Required data	Active route from `GET /api/fulfillment/routes/active` (alias `GET /api/fulfillment/routes/today`).
Expected result	Next incomplete stop highlighted; **Set as next** via `PATCH /api/fulfillment/routes/:id/next-stop { orderId }`; move up/down via `POST /api/fulfillment/routes/:id/stops/reorder { stop_ids: uuid[] }`.
Possible errors	Cannot reorder completed/failed stops; route not owned by driver.
Validation checklist	[ ] Next stop badge on correct card. [ ] Reorder persists after refresh. [ ] Route stop status updates sync with order delivery status.

API: `GET /api/fulfillment/routes/active`, `PATCH …/next-stop`, `POST …/stops/reorder`, `PATCH …/routes/:routeId/stops/:stopId` for stop-level status.

Step 7 — GPS permission and live location sharing

Field	Detail
Goal	Share live position during active deliveries for supplier maps and restaurant ETA.
Who	Driver on trackable assignment (`assigned`, `picked_up`, `out_for_delivery` per `isTrackableDeliveryStatus`).
Navigation path	`/app/driver-deliveries` — header GPS banner (`DriverDeliveriesHeader` + `useDriverLocationTracking`)
Required data	Browser location permission; `VITE_GPS_TRACKING_ENABLED` not `false`; device GPS on.
Expected result	`navigator.geolocation.watchPosition` sends pings every `VITE_GPS_UPDATE_INTERVAL_SECONDS` (default 15s) via `POST /api/orders/:id/location` with lat/lng/accuracy; banner shows **Location active**.
Possible errors	Permission denied → banner **Location permission needed**; unsupported browser; server `GPS_TRACKING_ENABLED=false`; send failure → **Location not updating**.
Validation checklist	[ ] Browser prompts for location on first visit. [ ] Banner green/active during delivery. [ ] Supplier **View tracking** drawer shows live/stale marker. [ ] `driver_location_ping` rows created server-side.

API: `POST /api/orders/:id/location` — body: `latitude`, `longitude`, `accuracyMeters`, optional `speedMps`, `headingDegrees`, `recordedAt`.

Client env: `VITE_GPS_TRACKING_ENABLED`, `VITE_GPS_UPDATE_INTERVAL_SECONDS`.

Step 8 — Open Maps for turn-by-turn navigation

Field	Detail
Goal	Navigate to restaurant delivery area using external maps (Supplify does not provide turn-by-turn in-app).
Who	Driver.
Navigation path	Each delivery card → **Open Maps** link
Required data	`deliveryArea` text or restaurant name for query string.
Expected result	Opens `https://maps.google.com/?q={encoded destination}` in new tab; min 48px touch target for mobile.
Possible errors	"Delivery area not set" if supplier/restaurant omitted address text (coords still help supplier-side ETA).
Validation checklist	[ ] Link opens maps app/site. [ ] Works on mobile Safari/Chrome.

Step 9 — Proof of delivery (POD)

Field	Detail
Goal	Attach delivery evidence (photo, recipient name, GPS at delivery).
Who	Driver or supplier fulfillment manager on assigned order.
Navigation path	Order actions on driver card after delivery; supplier may upload on `/app/fulfillment` dispatch board (“POD on file” / “No POD”)
Required data	`file_key` (uploaded asset), optional `recipient_name`, `notes`, `latitude` / `longitude`, `driver_assignment_id`.
Expected result	`POST /api/orders/:id/proof-of-delivery` creates `proof_of_delivery` row; `delivery_gps_lat/lng` stored when coordinates sent; `hasPod` true on board.
Possible errors	Upload/storage failure; order not in delivered state for auto-assignment lookup.
Validation checklist	[ ] POD submits after photo upload. [ ] Supplier dispatch shows “POD on file”. [ ] Restaurant can confirm POD where receiving workflow supports it (`confirmProofOfDelivery`).

API: `POST /api/orders/:id/proof-of-delivery`, `GET /api/orders/:id/proof-of-delivery`.

Step 10 — Failed delivery and reschedule

Field	Detail
Goal	Record exceptions when delivery cannot complete.
Who	Driver.
Navigation path	`/app/driver-deliveries` → **Problem** (failed) or **Reschedule**
Required data	Notes strongly recommended — used as `failure_reason` when status is `failed` (defaults to “Delivery failed” if empty).
Expected result	`failed` notifies supplier via `notifyDriverDeliveryMilestone`; order remains visible in supplier exceptions; `rescheduled` sets warning state for replanning. Route stop updated when on route via `handleRouteStopStatus`.
Possible errors	Missing permission; cannot fail already terminal stop.
Validation checklist	[ ] Failed stop shows danger badge. [ ] Supplier fulfillment exceptions list includes issue. [ ] Notes visible on order timeline.

API: `PATCH /api/orders/:id/delivery-status` with `status: "failed"` and `failure_reason`; route stop `PATCH` with `status: "FAILED"`.

Step 11 — Privacy rules (driver, restaurant, supplier)

Field	Detail
Goal	Understand what each party can see about location and identity.
Who	Driver (read); compliance-aware ops.
Navigation path	N/A — behavior enforced server-side
Required data	N/A
Expected result	**Restaurant:** live map only after `picked_up` / `out_for_delivery`; driver pin only (no destination coordinates on restaurant map); driver phone hidden unless `GPS_RESTAURANT_SHOW_DRIVER_PHONE=true`; no route stop list or ping history. **Supplier:** full tracking drawer with destination pin and GPS stale/live states. **Driver:** shares pings only for assigned active orders; no email per ping.
Possible errors	N/A
Validation checklist	[ ] Restaurant cannot see map before dispatch. [ ] Driver name visibility follows `GPS_RESTAURANT_SHOW_DRIVER_NAME`. [ ] Pings stop when no trackable deliveries.

Reference: `docs/features/drivers-and-gps-tracking.md` — Privacy section; `restaurant-tracking-payload.js`.

Step 12 — Sticky action bar and mobile UX

Field	Detail
Goal	Complete deliveries one-handed on phone.
Who	Driver on mobile viewport.
Navigation path	`/app/driver-deliveries` — `DriverStickyActionBar` at bottom
Required data	Next standalone order or next route stop computed client-side.
Expected result	Primary action for next delivery always visible; large touch targets (48px min height on buttons).
Possible errors	Bar hidden when no active deliveries.
Validation checklist	[ ] Sticky bar shows correct next stop. [ ] Action matches card primary button. [ ] Scroll does not hide critical actions.

Step 13 — Show completed deliveries

Field	Detail
Goal	Review finished work for the day.
Who	Driver.
Navigation path	`/app/driver-deliveries` → toggle **Show completed**
Required data	None.
Expected result	Terminal orders (`delivered` , `failed` , `rescheduled`) listed; counts in header (`activeCount` / `doneCount`).
Possible errors	None.
Validation checklist	[ ] Completed section expands. [ ] Done count matches deliveries finished.

Step 14 — PWA installation and home-screen use

Field	Detail
Goal	Install Supplify as home-screen app for faster daily access.
Who	Driver on supported mobile browser.
Navigation path	Browser menu → **Add to Home Screen** / **Install app** (after visiting `https://{your-host}/login` and logging in)
Required data	HTTPS origin; valid service worker if configured in web build.
Expected result	Standalone window opens to last session; driver lands on deliveries after auth.
Possible errors	iOS requires Safari for add-to-homescreen; third-party cookies/session may expire — re-login needed.
Validation checklist	[ ] Icon on home screen launches app shell. [ ] Login session persists reasonable duration. [ ] GPS permission survives per browser rules.

Step 15 — PWA and field troubleshooting

Field	Detail
Goal	Resolve common driver-side failures without supplier IT.
Who	Driver or dispatcher coaching driver.
Navigation path	`/app/driver-deliveries` + device settings
Required data	Symptom, order id, browser, OS version.
Expected result	Issue classified and fixed per table below.
Possible errors	N/A
Validation checklist	[ ] Hard refresh retried. [ ] Location permission re-granted. [ ] Supplier notified if server-side assignment missing.

Troubleshooting matrix

Symptom	Cause	Fix
Blank deliveries page	No assignments	Confirm supplier assigned you on `/app/fulfillment`
`403` on status update	Not linked driver	Supplier **Settings** → **Drivers** link user
GPS banner “permission needed”	Browser blocked location	Site settings → Allow location; iOS: Settings → Safari → Location
GPS “not updating”	Network/API error	Check mobile data; retry; verify `POST …/location` in network tab
Tracking stale on supplier map	Ping older than `GPS_STALE_AFTER_SECONDS` (300s default)	Keep app foreground; check interval env
Build route missing	<2 standalone deliveries	Wait for more assignments or use supplier-planned route
Actions disabled	`updating` in flight	Wait for prior request; refresh page
Logged out unexpectedly	Session timeout	`/login` again; use PWA add-to-homescreen after login
Wrong nav (full supplier menu)	User has non-driver roles	Use driver-only account or supplier adjusts roles
Maps opens wrong place	Missing delivery area text	Use restaurant name; ask supplier to fix address

Escalation to supplier dispatch

Provide: driver name, order uuid (`formatOrderRef` on card), current status shown in UI, screenshot of GPS banner, and time of failure. Supplier verifies on `/app/fulfillment` → **View tracking** and `GET /api/orders/:id/tracking` .

Server env (supplier ops): `GPS_TRACKING_ENABLED` , `GPS_STALE_AFTER_SECONDS` , `GPS_MIN_ACCURACY_METERS` , `GPS_ALLOW_RESTAURANT_LIVE_TRACKING` .

API quick reference (driver)

Method	Path	Purpose
GET	`/api/supplier/deliveries/board`	Driver delivery list
GET	`/api/fulfillment/routes/active`	Today's route + stops
POST	`/api/fulfillment/routes/build-from-assignments`	Build route from assignments
PATCH	`/api/orders/:id/delivery-status`	Status updates
POST	`/api/orders/:id/location`	GPS ping
GET	`/api/orders/:id/tracking`	Tracking read (driver assigned)
POST	`/api/orders/:id/proof-of-delivery`	Submit POD
PATCH	`/api/fulfillment/routes/:id/stops/reorder`	Reorder stops
PATCH	`/api/fulfillment/routes/:id/next-stop`	Set next stop

Plan requirement: Supplier must be on plan with `driver_management` (Gold+) for driver CRUD and assignments; driver portal itself requires linked driver user regardless of driver device.

Part VI — Platform Admin Onboarding Guide

Guide for **Supplify platform administrators** managing tenants, billing, support, diagnostics, and operational health. UI routes live under `/app/admin*`; APIs under `/api/admin-dashboard` and selected `/api/admin` endpoints.

Primary persona: User with `role: ADMIN` and granular `adminPermissions` (`ADMIN_ACCESS`, `ADMIN_TENANTS`, `ADMIN_PLANS`, `ADMIN_FINANCE`, `ADMIN_SUPPORT`, `ADMIN_GROWTH`).

Step 1 — Admin login and permission model

Field	Detail
Goal	Access admin consoles with correct scoped permissions.
Who	Platform admin (seeded e.g. `admin@supplify.com` via `pnpm run seed:demo-users`).
Navigation path	`/login` → `/app/admin` (default landing)
Required data	Admin Keycloak credentials; ADMIN realm role.
Expected result	Sidebar shows **ADMIN** section (Admin Dashboard, Supplier Admin, Restaurant Admin, Settings); `GET /api/auth/me` → `role: "ADMIN"`; tabs gated by `canAdminTab` in `AdminDashboardPage`.
Possible errors	Missing admin permissions hide tabs (fallback to first allowed tab); non-admin user cannot access `/app/admin`.
Validation checklist	[ ] `/app/admin` loads overview. [ ] Tabs match permission set. [ ] `GET /api/admin-dashboard/overview` returns 200.

Permission map (UI tabs):

Tab	Permission
Overview, Activity, Health, Operations, Audit	`ADMIN_ACCESS`
Tenants	`ADMIN_TENANTS`
Users	`ADMIN_SUPPORT`
Plans, Subscriptions, Usage, Limits	`ADMIN_PLANS`
Finance	`ADMIN_FINANCE`
Features, Deals	`ADMIN_GROWTH`

Step 2 — Navigate admin portals (platform vs tenant-type)

Field	Detail
Goal	Use the correct portal for cross-tenant vs supplier-only vs restaurant-only work.
Who	Admin with `ADMIN_TENANTS` or `ADMIN_ACCESS`.
Navigation path	`/app/admin` (platform) · `/app/admin/suppliers` · `/app/admin/restaurants` · tab deep links e.g. `/app/admin/subscriptions`, `/app/admin/suppliers/audit`
Required data	None.
Expected result	Platform portal exposes full nav groups (Monitor, Accounts, Billing, Growth). Supplier/restaurant portals limit tabs to **Directory**, **Usage & quotas**, **Audit log** per `adminNavConfig.ts`.
Possible errors	Invalid tab segment redirects to default tab for portal.
Validation checklist	[ ] Portal switcher highlights correct portal. [ ] Supplier portal pins **Tenants** tab to supplier directory. [ ] URL bookmarking restores tab.

Step 3 — Platform overview and activity feed

Field	Detail
Goal	Monitor signups, conversions, and recent platform events.
Who	`ADMIN_ACCESS` .
Navigation path	`/app/admin` or `/app/admin/overview` → **Overview**; **Activity** → `/app/admin/activity`
Required data	Date filters on activity as exposed in UI.
Expected result	`GET /api/admin-dashboard/overview` returns KPI metrics; `GET /api/admin-dashboard/activity` returns feed (tenant creation, subscriptions, impersonation, etc.).
Possible errors	Empty feed on fresh environment (expected).
Validation checklist	[ ] Overview cards render counts. [ ] Activity shows events after test tenant signup. [ ] Conversion stats load via `GET /api/admin-dashboard/conversion-stats` .

Step 4 — Tenant directory (suppliers and restaurants)

Field	Detail
Goal	Search, filter, and act on any tenant row.
Who	`ADMIN_TENANTS` .
Navigation path	`/app/admin/tenants` or `/app/admin/suppliers` or `/app/admin/restaurants`
Required data	Search string; status filter (ACTIVE, TRIALING, PAST_DUE, SUSPENDED, CANCELLED, NONE).
Expected result	`GET /api/admin-dashboard/tenants/suppliers` and … `/tenants/restaurants` paginate with plan, revenue/spend, subscription id; client-side filter on loaded pages.
Possible errors	`403` without `ADMIN_TENANTS` ; empty list if DB not seeded.
Validation checklist	[ ] Search matches name/email. [ ] Status filter works. [ ] Row actions visible (impersonate, change plan, diagnostics, password reset).

API: `GET /api/admin-dashboard/tenants/suppliers` , `GET /api/admin-dashboard/tenants/restaurants` , `GET /api/admin-dashboard/tenants/search?q=` .

Step 5 — Create supplier tenant (admin API)

Field	Detail
Goal	Provision supplier without self-service registration (migrations, demos, enterprise onboarding).
Who	`ADMIN` role (API-only today — `POST /api/suppliers` admin route).
Navigation path	API: `POST /api/suppliers` (no first-class wizard in tenants UI; use API client or internal tooling)
Required data	`name` , `slug` , `contactEmail` , optional `vatNo` , `phone` , `address` .
Expected result	`201` with `supplier` row; `createPendingActivationSubscription` (free, `pending_activation`); `ensureTenantSystemRoles` . Owner must still be linked via separate user invite/workspace binding.
Possible errors	Duplicate slug; validation `400` ; missing admin auth `403` .
Validation checklist	[ ] Supplier appears in admin tenants list. [ ] Subscription row exists with pending activation. [ ] Tenant searchable via `tenants/search` .

Note: Restaurant admin create may follow self-service `/register/complete` or future admin API — supplier create is explicitly in `apps/api/src/routes/suppliers/admin.js` .

Step 6 — Subscriptions list and filters

Field	Detail
Goal	View all subscription rows across tenants for billing operations.
Who	`ADMIN_PLANS` .
Navigation path	`/app/admin/subscriptions`
Required data	Optional query filters `status` , `tenantType` (SUPPLIER
Expected result	`GET /api/admin-dashboard/subscriptions` returns deduped active/trialing preference per tenant with plan metadata.
Possible errors	Large lists slow without filters.
Validation checklist	[ ] Suspended/past-due counts in summary strip. [ ] Tenant name/email columns populated. [ ] Row links to change-plan dialog when subscription id present.

Step 7 — Unlock pending activation accounts

Field	Detail
Goal	Clear `pending_activation` when tenant cannot self-activate (support scenario).
Who	`ADMIN_PLANS` .
Navigation path	`/app/admin/subscriptions` → row action **Unlock**
Required data	`subscription_id` ; optional reason in audit.
Expected result	`POST /api/admin-dashboard/subscriptions/:id/unlock` calls `unlockSubscriptionAccount` ; tenant can write immediately.
Possible errors	Subscription not found; not in lockable state.
Validation checklist	[ ] Tenant user reaches `/app` without `/app/activate` redirect. [ ] `GET /api/billing/status` shows unlocked. [ ] Audit log entry created.

Classification: Safe support action (reversible by re-lock only through billing rules; no data deletion).

Step 8 — Extend free trial / sandbox

Field	Detail
Goal	Extend expired or expiring free sandbox for a tenant.
Who	`ADMIN_PLANS` .
Navigation path	`/app/admin/subscriptions` → **Extend trial** (expired trial rows)
Required data	`days` between 7–90 (`clampFreeTrialDays` platform settings).
Expected result	`POST /api/admin-dashboard/subscriptions/:id/extend-free-trial` calls `extendFreeSandboxTrial` ; writes restored until new expiry.
Possible errors	Not on free/trial plan; days out of range.
Validation checklist	[ ] Trial end date updated in UI. [ ] Tenant writes succeed after expiry had blocked them. [ ] Event in admin activity feed.

Classification: Safe — extends timeboxed access; audited.

Step 9 — Change plan (preview and apply)

Field	Detail
Goal	Move tenant between Free/Silver/Gold/Platinum/Enterprise with impact preview.
Who	`ADMIN_PLANS` .
Navigation path	Tenants or Subscriptions tab → **Change plan** (`AdminChangePlanDialog`)
Required data	Target `planId` ; optional `force` , `reason` , `applyAtPeriodEnd` , `allowExceedance` for over-limit tenants.
Expected result	`POST /api/admin-dashboard/subscriptions/:id/preview-change` shows usage vs limits diff; `PATCH /api/admin-dashboard/subscriptions/:id` applies change and invalidates entitlement cache.
Possible errors	Downgrade blocked by usage unless `force: true` ; enterprise validation failures.
Validation checklist	[ ] Preview lists feature/limit deltas. [ ] Tenant nav reflects new entitlements after cache refresh. [ ] `GET /api/admin-dashboard/tenants/:type/:id/entitlements` matches.

Classification: Moderate — `force: true` is **dangerous** (can leave tenant over limits or remove critical features mid-operation).

Step 10 — Impersonate tenant (support)

Field	Detail
Goal	Reproduce tenant issues in-app as their workspace.
Who	`ADMIN_SUPPORT` / tenants actions.
Navigation path	`/app/admin/suppliers` or `/app/admin/restaurants` → row **Impersonate**
Required data	`tenantId` , `tenantType` (`RESTAURANT`
Expected result	`POST /api/admin-dashboard/impersonate` sets signed cookie; redirect to `/app/dashboard` ; banner shows impersonation; `GET /api/admin-dashboard/impersonate` returns status. Effective tenant from `getEffectiveTenant(req)` — billing lock **still applies** when impersonating locked tenant.
Possible errors	`TENANT_SUSPENDED` requires confirmation; cannot impersonate admin email; `403` forbidden.
Validation checklist	[ ] Banner visible while impersonating. [ ] Sidebar matches tenant type. [ ] `POST /api/admin-dashboard/impersonate/stop` ends session. [ ] Actions audited in audit log.

Classification: Moderate — read/write as tenant; stop impersonation when done. Not a blank check — permissions use effective tenant RBAC.

Step 11 — Stop impersonation

Field	Detail
Goal	Return to platform admin context.
Who	Impersonating admin.
Navigation path	Impersonation banner → **Stop** (or API)
Required data	Active impersonation cookie.
Expected result	`POST /api/admin-dashboard/impersonate/stop` clears cookie; redirect admin shell; platform stats on `GET /api/admin/dashboard` again.
Possible errors	Already stopped — idempotent.
Validation checklist	[ ] Admin sidebar returns. [ ] `/app/admin` accessible. [ ] Audit entry for stop.

Step 12 — User support and password reset

Field	Detail
Goal	Help users who cannot log in (Keycloak password reset).
Who	`ADMIN_SUPPORT` .
Navigation path	`/app/admin/users` → user row → reset dialog (`AdminResetPasswordDialog`)
Required data	Target user id/email; new temporary password per policy.
Expected result	`POST /api/admin-dashboard/users/reset-password` via `adminResetUserPassword` ; user can log in at `/login` .
Possible errors	User not found; Keycloak admin API failure.
Validation checklist	[ ] User confirms login with new password. [ ] Audit log records reset. [ ] User changes password in Keycloak if required.

Classification: Moderate — security-sensitive; verify identity out-of-band before reset.

Step 13 — Tenant diagnostics drawer

Field	Detail
Goal	Read-only operational snapshot before deep support.
Who	Admin on tenants tab.
Navigation path	Tenants row → **Diagnostics** (stethoscope) → `AdminTenantDiagnosticsDrawer`
Required data	`tenantId` , `tenantType` .
Expected result	`GET /api/admin-dashboard/tenants/:tenantType/:id/operational-snapshot` plus entitlements and usage endpoints show: subscription status, trial end, `writeBlocked` , effective feature flags, supplier GPS today counts / fulfillment issues / pending deals, restaurant expiry & quick list stats & tracking privacy flags, email provider health, recent email failures, usage vs limits.
Possible errors	Drawer loading timeout on large tenant.
Validation checklist	[ ] Subscription section matches subscriptions tab. [ ] Write blocked flag matches tenant complaint. [ ] Links to Limits/Features tabs work.

API: `GET …/operational-snapshot` , `GET …/entitlements` , `GET …/tenants/suppliers/:id/usage` , `GET …/tenants/restaurants/:id/usage` .

Step 14 — Operations panel (email, inventory, fulfillment, GPS)

Field	Detail
Goal	Platform-wide operational triage.
Who	`ADMIN_ACCESS` .
Navigation path	`/app/admin/operations` with sub-tabs: summary, email, inventory, fulfillment, gps
Required data	None; optional filters per sub-panel.
Expected result	`GET /api/admin-dashboard/operational-summary` ; `…/operational/email-logs` ; `…/operational/fulfillment-issues` ; `…/operational/active-deliveries` populate `AdminOperationsPanel` .
Possible errors	Email provider misconfiguration surfaces as danger alerts.
Validation checklist	[ ] Summary shows actionable counts. [ ] Email failures list recent rows. [ ] Active deliveries map data consistent with supplier fulfillment.

Step 15 — Health check and platform settings

Field	Detail
Goal	Verify API dependencies and tune global trial defaults.
Who	`ADMIN_ACCESS` (settings patch); health read for all admin access.
Navigation path	`/app/admin/health`; platform settings on overview or dedicated settings surfaces
Required data	For patch: `freeTrialDays` (7–90), other keys in platform settings schema.
Expected result	`GET /api/admin-dashboard/health` returns component status; `GET/PATCH /api/admin-dashboard/platform-settings` for defaults; growth settings under `ADMIN_GROWTH`.
Possible errors	Validation on out-of-range trial days.
Validation checklist	[ ] Health page not degraded in prod. [ ] Trial default matches new registrations after patch.

Classification: Moderate — platform settings affect all new tenants.

Step 16 — Plans catalog management

Field	Detail
Goal	Maintain `subscription_plan` rows (limits, features, pricing).
Who	`ADMIN_PLANS`.
Navigation path	`/app/admin/plans`
Required data	Plan code, tenant type, limits JSON, features JSON, prices; enterprise activation rules.
Expected result	`GET /api/admin-dashboard/plans`; `POST /api/admin-dashboard/plans`; `PATCH /api/admin-dashboard/plans/:id` with validation (`validatePlanLimitsAndFeatures`, tier ladder warnings).
Possible errors	Invalid limit keys; enterprise plan validation failure.
Validation checklist	[ ] `pnpm run seed:tier-catalog` baseline present. [ ] Edit reflects in tenant entitlements after cache invalidation. [ ] Tier ladder warnings shown in UI when applicable.

Classification: Dangerous — misconfigured plan affects all subscribers on that plan.

Step 17 — Limits and overrides

Field	Detail
Goal	Temporarily raise caps or set per-tenant limit exceptions.
Who	`ADMIN_PLANS`.
Navigation path	`/app/admin/limits` (`AdminLimitsTab`)
Required data	Tenant or plan id, `limitKey`, `override_value`, **reason** (required in UI).
Expected result	`POST /api/admin-dashboard/tenants/:tenantType/:id/override-limit`; plan-level `POST … /plans/:planId/override-limit`; effective value via `GET … /effective-limit/:limitKey`.
Possible errors	Unknown limit key for tenant type; missing reason.
Validation checklist	[ ] Override appears in list. [ ] Tenant can perform action previously blocked. [ ] Delete override restores plan default.

Safe actions: Read effective limits; add small temporary override with documented reason.

Dangerous actions: Large or permanent overrides without expiry; deleting overrides during active billing dispute.

Step 18 — Feature flags (global and per-tenant)

Field	Detail
Goal	Toggle platform features without code deploy.
Who	`ADMIN_GROWTH` (features tab); tenant overrides may need `ADMIN_PLANS` + features routes.
Navigation path	`/app/admin/features`
Required data	`featureKey`, enabled boolean; tenant override path includes tenant id.
Expected result	`GET /api/admin-dashboard/feature-flags`; `PATCH /api/admin-dashboard/feature-flags/:featureKey`; per-tenant `PUT/DELETE … /tenants/:tenantType/:id/feature-overrides/:featureKey`.
Possible errors	Unknown feature key rejected by `getAllowedFeatureKeys`.
Validation checklist	[ ] Global flag toggles behavior in staging. [ ] Tenant override wins over global in diagnostics drawer. [ ] Clear override restores inheritance.

Classification: Dangerous — enabling experimental flags in production; always verify in impersonation session first.

Step 19 — Finance overview

Field	Detail
Goal	Monitor MRR, overdue amounts, and billing health.
Who	`ADMIN_FINANCE` .
Navigation path	`/app/admin/finance`
Required data	None.
Expected result	`GET /api/admin-dashboard/financial-overview` returns aggregates for executive summary.
Possible errors	Stripe/gateway misconfig shows zeros or errors.
Validation checklist	[ ] KPIs load. [ ] Overdue count tone danger when >0. [ ] Cross-check with subscriptions tab.

Step 20 — Deals moderation and growth

Field	Detail
Goal	Approve/reject supplier deals requiring platform moderation.
Who	`ADMIN_GROWTH` .
Navigation path	`/app/admin/deals` (`AdminDealsPanel`)
Required data	Deal id; approve/reject/pause actions.
Expected result	Admin promotion routes e.g. `POST /api/promotions/admin/:id/approve` , `reject` , `pause` ; pending queue `GET /api/promotions/admin/pending` .
Possible errors	Deal already terminal state.
Validation checklist	[ ] Pending queue matches supplier submissions. [ ] Approve makes deal visible on restaurant `/app/deals` . [ ] Reject notifies supplier.

Classification: Safe approve/reject with audit; **moderate** pause active paid deals.

Step 21 — Audit log (compliance)

Field	Detail
Goal	Immutable trace of admin actions.
Who	`ADMIN_ACCESS` .
Navigation path	`/app/admin/audit`
Required data	Filters: actor, action, resource, date range.
Expected result	`GET /api/admin-dashboard/audit-logs` (paginated); legacy `GET /api/admin/audit` on `admin_audit_log` table.
Possible errors	Large date range slow — use filters.
Validation checklist	[ ] Impersonation events tagged. [ ] Plan changes include actor. [ ] Password resets logged without exposing new password.

Step 22 — Safe vs dangerous actions (reference)

Generally safe (read or low blast radius)

Action	API / UI
View overview, health, diagnostics	GET overview, health, operational-snapshot
Search tenants	`GET … /tenants/search`
Impersonate read-only investigation	Impersonate + navigate (avoid writes unless intended)
Unlock pending activation	`POST … /subscriptions/:id/unlock`
Extend free trial (7–90 days)	`POST … /extend-free-trial`
Audit log export/review	`GET … /audit-logs`
Approve/reject pending deals	promotions admin routes

Moderate (requires ticket + confirmation)

Action	Risk
Password reset	Account takeover if misdirected
Impersonation with writes	Changes attributed to tenant users
Plan change without force	May fail — use preview first
Per-tenant limit override	Billing/usage skew
Platform settings patch	Affects new signups globally

Dangerous (dual control / change window)

Action	Risk
`PATCH` subscription with `force: true`	Downgrade removing features mid-order; data over limits
Plan catalog edit on live plans	All tenants inherit bad limits/features
Global feature flag enable	Unvetted behavior platform-wide
Tenant feature override delete	Unexpected feature loss
`POST /api/suppliers` without owner linkage	Orphan tenant rows
Suspend/cancel subscription without comms	Production outage for tenant

Billing impersonation note: Admins **not** impersonating skip billing middleware; impersonating **does** enforce tenant lock (`billingAccess.test.js`) — unlock or extend trial before expecting writes on locked tenant.

Step 23 — Support workflow (end-to-end)

Field	Detail
Goal	Standard triage path from ticket to resolution.
Who	Support admin (`ADMIN_SUPPORT + ADMIN_TENANTS`).
Navigation path	Tenants search → Diagnostics → (optional) Impersonate → Subscriptions/limits → Stop impersonate → Audit
Required data	Tenant name/email, issue type, order id if logistics, screenshots, `requestId` from API errors.
Expected result	Issue classified: activation, plan limit, feature flag, logistics (escalate to supplier), auth (password reset). Document resolution in external ticket with audit reference id.
Possible errors	Skipping diagnostics leads to wrong plan changes.
Validation checklist	[ ] Tenant identified in search. [ ] Diagnostics `writeBlocked` explains 402. [ ] Impersonation reproduces issue. [ ] Fix verified under tenant context. [ ] Impersonation stopped. [ ] Audit entry exists.

API mount reference

Prefix	Router
`/api/admin-dashboard`	`apps/api/src/routes/admin-dashboard/index.js`
`/api/admin`	`apps/api/src/routes/admin.routes.js` (audit, role-aware dashboard stats)

Web route reference

Path	Purpose
`/app/admin`	Platform dashboard
`/app/admin/:tab`	Tab deep link (overview, tenants, subscriptions, …)
`/app/admin/suppliers`	Supplier tenant portal
`/app/admin/suppliers/:tab`	Supplier portal tab
`/app/admin/restaurants`	Restaurant tenant portal
`/app/admin/restaurants/:tab`	Restaurant portal tab
`/app/settings`	Admin personal settings (notifications, Keycloak link)

QA cross-reference: `docs/qa/regression-checklist.md` Part 0 (setup), admin sections; stub card `4242424242424242` for billing tests.

Part VII — Technical Architecture *(Internal Technical Reference)*

Supplify is a **pnpm monorepo** with a React SPA (`apps/web`), an Express API (`apps/api`), and shared infrastructure (PostgreSQL, Redis, MinIO/S3, Keycloak). Production runs on **Railway** as separate Docker services; local development can use native hot-reload (`pnpm dev`) or the full **Docker Compose** stack.

System overview

Layer	Technology	Location
Frontend	Vite 5, React 18, Redux Toolkit + RTK Query	`apps/web/`
API	Node 22, Express 4, ESM	`apps/api/src/`
Auth	Keycloak OIDC (authorization code + refresh)	`apps/api/src/lib/auth.js`
Cache / pub-sub	Redis (`ioredis`)	`apps/api/src/lib/cache.js` , `socket-redis-adapter.js`
Object storage	Local filesystem or S3-compatible (MinIO)	`apps/api/src/services/storage/`
Real-time	Socket.IO on shared HTTP server	`apps/api/src/lib/socket.js`
DB	PostgreSQL 16, 175 SQL migrations	`apps/api/db/migrations/`

Frontend (Vite / React / RTK)

Build & dev server

- **Bundler:** Vite with `@vitejs/plugin-react` (`apps/web/vite.config.ts`).
- **Dev port:** `5173` ; proxies `/api` , `/auth` , and `/socket.io` to `http://localhost:4000` .
- **Production build:** manual chunk splitting (`react-vendor` , `redux-vendor` , `charts` , etc.) to keep route-level code lazy.

State management

Redux store (`apps/web/src/store/index.ts`):

Slice / API	Purpose
`auth`	Current user from `/auth/me`
`cart`	Restaurant ordering cart
`monetization`	Plan limits / upsell state
`billing`	Billing UI state
`api` (RTK Query)	All HTTP — injected endpoint modules

RTK Query base client (`apps/web/src/services/api/base.ts`):

- `credentials: 'include'` — sends HttpOnly auth cookies.
- Unwraps API envelope `{ ok, data, error }` .
- Redirects to `/login` on `401` / `JWT_EXPIRED` .

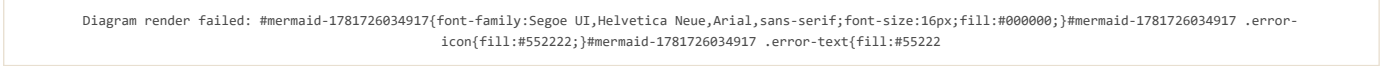
Endpoint modules are side-effect imports in `apps/web/src/services/api/index.ts` (orders, inventory, admin, staff portal, billing, etc.).

Routing & auth shell

- `AuthGuard` (`apps/web/src/components/AuthGuard.tsx`) calls `useGetMeQuery` , hydrates `auth` slice, and gates `/app/*` .
- `STAFF_PORTAL` users are redirected away from `/app` to `/staff/dashboard` .
- Permission-gated UI uses `usePermissions()` (`apps/web/src/hooks/usePermissions.ts`) — mirrors backend `hasPermission` semantics including `_MANAGE` supersets.

Backend — Express middleware chain

Entry point: `apps/api/src/server.js` . Middleware order is fixed and security-sensitive.



Ordered stack (summary)

#	Middleware	Scope	Role
1	`requestTimingMiddleware`	global	Slow-request structured logging
2	`compression()`	global	Gzip JSON responses
3	`helmet()`	global	CSP, HSTS (prod), CORP cross-origin for images
4	`cors()`	global	`WEB_ORIGINS`, credentials, `X-CSRF-Token`, `X-Branch-Id`
5	Rate limiters	path-prefix	`/auth`, `/api/public`, global, `/api/orders`, `/api/chat`, etc.
6	Body parsers	global	JSON/urlencoded, 10 MB
7	`cookieParser()`	global	Auth + impersonation + tenant cookies
8	`session()`	`/auth` only	OAuth `state` in PostgreSQL session store
9	`requestContext`	global	`requestId`
10	`requestLogger`	global	Optional (`ENABLE_REQUEST_LOGGING`)
11	`impersonationContext`	global	Verify `impersonation_token` cookie
12	`activeTenantContext`	global	Active tenant / branch cookie
13	`billingAccessMiddleware`	global	Block locked tenants except billing reads
14	`csrfProtection`	global	Skip `/api/public/*`
15	Static `/uploads`	conditional	`STORAGE_DRIVER=local`
16	`requireStartupMigrationsReady`	`/api/*`, `/auth/*` routes	503 while migrations run
17	Route mounts	per-prefix	50+ routers (see `server.js` imports)
18	404 + `errorHandler`	global	Structured `{ ok, error, requestId }`

Startup lifecycle

On `server.listen`:

1. Memory monitor + DB pool warmup / keepalive.
2. Keycloak OIDC config pre-warm.
3. `runStartupSchemaTasks()` — migrations + `ensureOrderCancellationColumns`, then MinIO bucket readiness.
4. `registerCronJobs({ trackInterval })` — 18 in-process timers.
5. Dev-only: enable Keycloak direct-access grants for invite login.

Graceful shutdown (`SIGTERM` / `SIGINT`): clear cron timers → close HTTP → `closePool()` → `disconnectCache()`.

Redis

Config: `REDIS_URL` via `resolveRedisUrl()` (`apps/api/src/config/resolve-redis-url.js`). Railway warning logged if a public proxy URL is used instead of internal service reference.

Uses:

Use case	Module
Cross-request cache	`apps/api/src/lib/cache.js` — permissions, user-by-sub, tenant context; falls back to in-memory (500 entries) if unset
Rate-limit store	`apps/api/src/lib/rate-limit-store.js`
Socket.IO adapter	`apps/api/src/lib/socket-redis-adapter.js` — multi-instance fan-out
Singleflight dedup	`apps/api/src/lib/singleflight.js`

`/health` exposes `redis.connected` when `MEMORY_HEALTH_EXPOSE` is enabled.

MinIO / object storage

Driver selection (`apps/api/src/config/env.js`):

- `STORAGE_DRIVER=local` — default when no S3 endpoint.
- `STORAGE_DRIVER=s3` — when `STORAGE_ENDPOINT` / `S3_ENDPOINT` is set (MinIO locally, Railway Storage in prod).

Service: `apps/api/src/services/storage/storage.service.js` delegates to `localStorageProvider` or `s3CompatibleProvider`.

Operation	Used for
`ensureStorageReady()`	Boot-time bucket creation
`createPresignedUpload()`	Browser-direct uploads
`getObjectStream()`	Private bucket reads via `/api/files/object`
`putObject` / `deleteObject`	Import workers, internal copies

Docker Compose maps MinIO to `http://minio:9000` with `S3_ACCESS_KEY` / `S3_SECRET_KEY` (see `docker-compose.yml` `x-api-environment`).

Socket.IO

Initialized on the same `http.Server` as Express (`initializeSocket(server)`).

- **Auth:** `resolveSocketUserFromCookieHeader` — same `access_token` cookie as REST.
- **Rooms:** `user_{userId}`, `restaurant_{restaurantId}`.
- **Events:** `notification_new`, `consumer_order_new`, chat message persistence via `chatSocket.service.js`.
- **Scale-out:** Redis adapter when `REDIS_URL` is set.

Vite dev proxy and production nginx must forward WebSocket upgrades on `/socket.io`.

Cron jobs (18)

All jobs register in `apps/api/src/lib/register-cron-jobs.js` and execute via `runCronJob` (`apps/api/src/lib/cron-runner.js`):

- Skipped when `NODE_ENV=test`.
- Gated by `CRONS_ENABLED` (per-tick).
- **Postgres advisory lock** per job name — one winner per cluster per tick.
- In-memory guard prevents concurrent duplicate runs in one process.

#	`CRON_JOBS` key	Interval (default)	Handler
1	`scheduled_orders`	`CRON_SCHEDULED_ORDERS_INTERVAL_MS` (1h prod / 5m dev)	`executeScheduledOrders`
2	`invoice_overdue`	24h	`checkOverdueInvoices`
3	`subscription_billing`	1h	`runSubscriptionBillingJob`
4	`waitlist_offers`	15m	`checkExpiredWaitlistOffers`
5	`promotions_expiry`	30m	`runDeactivateExpiredPromotionsJob`
6	`invitation_expiry`	1h	branch + restaurant invitation expiry
7	`free_sandbox_expiry`	1h	`runFreeSandboxExpiryJob`
8	`trial_ending_soon`	1h	`runTrialEndingSoonJob`
9	`fulfillment_exceptions`	30m	`runFulfillmentExceptionChecks`
10	`delivery_rollover`	`CRON_DELIVERY_ROLLOVER_INTERVAL_MS` (1h)	`runDeliveryRolloverCron`
11	`operational_reminders`	`CRON_OPERATIONAL_REMINDERS_INTERVAL_MS` (24h)	`runOperationalRemindersJob`
12	`driver_location_retention`	24h	`runDriverLocationRetentionJob`
13	`email_retry`	`CRON_EMAIL_RETRY_INTERVAL_MS` (1h)	`runEmailRetryJob`
14	`email_digest`	`CRON_EMAIL_DIGEST_INTERVAL_MS` (24h)	`runEmailDigestJob`
15	`stale_gps_alerts`	`CRON_STALE_GPS_INTERVAL_MS` (15m)	`runStaleGpsAlertsJob`
16	`log_retention`	`CRON_LOG_RETENTION_INTERVAL_MS` (24h)	`runLogRetentionJob`
17	`reorder_forecast`	24h	`runReorderForecastJob`
18	`growth_program_maintenance`	1h	`runGrowthProgramMaintenanceJob`

Not in this registry: bulk product image import uses `image-import-worker.js` + Postgres advisory locks (see `docs/features/bulk-product-image-import.md`).

Set `CRONS_ENABLED=false` on a web-tier API replica if you split workers later.

Deployment

Railway

Service	Dockerfile	Config	Health
API	`apps/api/Dockerfile`	`apps/api/railway.json`	`GET /health` (120s timeout)
Web	`apps/web/Dockerfile`	`apps/web/railway.json`	static
Keycloak	`deploy/railway/keycloak/Dockerfile`	per-env `railway.json`	Keycloak health

API start command: `node apps/api/src/server.js`. Build context is **repo root** (not `apps/api`).

Runtime env defaults ship in `deploy/railway/<env>/api.env` and `web.env`. `loadRailwayApiEnvDefaults()` merges these when `RAILWAY_ENVIRONMENT` is detected.

Docker Compose (local full stack)

`docker-compose.yml` — services: `postgres`, `redis`, `minio`, `mailpit`, `keycloak`, `api`, `web`, `nginx`.

- API env block `x-api-environment` wires internal hostnames (`postgres`, `redis`, `minio`, `keycloak`).
- Nginx terminates same-origin `/api` + `/auth` for cookie compatibility.
- Commands: `pnpm dev:docker` or `scripts/run-local.cmd up`.

Native dev

```
pnpm setup && pnpm dev   # infra via Docker or local; Vite :5173 + API :4000
pnpm local:infra         # postgres, redis, minio, keycloak only
```

Environment variables (sanitized reference)

Secrets shown as `<set-in-vault>` — never commit real values.

Core

Variable	Required	Default / notes
`NODE_ENV`	no	`development`
`APP_ENV`	no	derived: `dev` / `staging` / `prod`
`PORT`	no	`4000` (Railway injects)
`DATABASE_URL`	**yes** (prod)	Postgres connection string
`DATABASE_MIGRATION_URL`	no	Direct URL for DDL; falls back to `DATABASE_PRIVATE_URL`
`DATABASE_SSL`	no	`false`
`SESSION_SECRET`	**yes** (prod)	OAuth session signing
`TRUST_PROXY`	no	`true`

Auth (Keycloak)

Variable	Required	Default
`KEYCLOAK_BASE_URL`	yes	`http://localhost:8080` (internal)
`KEYCLOAK_PUBLIC_URL`	yes	browser-facing issuer base
`KEYCLOAK_REALM`	no	`Supplify`
`KEYCLOAK_CLIENT_ID`	no	`supplify-api`
`KEYCLOAK_CLIENT_SECRET`	**yes** (prod)	`<set-in-vault>`
`KEYCLOAK_ADMIN` / `KEYCLOAK_ADMIN_PASSWORD`	dev	admin bootstrap
`OAUTH_CALLBACK_BASE_URL`	prod	Web origin for `/auth/callback`
`COOKIE_SECURE`	no	`true` in production
`COOKIE_SAME_SITE`	no	`lax`
`COOKIE_DOMAIN`	optional	cross-subdomain cookies
`IMPERSONATION_SECRET`	yes	defaults to `SESSION_SECRET`
`CONSUMER_AUTH_SECRET`	yes	B2C diner JWT (separate from Keycloak)

Web / CORS

Variable	Required	Default
`WEB_ORIGIN`	yes (prod)	primary SPA URL
`WEB_ORIGINS`	no	comma-separated allowlist
`PUBLIC_API_URL`	no	derived from `RAILWAY_PUBLIC_DOMAIN`
`PUBLIC_FRONTEND_URL`	no	alias of `WEB_ORIGIN`
`STAFF_PORTAL_BASE_URL`	no	staff magic-link base
`VITE_API_URL`	web build	baked at Docker build time

Redis

Variable	Required	Default
`REDIS_URL`	recommended (prod)	unset → in-memory cache only

Storage (MinIO / S3)

Variable	Required	Default
`STORAGE_DRIVER`	no	`local` or `s3`
`STORAGE_ENDPOINT`	s3	`http://localhost:9000`
`STORAGE_BUCKET`	no	`supplify`
`STORAGE_ACCESS_KEY_ID`	s3	`minioadmin` (dev)
`STORAGE_SECRET_ACCESS_KEY`	s3	`<set-in-vault>`
`STORAGE_PUBLIC_READ`	no	`true`
`STORAGE_S3_FORCE_PATH_STYLE`	no	`true` for MinIO

Crons & ops

Variable	Default
`CRONS_ENABLED`	`true`
`CRON_SCHEDULED_ORDERS_INTERVAL_MS`	1h prod / 5m dev
`DELIVERY_ROLLOVER_ENABLED`	`false`
`RATE_LIMIT_ENABLED`	`true` except `APP_ENV=dev`
`RUN_MIGRATIONS_ON_START`	`false`
`MEMORY_HEALTH_EXPOSE`	`true` in dev

Email / payments (abbreviated)

Variable	Purpose
`EMAIL_ENABLED`, `SMTP_HOST`, `SMTP_*`	Transactional email
`PAYMENTS_MODE`, `PAYMENTS_*`	Billing gateway
`VAPID_*`	Web push

Full list: `apps/api/src/config/env.js`.

Implementation evidence

Claim	Source
175 SQL migrations	`apps/api/db/migrations/*.sql` (count verified 2026-06-17)
18 cron jobs registered	`register-cron-jobs.js` `jobs.length` + `cron-runner.js` `CRON_JOBS`
Middleware order	`apps/api/src/server.js` lines 149–442
RTK store shape	`apps/web/src/store/index.ts`
Redis fallback	`apps/api/src/lib/cache.js`
Socket Redis adapter	`apps/api/src/lib/socket.js` → `attachSocketRedisAdapter`
Railway API healthcheck	`apps/api/railway.json` → `/health`
Docker Compose service map	`docker-compose.yml`
554 API routes (inventory)	`docs/audits/route-inventory.json`

Key files

```
apps/api/src/server.js              # HTTP server + middleware + routes
apps/api/src/lib/register-cron-jobs.js
apps/api/src/config/env.js          # canonical env schema
apps/web/vite.config.ts
apps/web/src/services/api/base.ts   # RTK Query + cookies
docker-compose.yml
apps/api/Dockerfile
apps/web/Dockerfile
deploy/railway/
```

Related docs

- 09-authentication-rbac.md — OIDC, cookies, permissions
- 08-database-guide.md — schema and migrations
- docs/operations/cron-jobs.md — cron operations runbook

Part VIII — Database Guide *(Internal Technical Reference)*

Supplify uses **PostgreSQL 16** with a **numbered SQL migration** pipeline (`apps/api/db/migrations/`). Schema changes are forward-only; **175 migrations** exist as of migration `0175_free_trial_supplier_growth_parity.sql` . Application code uses the `pg` pool (`apps/api/src/lib/db.js`) with optional statement timeouts and Railway-oriented pool keepalive.

Migration system

Concept	Detail
Tracker table	`schema_migrations(version, applied_at)` — created in `0000_schema_migrations.sql`
Runner	`apps/api/scripts/migrate.js` — `pnpm db:migrate`
Startup	`runFullStartupMigrations()` after HTTP listen; `/ready` returns `503 migrating` until complete
DDL URL	`DATABASE_MIGRATION_URL` bypasses poolers that block `ALTER TABLE`
Naming	`NNNN_snake_case_description.sql`

Do not edit applied migrations. Add a new file and run migrate.

Schemas & naming conventions

- **Single database**, `public` schema — no per-tenant Postgres schemas.
- **Tenant isolation** is logical: `restaurant_id` , `supplier_id` , `tenant_id` + `tenant_type` columns and query filters.
- **UUID primary keys** via `gen_random_uuid()` (`pgcrypto`).
- **Timestamps:** `created_at` , `updated_at` on most business tables.
- **JSONB** for flexible plan limits/features, addresses, metadata.
- **Enums** for stable lifecycles (`order_status` , staff PTO/swap statuses).

Core identity & tenancy

Table	Purpose
`app_user`	Platform user; `keycloak_sub` , `email` , `role` (`ADMIN` / `SUPPLIER` / `RESTAURANT` / `STAFF_PORTAL`)
`supplier`	Supplier tenant; optional `organization_id` for multi-branch
`restaurant`	Restaurant tenant; org linkage via `restaurant_organizations` (later migrations)
`supplier_organizations`	Parent org for supplier branches (`0082_supplier_branch_accounts.sql`)
`user_workspace_membership`	One active workspace per user boundary (`0104_user_workspace_membership.sql`)

RBAC (tenant-scoped)

Table	Purpose
`tenant_roles`	Named roles per `(tenant_id, tenant_type)` ; `is_system` for defaults
`tenant_role_permissions`	`permission` text codes (see `permission-keys.js`)
`tenant_user_roles`	User ↔ role assignment within a tenant
`role` / `permission` / `role_permission`	Legacy global role catalog + **admin** roles (`0042_rbac_seed_roles_permissions.sql`)
`user_role`	Legacy user ↔ global role (merged at permission resolution)

Catalog & supplier inventory

Table	Purpose
`catalog`	Supplier catalog container
`product`	SKU, names, category; scoped by `supplier_id`
`price`	Time-bounded product pricing
`inventory`	Supplier stock (`product_id` PK, `available_qty`)

Restaurant inventory

Table	Purpose
`restaurant_inventory`	Par levels per `(restaurant_id, product_id)` — `0004_restaurant_inventory.sql`
`restaurant_inventory_lot`	Batch/expiry tracking (`0133_restaurant_inventory_lots.sql`)
`restaurant_inventory_settings`	Tenant-level inventory prefs

Orders & commercial flow

Table	Purpose
`customer_order`	Restaurant purchase order; `status order_status`
`order_item`	Lines with `supplier_id` , qty, pricing
`invoice` / `invoice_line_item`	AR/AP documents — `0009_finance_billing.sql`
`credit_note`	Adjustments linked to invoices/orders
`subscription_plan`	Plan catalog (limits/features JSONB)
`subscription`	Tenant subscription state
`tenant_subscription_addon`	Add-on entitlements

Fulfillment & delivery

Table	Purpose
`delivery_wave` / `pick_list`	Batch picking (`0006_fulfillment_logistics.sql`)
`delivery_route` / `route_stop`	Route planning
`proof_of_delivery`	POD capture
`drivers` / `driver_assignments`	Driver lifecycle (`0088_drivers_fulfillment.sql`)
`fulfillment_exceptions`	Operational exception queue
`warehouse`	Supplier warehouses (`0081_warehouse_fulfillment.sql`)

Tenant isolation patterns

API enforcement (not RLS):

1. `resolveTenantContext` attaches `tenantId` + `tenantType` from cookies, membership, or impersonation.
2. Route handlers filter `WHERE restaurant_id = $tenant` or `supplier_id = $tenant` .
3. `requireOwnership` / branch-org guards for multi-location suppliers.
4. Admin routes use `resolveAdminContext` ; impersonation uses `getEffectiveTenant()` .

Subscription gating: `billingAccessMiddleware` blocks locked tenants except billing/subscription read endpoints.

Status fields

`order_status` enum

Evolved across `0001_init.sql` , `0028_order_status_enhancements.sql` , `0069_approvals_budgets.sql` , `0110_order_status_received_with_dispute.sql` :

Value	Typical meaning
DRAFT	Cart / not placed
PENDING_APPROVAL	Internal approval workflow
PLACED	Submitted to supplier
CONFIRMED	Supplier accepted
FULFILLING	Pick/pack/ship in progress
DELIVERED	Delivered to restaurant
RECEIVED_PARTIAL	Partial receiving
RECEIVED_FULL	Fully received
RECEIVED_WITH_DISPUTE	Receiving dispute open
INVOICED	Invoice generated
COMPLETED	Closed
CANCELLED	Cancelled

Invoice `status` (TEXT CHECK)

`DRAFT` → `ISSUED` → `PARTIALLY_PAID` / `PAID` / `OVERDUE` / `VOID`

Subscription `status`

`TRIALING`, `ACTIVE`, `SUSPENDED`, `CANCELLED`, `PAST_DUE`

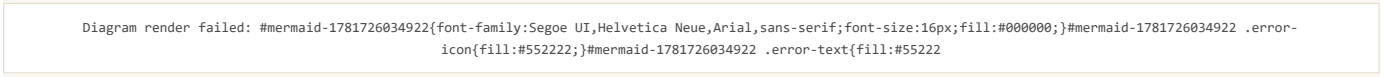
Driver assignment `status`

`assigned`, `picked_up`, `out_for_delivery`, `delivered`, `failed`, `reassigned`

Fulfillment exception `status`

`open`, `resolved`, `ignored`

Entity-relationship diagram (core commercial domain)



Key relationships (query mental model)

1. **Order → suppliers:** One restaurant order may span multiple suppliers via `order_item.supplier_id`.
2. **Invoice → order:** Optional `invoice.order_id`; line items may reference `order_item_id`.
3. **Inventory types:** `inventory` = supplier sellable stock; `restaurant_inventory` = restaurant on-hand after receiving.
4. **Subscription:** Polymorphic `tenant_id` + `tenant_type IN ('SUPPLIER','RESTAURANT')` — not a FK to keep one table.
5. **Delivery:** `driver_assignments` is the operational delivery record; `delivery_route` / `route_stop` support planning; `proof_of_delivery` stores confirmation artifacts.

Seeds & demo data

Seeds are **scripts**, not migrations. Entry points from root `package.json`:

Command	Purpose
`pnpm db:seed`	Base API seed (`@supplify/api db:seed`)
`pnpm seed:demo-users`	Keycloak + `app_user` demo accounts
`pnpm seed:demo-tenants`	Restaurant/supplier tenants
`pnpm seed:plan-tiers` / `seed:tier-catalog`	Subscription plan catalog
`pnpm seed:billing`	Billing fixtures
`pnpm seed:full`	Full demo stack
`pnpm seed:prodlike`	Production-like minimal data
`pnpm seed:features`	Feature-flag / plan feature alignment
`pnpm db:migrate-users-to-roles`	Backfill `tenant_user_roles` from legacy

Supporting modules: `apps/api/scripts/seed/businessDemoData.js`, `tierDefinitions.js`, `wipe-commercial-data.js`.

System role matrix sync: `ensureTenantSystemRoles()` in `apps/api/src/lib/tenant-roles.js` applies `RESTAURANT_SYSTEM_ROLES` / `SUPPLIER_SYSTEM_ROLES` from `role-matrix.js` when tenants are created or on admin sync.

Indexes & performance

Hot-path indexes added in later migrations, e.g.:

- `0139_railway_hot_path_indexes.sql` — restaurant inventory, orders
- `0142_order_create_hot_path_indexes.sql`
- `0091_performance_indexes.sql`

Use `EXPLAIN ANALYZE` on slow list endpoints; check `SLOW_REQUEST_MS` logs for stage breakdown.

Implementation evidence

Claim	Source
175 migrations	`apps/api/db/migrations/*.sql`
Initial schema	`0001_init.sql`
`order_status` extensions	`0028`, `0069`, `0110`
Invoice schema	`0009_finance_billing.sql`
Subscription schema	`0022_subscription_system.sql`
Fulfillment/delivery	`0006_fulfillment_logistics.sql`, `0088_drivers_fulfillment.sql`
Restaurant inventory	`0004_restaurant_inventory.sql`
Tenant RBAC tables	`0078_tenant_named_roles.sql`, `0041_rbac_tenant_roles.sql`
Tenant isolation in API	`apps/api/src/lib/rbac.js` `resolveTenantContext`
Migration on boot	`apps/api/src/lib/startup-migrations.js`, `server.js` `runStartupSchemaTasks`

Operational commands

```
pnpm db:migrate                    # apply pending migrations
pnpm db:seed                       # development seed
psql $DATABASE_URL -c "SELECT version FROM schema_migrations ORDER BY version DESC LIMIT 5;"
```

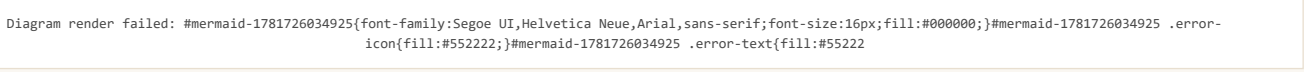
Related docs

- 07-technical-architecture.md — Postgres pool, Redis, deployment
- 09-authentication-rbac.md — `tenant_roles` permission model
- docs/operations/cron-jobs.md — jobs that mutate subscription/invoice state

Part IX — Authentication & RBAC *(Internal Technical Reference)*

Supplify authenticates users with **Keycloak OIDC** (authorization code flow). The API stores platform identity in `app_user` and enforces **tenant-scoped RBAC** via permission keys in `tenant_role_permissions`. Authorization is **mandatory on the backend** (`requirePermission`); the React app mirrors checks for UX only.

Keycloak OIDC flow



Endpoints (`apps/api/src/routes/auth.routes.js`)

Route	Auth	Purpose
`GET /auth/login`	session	Start OIDC; clears impersonation/active-tenant cookies
`GET /auth/register`	session	Keycloak self-registration
`GET /auth/callback`	session	Code exchange + cookie write
`GET /auth/logout`	public	Clear cookies + Keycloak end-session
`GET /auth/session`	optional	Lightweight session probe
`GET /auth/me`	required	Full user + RBAC payload
`POST /auth/refresh`	cookie	Refresh access token

Token verification

- `getKeycloakConfig()` loads `.well-known/openid-configuration` (`apps/api/src/lib/auth.js`).
- Access tokens verified with cached `createRemoteJWKSet` per realm JWKS URI.
- Issuer normalized for trailing-slash mismatches.

- Keep-alive HTTP agents reduce Railway ↔ Keycloak latency.

OAuth callback origin

`callbackOrigin(req)` prefers `OAUTH_CALLBACK_BASE_URL`, then `X-Forwarded-Host`, then request host — so cookies are **first-party** on the web domain (critical for mobile Chrome).

Token & cookie flow

HttpOnly cookies (web)

Set in `setAuthCookies()` (`apps/api/src/lib/rbac.js`):

Cookie	TTL	Content
`access_token`	1 hour	Keycloak JWT
`refresh_token`	7 days	Keycloak refresh token

Options: `httpOnly`, `secure` (`COOKIE_SECURE`), `sameSite` (`COOKIE_SAME_SITE`), optional `domain` (`COOKIE_DOMAIN`).

Request authentication

`requireAuth` middleware:

1. Read `access_token` from cookie (or `Authorization: Bearer` for mobile).
2. Verify JWT; on expiry attempt `refreshAccessToken` and re-set cookies.
3. Load `app_user` by `keycloak_sub` (Redis-cached).
4. `assertStaffPortalRouteAccess` — block `STAFF_PORTAL` from non-allowlisted paths.

Other security cookies

Cookie	Purpose
`impersonation_token`	Signed JWT for admin view-as (`impersonation.js`)
`active_tenant`	Multi-branch / workspace switch (`tenant-switch.js`)
Session cookie	`/auth` only — OAuth `state` in PostgreSQL session store

CSRF

`csrfProtection` on state-changing API calls; skipped for `/api/public/*`. Frontend sends `X-CSRF-Token` (header-based — safe with compression).

Permission keys (52)

Canonical constants: `apps/api/src/lib/permission-keys.js` (`PERMISSION_KEYS`).

Domain	Keys
Orders	`ORDERS_VIEW`, `ORDERS_CREATE`, `ORDERS_EDIT`, `ORDERS_MANAGE`
Invoices	`INVOICES_VIEW`, `INVOICES_CREATE`, `INVOICES_EDIT`, `INVOICES_MANAGE`
Inventory	`INVENTORY_VIEW`, `INVENTORY_EDIT`, `INVENTORY_MANAGE`
Reservations	`RESERVATIONS_VIEW`, `RESERVATIONS_CREATE`, `RESERVATIONS_EDIT`, `RESERVATIONS_MANAGE`
Staff / team	`STAFF_VIEW`, `STAFF_INVITE`, `STAFF_EDIT`, `STAFF_MANAGE`
Settings	`SETTINGS_VIEW`, `SETTINGS_EDIT`, `SETTINGS_MANAGE`
Chat	`CHAT_VIEW`, `CHAT_SEND`, `CHAT_MANAGE`
Subscriptions	`SUBSCRIPTIONS_VIEW`, `SUBSCRIPTIONS_MANAGE`
Catalog	`CATALOG_VIEW`, `CATALOG_EDIT`, `CATALOG_MANAGE`
Warehouses	`WAREHOUSES_VIEW`, `WAREHOUSES_EDIT`, `WAREHOUSES_MANAGE`
Receiving	`RECEIVING_VIEW`, `RECEIVING_MANAGE`
Payments	`PAYMENTS_VIEW`, `PAYMENTS_MANAGE`
Fulfillment	`FULFILLMENT_VIEW`, `FULFILLMENT_MANAGE`
Promotions	`PROMOTIONS_VIEW`, `PROMOTIONS_MANAGE`
Customers	`CUSTOMERS_IMPORT`, `CUSTOMERS_MANAGE`
Growth	`GROWTH_VIEW`
Driver	`DRIVER_DELIVERIES_VIEW`, `DRIVER_DELIVERIES_MANAGE`
Admin	`ADMIN_ACCESS`, `ADMIN_TENANTS`, `ADMIN_PLANS`, `ADMIN_SUPPORT`, `ADMIN_FINANCE`, `ADMIN_GROWTH`

`hasPermission(permissions, required)` (`permissions.js`): exact match, or holder has domain `_MANAGE` when checking `_VIEW` / `_EDIT` / etc.

Caching: Redis key `perm:{userId}:{tenantId}:{tenantType}` — TTL 180s; invalidated on role changes.

Restaurant system roles (7)

Defined in `apps/api/src/lib/role-matrix.js` (`RESTAURANT_SYSTEM_ROLES`). Synced to DB per tenant via `ensureTenantSystemRoles()` .

Role	Intent
Owner	`permissions: 'ALL'` — full restaurant workspace
Restaurant Manager	Orders, receiving, catalog read, chat; no billing/team admin
Purchaser	Browse catalog, create/edit orders
Receiving Staff	Receive goods, disputes; no order create
Accountant	Invoices, payments, subscriptions view
Viewer	All `*_VIEW` for restaurant workspace; zero writes
FOH Staff	Reservations only

Restaurant permission matrix

Legend: ✓ = granted, — = denied. Owner has all keys (omitted for brevity).

Permission	Manager	Purchaser	Receiving	Accountant	Viewer	FOH
`ORDERS_VIEW`	✓	✓	✓	✓	✓	—
`ORDERS_CREATE`	✓	✓	—	—	—	—
`ORDERS_EDIT`	✓	✓	—	—	—	—
`ORDERS_MANAGE`	✓	—	—	—	—	—
`RECEIVING_VIEW`	✓	—	✓	—	✓	—
`RECEIVING_MANAGE`	✓	—	✓	—	—	—
`CATALOG_VIEW`	✓	✓	—	—	✓	—
`INVENTORY_VIEW`	✓	✓	—	—	✓	—
`INVOICES_VIEW`	✓	—	—	✓	✓	—
`INVOICES_*` (write)	—	—	—	✓	—	—
`PAYMENTS_*`	—	—	—	✓	✓ view	—
`SUBSCRIPTIONS_VIEW`	—	—	—	✓	✓	—
`SUBSCRIPTIONS_MANAGE`	—	—	—	—	—	—
`STAFF_VIEW`	—	—	—	—	✓	—
`STAFF_INVITE` / `STAFF_MANAGE`	—	—	—	—	—	—
`SETTINGS_VIEW`	✓	—	—	—	✓	—
`SETTINGS_MANAGE`	—	—	—	—	—	—
`CHAT_VIEW` / `CHAT_SEND`	✓	✓	—	—	view only	—
`RESERVATIONS_*`	view/create/edit	—	—	—	view	✓

Verified by `apps/api/src/lib/tenant-role-matrix.test.js` .

Supplier system roles (9)

`SUPPLIER_SYSTEM_ROLES` in `role-matrix.js` :

Role	Intent
Owner	Full supplier workspace
Supplier Manager	Orders, catalog, fulfillment, growth view; no billing/team admin
Warehouse Manager	Warehouses, fulfillment, inventory
Order Fulfillment Staff	Fulfillment board; no decline/billing
Driver	`DRIVER_DELIVERIES_*` only
Catalog Manager	Catalog + inventory edit
Promotions Manager	Promotions + order manage for deals
Accountant	Finance keys only
Viewer	All supplier `*_VIEW` ; no mutations

Supplier permission matrix (selected)

Permission	Sup. Manager	WH Manager	Fulfill. Staff	Driver	Catalog Mgr	Promo Mgr	Accountant	Viewer
`ORDERS_VIEW`	✓	✓	✓	—	✓	✓	✓	✓
`ORDERS_EDIT`	✓	—	✓	—	—	—	—	—
`ORDERS_MANAGE`	✓	—	—	—	—	✓	—	—
`CATALOG_VIEW`	✓	—	—	—	✓	✓	—	✓
`CATALOG_EDIT / MANAGE`	✓	—	—	—	✓	—	—	—
`FULFILLMENT_*`	✓	✓	✓	—	—	—	—	view
`WAREHOUSES_*`	view	✓ edit	view	—	—	—	—	view
`DRIVER_DELIVERIES_*`	—	—	—	✓	—	—	—	—
`PROMOTIONS_*`	view	—	—	—	—	✓	—	—
`INVOICES_* / PAYMENTS_*`	view	—	—	—	—	—	✓	view
`STAFF_*`	—	—	—	—	—	—	—	view
`SETTINGS_MANAGE`	—	—	—	—	—	—	—	—
`GROWTH_VIEW`	✓	—	—	—	—	—	—	—
`CUSTOMERS_IMPORT`	✓	—	—	—	—	—	—	—

Admin permissions

Platform admins (`app_user.role = 'ADMIN'`) use the **legacy `role / permission` tables** (`0042_rbac_seed_roles_permissions.sql`).

Admin role code	Permissions
`SUPER_ADMIN`	All `ADMIN_*` keys
`SUPPORT_ADMIN`	`ADMIN_ACCESS` , `ADMIN_TENANTS` , `ADMIN_SUPPORT`
`FINANCE_ADMIN`	`ADMIN_ACCESS` , `ADMIN_TENANTS` , `ADMIN_FINANCE`
`GROWTH_ADMIN`	`ADMIN_ACCESS` , `ADMIN_GROWTH`

Admin dashboard route mapping

`resolveAdminDashboardPermission()` (`route-permissions.js`):

Path prefix	Required permission
`/financial-overview`	`ADMIN_FINANCE`
`/plans` , `/subscriptions` , `/usage` , limits	`ADMIN_PLANS`
`/tenants`	`ADMIN_TENANTS`
`/users` , `/impersonate`	`ADMIN_SUPPORT`
`/feature-flags` , overrides	`ADMIN_GROWTH`
default	`ADMIN_ACCESS`

`ALLOW_AUTO_SUPER_ADMIN` (default `false`): first ADMIN without roles can receive `SUPER_ADMIN` in dev only.

Staff portal (`STAFF_PORTAL`)

Operational restaurant staff (scheduling, check-in) are **separate from Team RBAC**.

Constant	Value
`STAFF_PORTAL_APP_ROLE`	`STAFF_PORTAL` (`staff-portal-auth.js`)
Keycloak realm role	`staff_portal`
Login redirect	`/staff/dashboard` (`auth.routes.js` callback)
Data link	`staff_member.user_id + portal_access_enabled`

API path allowlist (staff-only users)

```
/auth/me, /auth/logout, /auth/refresh, /auth/session
/api/staff/self/*
```

Any other route returns `403 STAFF_PORTAL_FORBIDDEN` . Platform routes use `requirePlatformAppAccess` .

Magic links: `POST /api/public/staff/request-link` (rate-limited `rl:staff-link`). Base URL: `STAFF_PORTAL_BASE_URL` .

Staff portal does **not** receive restaurant/supplier Keycloak realm roles — dual-link users with a platform role can access both portals.

Impersonation

Admins with `ADMIN_SUPPORT` (or `SUPER_ADMIN`) can "view as" a tenant.

Diagram render failed: #mermaid-1781726034926{font-family:Segoe UI,Helvetica Neue,Arial,sans-serif;font-size:16px;fill:#000000;}#mermaid-1781726034926 .error-icon{fill:#552222;}#mermaid-1781726034926 .error-text{fill:#55222

Property	Detail
Cookie	`impersonation_token`
Signing	`IMPERSONATION_SECRET`, max age `IMPERSONATION_MAX_DURATION_MINUTES` (default 60)
Payload	`adminUserId`, `tenantId`, `tenantType`, `viewAsRoleId?`
Trust	`getEffectiveTenant` requires `req.userData.id === adminUserId`
Permissions	View-as role permissions, or Owner fallback — **no blanket bypass** in `requirePermission`
Cleared on	logout, login, successful OAuth callback

Frontend: `useImpersonation()` + `usePermissions()` — impersonating admin uses `tenantPermissions` from `/auth/me` when hydrated.

Frontend enforcement

Mechanism	File
Route guard	`AuthGuard.tsx` — session, register completion, legal reaccept, `STAFF_PORTAL` redirect
Permission hooks	`usePermissions.ts` — `can()`, `canAny()`, `isWorkspaceViewer`
Nav / buttons	`Sidebar.tsx`, feature pages (`OrdersPage`, `FulfillmentPage`, etc.)
Admin fallback	`ADMIN_FALLBACK_PERMISSIONS` when `adminPermissions` empty (dev partial seed)
Owner shortcut	`isTenantOwner(user)` → all permissions in UI

Important: Hidden UI ≠ security. All mutations must pass API `requirePermission` / `requireAnyPermission`.

Backend enforcement

Layer	Behavior
`requireAuth`	JWT + user load + staff portal gate
`requireRole('RESTAURANT' | 'SUPPLIER' | 'ADMIN')`	Persona check
`resolveTenantContext`	Attach `tenantContext.permissions`
`resolveAdminContext`	Attach `adminContext.permissions`
`requirePermission(key)`	Owner role bypass; else tenant or admin perms
`requireAnyPermission(...)`	OR of keys
Domain guards	`ordersRouterMutationGuard`, `staffMutationGuard`, `adminDashboardPermissionGuard`, etc. (`route-permissions.js`)
Plan / billing	`billingAccessMiddleware`, subscription feature gates

Example pattern (`orders.routes.js`):

```
router.use(requireAuth, requireRole('RESTAURANT', 'SUPPLIER', 'ADMIN'))
router.use(resolveTenantContext)
router.use(requirePermission('ORDERS_VIEW'))
router.use(ordersRouterMutationGuard) // POST → ORDERS_CREATE | ORDERS_MANAGE
```

Permission resolution algorithm

`getPermissionsForUser(userId, tenantId, tenantType)` (`permissions.js`):

1. Check Redis cache.
2. If user has `tenant_user_roles` row → use `tenant_role_permissions` for that role (**named assignment**).
3. Else merge legacy `user_role` → `role_permission` with optional org/branch expansion (supplier org roles).
4. **Never reduce access** when merging legacy + named unless named assignment exists (then legacy union is skipped for invited staff).
5. ADMIN impersonation: separate path via `getImpersonationEffectivePermissions`.

Custom roles: tenants may create non-system roles via `POST /api/roles` (`tenant-roles.routes.js`) with subset-of-assigner validation.

Implementation evidence

Claim	Source
52 permission keys	`Object.keys(PERMISSION_KEYS).length` in `permission-keys.js`
7 restaurant + 9 supplier roles	`role-matrix.js` array lengths
OIDC callback + cookies	`auth.routes.js`, `auth.js`, `rbac.js` `setAuthCookies`
Role matrix tests	`tenant-role-matrix.test.js`
Staff portal gate	`staff-portal-auth.js`, `staff-portal-access.test.js`
Impersonation	`impersonation.js`, `impersonationContext.js`
Admin role seed	`0042_rbac_seed_roles_permissions.sql`
`requirePermission`	`rbac.js` lines 943–987
Frontend parity	`usePermissions.ts`, `AuthGuard.tsx`

Key files

```
apps/api/src/routes/auth.routes.js
apps/api/src/lib/auth.js
apps/api/src/lib/rbac.js
apps/api/src/lib/permissions.js
apps/api/src/lib/permission-keys.js
apps/api/src/lib/role-matrix.js
apps/api/src/lib/route-permissions.js
apps/api/src/lib/staff-portal-auth.js
apps/api/src/lib/impersonation.js
apps/api/src/middlewares/impersonationContext.js
apps/web/src/hooks/usePermissions.ts
apps/web/src/components/AuthGuard.tsx
```

Related docs

- 07-technical-architecture.md — session store, Redis, middleware order
- 08-database-guide.md — `tenant_roles` tables
- docs/features/staff-portal-access.md — staff portal product notes
- docs/architecture/security-baseline.md — CSRF, public routes

Part X — Subscriptions and Plans *(Internal Technical Reference)*

Supplify monetization is **plan-driven**: every restaurant and supplier workspace has a `subscription` row joined to `subscription_plan` (limits + features JSON). Self-serve tiers are **Free Trial**, **Silver**, **Gold**, and **Platinum**. The legacy **Enterprise** tier was deactivated in migration `0066_remove_enterprise_tier.sql` and is not selectable; `normalizePlanCode()` maps `enterprise` → `platinum` for comparisons only.

Canonical plan codes: `free`, `silver`, `gold`, `platinum` (`apps/api/src/lib/plan-codes.js`).

Plan catalog summary

Plan	Display name	Restaurant monthly / yearly	Supplier monthly / yearly	Notes
`free`	Free Trial	$0 / $0	$0 / $0	Time-limited sandbox; Gold **feature** gates, Free **limits**
`silver`	Silver	$49 / $490	$49 / $490	First paid tier (`0117_silver_tier_limits_features.sql`)
`gold`	Gold	$149 / $1,490	$149 / $1,490	Core production tier (`0119_gold_tier_limits_features.sql`)
`platinum`	Platinum	$349 / $3,490	$349 / $3,490	High-capacity tier (`0120_platinum_tier_limits_features.sql`)

Prices are stored on `subscription_plan.price_per_month` and `price_per_year`. Migrations **0117**, **0119**, and **0120** set limits, features, and confirm pricing for Silver/Gold/Platinum. Free-tier limits are maintained in `0145_plan_catalog_audit_sync.sql` (and runtime fallbacks in `limit-resolution.js`).

Free Trial: Gold features, Free limits

Free Trial is **not** a stripped-down feature tier. Runtime enforcement uses **Gold feature JSON** while **Free limit caps** apply:

1. **DB sync** — Migrations `0112_free_gold_feature_parity.sql`, `0145_plan_catalog_audit_sync.sql`, and `0175_free_trial_supplier_growth_parity.sql` copy Gold `features` onto Free rows.
2. **Runtime override** — `resolveEffectivePlanFeatures()` (`apps/api/src/lib/subscription/free-trial-plan-features.js`) loads cached Gold features whenever `plan_code === 'free'`, so API gates stay correct even if catalog rows drift.

```
// free-trial-plan-features.js — Free Trial uses Gold feature gates
if (planCode !== 'free' || !tenantType) return raw
return getGoldPlanFeatures(tenantType)
```

Sandbox expiry — Free workspaces get `subscription.free_sandbox_expires_at` (`0113_free_sandbox_expiry.sql` , default 7 days from `platform_setting.free_sandbox_days`). After expiry, `billingAccessMiddleware` locks writes (402); most GETs remain read-only except sensitive exports/reports.

Hidden limit — `scheduled_order_grace_per_day` lets scheduled quick-lists overflow the daily order cap once per day on Free; it is enforced but hidden from the entitlements UI (`HIDDEN_ENTITLEMENT_LIMIT_KEYS`).

Feature keys

Canonical keys live in `apps/api/src/lib/feature-keys.js` .

Restaurant (26 keys)

`chat` , `order_calendar` , `reports` , `smart_reorder` , `multi_branch` , `receiving_quality` , `disputes_returns` , `finance_invoices` , `quick_lists` , `inventory_management` , `waste_tracking` , `advanced_roles` , `notifications` , `api_integrations` , `support_sla` , `custom_branding` , `feature_flags_access` , `supplier_reviews` , `push_notifications` , `order_amendments` , `tenant_audit_log` , `waitlist_auto_promo` , `supplier_deals` , `supplier_deals_redeem` , `fulfillment_tools` , `ai_platform`

Supplier (24 keys)

`chat` , `order_calendar` , `reports` , `multi_branch` , `warehouses` , `multi_warehouse` , `fulfillment_tools` , `fulfillment` , `driver_management` , `disputes_returns` , `finance_invoices` , `quick_lists` , `inventory_management` , `advanced_roles` , `notifications` , `api_integrations` , `support_sla` , `custom_branding` , `feature_flags_access` , `promotions` , `push_notifications` , `order_amendments` , `tenant_audit_log` , `supplier_growth`

Enabled semantics — A feature is **on** when its plan value is `true` or a non-empty tier string (not `false` , `disabled` , or `""`). Same rule in API (`evaluatePlanFeatureValue`) and web (`planLimits.ts`).

Limit keys

From `apps/api/src/lib/limit-resolution.js` .

Restaurant limits (15 keys)

Key	Meaning
`branches`	Active branch locations
`users`	Primary contact + `restaurant_team` rows
`orders_per_day`	`PLACED` orders today
`suppliers_per_restaurant`	`supplier_follow` count
`restaurant_inventory_skus`	Distinct SKUs in `restaurant_inventory`
`chats_per_day`	Daily chat meter
`open_conversations`	Non-archived conversations
`storage_mb`	Cumulative file storage
`quick_lists`	Quick list count
`quick_list_items`	Items across all quick lists
`scheduled_quick_lists`	Lists with `is_scheduled = true`
`scheduled_order_grace_per_day`	Hidden overflow for scheduled orders (Free)
`deal_redemptions_per_day`	Supplier deal redemptions
`ai_requests_per_day`	LLM reorder assistant calls (`0167`)

Supplier limits (9 keys)

Key	Meaning
`branches`	Active branch locations
`warehouses`	Active warehouses (`0` = feature off)
`users`	Supplier contact (always 1)
`supplier_products_skus`	Products in catalog
`chats_per_day`	Daily chat meter
`open_conversations`	Non-archived conversations
`storage_mb`	Cumulative file storage
`promotions`	Non-expired promotions

Unlimited — Limit value `-1` or `null` in plan JSON means no cap (`formatPlanLimitDisplay` → `unlimited`).

Resolution order — `resolveEffectiveLimit()` / `resolveAllEffectiveLimits()` : plan default → plan override (`plan_limit_override` , increase-only) → tenant override (`tenant_limit_override` , increase-only) → branch/warehouse addons (`subscription-addons.js`).

Free fallbacks — If plan JSON omits keys on Free, `FREE_TIER_LIMIT_PATCHES` and `fillMissingFreeTierLimits()` apply canonical caps before enforcement.

Plan limit tables

Values from migrations **0117**, **0119**, **0120**, **0145** (Free), and **0167** (`ai_requests_per_day`). `-1` = unlimited.

Restaurant limits

Limit	Free Trial	Silver	Gold	Platinum
`branches`	1	1	3	∞
`users`	1	3	15	∞
`orders_per_day`	3	20	100	∞
`suppliers_per_restaurant`	1	5	30	∞
`restaurant_inventory_skus`	10	250	3,000	∞
`chats_per_day`	3	30	500	∞
`open_conversations`	1	5	30	∞
`storage_mb`	50	500	10,240	30,720
`quick_lists`	1	10	50	∞
`quick_list_items`	1	100	500	∞
`scheduled_quick_lists`	1	3	15	∞
`deal_redemptions_per_day`	1	10	50	∞
`scheduled_order_grace_per_day`	1	0	0	0
`ai_requests_per_day`	0	0	20	100

Supplier limits

Limit	Free Trial	Silver	Gold	Platinum
`branches`	1	1	3	∞
`warehouses`	0	1	3	∞
`users`	1	3	15	∞
`supplier_products_skus`	10	250	3,000	∞
`chats_per_day`	3	30	500	∞
`open_conversations`	1	5	30	∞
`storage_mb`	50	500	10,240	30,720
`promotions`	1	3	25	∞

Restaurant feature × plan matrix

Free Trial column = effective gates (Gold features via parity). ✓ = enabled; — = disabled. Tier strings are summarized in the **Tier** column for paid plans.

Feature	Free Trial	Silver	Gold	Platinum
chat	✓	✓ multi_supplier	✓ group_chat_files	✓ real_time_media_read_receipts
order_calendar	✓	✓	✓	✓
reports	✓	✓ basic_kpis	✓ usage_cost_dashboards	✓ advanced_forecasting_custom_reports
smart_reorder	✓	—	✓ full_90day_trends	✓ ai_forecast_seasonality
multi_branch	✓	—	✓	✓ central_purchasing
receiving_quality	✓	✓ photos_enabled	✓ quality_scoring	✓ supplier_performance_reports
disputes_returns	✓	✓	✓	✓
finance_invoices	✓	✓ record_payments	✓ expense_analytics	✓ advanced_finance_dashboard
quick_lists	✓	✓ automated_weekly	✓ full_schedule	✓ ai_smart_automation
inventory_management	✓	✓ real_time	✓ multi_branch_tracking	✓ lot_expiry_tracking
waste_tracking	✓	✓ analytics_dashboard	✓ analytics_dashboard	✓ cost_percentage_vs_sales
advanced_roles	✓	—	✓	✓
notifications	✓	✓ in_app_and_email	✓ email_and_whatsapp	✓ email_whatsapp_webhook
api_integrations	✓	—	✓ api_key_access	✓ full_api_webhooks
support_sla	✓	✓ standard_72h	✓ priority_24h	✓ dedicated_same_day
custom_branding	✓	—	✓ logo_colors	✓ white_label_domain
feature_flags_access	✓	—	✓ addon_toggles	✓ all_experimental
supplier_reviews	✓	✓	✓	✓
push_notifications	✓	✓	✓	✓
order_amendments	✓	✓	✓	✓
tenant_audit_log	✓	—	✓	✓
waitlist_auto_promo	✓	—	✓	✓
supplier_deals	✓	✓	✓	✓
supplier_deals_redeem	✓	✓	✓	✓
fulfillment_tools	—	—	—	—
ai_platform	✓	—	✓	✓

> `waste_tracking` on Silver is `analytics_dashboard` per `0145` (overrides `0117` `manual_entry`). `fulfillment_tools` is intentionally off for restaurants on all tiers (`0117`–`0120`).

Supplier feature × plan matrix

Feature	Free Trial	Silver	Gold	Platinum
`chat`	✓	✓ multi_supplier	✓ group_chat_files	✓ real_time_media_read_receipts
`order_calendar`	✓	✓	✓	✓
`reports`	✓	✓ basic_kpis	✓ usage_cost_dashboards	✓ advanced_forecasting_custom_reports
`multi_branch`	✓	—	✓	✓
`warehouses`	✓	✓	✓	✓
`multi_warehouse`	✓	—	✓	✓
`fulfillment_tools`	✓	✓ manual_orders_invoices	✓ warehouse_pick_pack	✓ routing_full_suite
`fulfillment`	✓	✓	✓	✓
`driver_management`	✓	—	✓	✓
`disputes_returns`	✓	✓	✓	✓
`finance_invoices`	✓	✓ record_payments	✓ expense_analytics	✓ advanced_finance_dashboard
`quick_lists`	—	—	—	—
`inventory_management`	✓	✓ real_time	✓ multi_branch_tracking	✓ lot_expiry_tracking
`advanced_roles`	✓	—	✓	✓
`notifications`	✓	✓ in_app_and_email	✓ email_and_whatsapp	✓ email_whatsapp_webhook
`api_integrations`	✓	—	✓ api_key_access	✓ full_api_webhooks
`support_sla`	✓	✓ standard_72h	✓ priority_24h	✓ dedicated_same_day
`custom_branding`	✓	—	✓ logo_colors	✓ white_label_domain
`feature_flags_access`	✓	—	✓ addon_toggles	✓ all_experimental
`promotions`	✓	✓	✓	✓
`push_notifications`	✓	✓	✓	✓
`order_amendments`	✓	✓	✓	✓
`tenant_audit_log`	✓	—	✓	✓
`supplier_growth`	✓	✓	✓	✓

> Free Trial includes `supplier_growth` via Gold parity (`0175`). `quick_lists` is not seeded on supplier plan JSON (key exists in `feature-keys.js` but remains off). `finance_invoices` on Silver+ added by `0144_supplier_finance_invoices_plan_features.sql` .

Enforcement architecture

`billingAccessMiddleware` (402)

File: `apps/api/src/middlewares/billingAccess.js` . Mounted globally in `server.js` after auth context, before CSRF.

Behavior	Detail
Bypass paths	`/api/billing` , `/api/register` , `/auth` , `/health` , `/api/public`
Always-allowed GET	`/api/subscriptions/entitlements` , `/current` , `/plans`
Admin	Platform `ADMIN` bypasses locks unless **impersonating** a tenant
Locked account	Non-GET → **402** with `buildAccountLockedError`
Free Trial expired	GET allowed (read-only); writes → 402
Sensitive GET	`/api/reports/*` , any `*/export` , invoice PDF → 402 when locked
Failure	DB error → **503** `BILLING_CHECK_UNAVAILABLE`

`requireFeature` (403)

File: `apps/api/src/lib/subscription/entitlements.js` .

- Resolves `resolveEffectivePlanFeatures()` (Free → Gold features).
- Applies tenant/global overrides via `resolveFeatureEnabled()` (`feature-flags.js`).
- Feature aliases (e.g. `fulfillment` ↔ `fulfillment_tools`) resolved when primary key is off.
- Response: `FEATURE_NOT_AVAILABLE` + `buildFeatureNotAvailablePayload()` (recommended plans, upgrade URL).
- Records `BLOCKED_FEATURE` conversion event.

Example mounts — `disputes.routes.js` → `disputes_returns`; `receiving.routes.js` → `receiving_quality`; `order-amendments.routes.js` → `order_amendments`; `reports.routes.js` → `reports`.

`requireWithinLimit` (403)

Same file. Calls `checkLimit()` before handler; sets `req.planLimit` on success.

- Response: `LIMIT_EXCEEDED` + `buildLimitExceededPayload()`.
- Records `BLOCKED_LIMIT` conversion event.

Example mounts — `branches.routes.js` → `branches`; `warehouses.routes.js` → `warehouses`; `quick-lists.routes.js` → `quick_lists`; `promotions/supplier.js` → `promotions`.

Atomic daily meters — `orders_per_day`, `chats_per_day`, `ai_requests_per_day` use `checkAndIncrementUsage()` inside transactions to prevent race overshoot.

Entitlements API

`GET /api/subscriptions/entitlements` → `getEntitlements()` returns the canonical payload:

- `plan`, `features`, `planFeatures`, `featureSources`, `limits`, `baseLimits`, `usage`, `overrides`, `addons`, `locationLimits`, `freeSandbox`, `smartReorder` (restaurant).

Cached 300s (`entitlements.js`); usage refreshed every 60s on cache hit.

Frontend: `useEntitlements` and `planFeatureGates`

`useEntitlements`

File: `apps/web/src/hooks/useEntitlements.ts`.

- RTK Query `useGetEntitlementsQuery` → `GET /api/subscriptions/entitlements`.
- Skipped when impersonation context says tenant entitlements should not load (`shouldLoadTenantEntitlements`).
- Returns `{ entitlements, isLoading, error, refetch, user }`.

`planFeatureGates`

File: `apps/web/src/lib/planFeatureGates.ts`. Thin helpers over `isEntitlementFeatureEnabled()` from `planLimits.ts`:

Helper	Feature key(s)
`canUseGlobalReports`	`reports`
`canUseFinanceInvoices`	`finance_invoices`
`canUseSupplierDeals`	`supplier_deals`
`canUseFulfillment`	`fulfillment` **or** `fulfillment_tools`
`canUseQuickLists`	`quick_lists`
`canUseSupplierGrowth`	`supplier_growth`

Resolution rule (`planLimits.ts`) — checks `entitlements.features[key]` first, then `entitlements.planFeatures[key]` (important for Free Trial where `planFeatures` carries Gold tiers). Matches API `evaluatePlanFeatureValue`.

UI consumers — `Sidebar.tsx`, `DashboardWidgetGrid.tsx`, `FulfillmentPage.tsx`, `ReportsPage.tsx`, `InvoicesPage.tsx`, `ProductsPage.tsx`, `BranchDetailPage.tsx`, org overview pages, `BranchContext.tsx`.

When API returns 403/402, RTK base client surfaces monetization errors; `monetization` Redux slice drives upgrade modals.

Subscription lifecycle (brief)

Concern	Implementation
Default plan	`ensureTenantSubscription()` creates Free if none (`plans.js`)
Org billing	Child branches bill to org root via `resolveOrgBillingTenantId`
Pending downgrade	`pending_plan_id` + `pending_effective_at` applied on read
Cache invalidation	`invalidateTenantSubscriptionCache()` clears sub + entitlements + billing
Recommendations	`recommendPlan()` — deterministic upsell from blocked limits/features
Bronze alias	`bronze` → `silver` (`0116_rename_bronze_to_silver.sql`, `LEGACY_PLAN_CODE_ALIASES`)

Source files (quick index)

Area	Path
Feature keys	`apps/api/src/lib/feature-keys.js`
Limit keys & Free patches	`apps/api/src/lib/limit-resolution.js`
Plan codes	`apps/api/src/lib/plan-codes.js`
Free → Gold features	`apps/api/src/lib/subscription/free-trial-plan-features.js`
Entitlements + middleware	`apps/api/src/lib/subscription/entitlements.js`
Plans & recommendations	`apps/api/src/lib/subscription/plans.js`
Billing lock	`apps/api/src/middlewares/billingAccess.js`
Feature flag resolution	`apps/api/src/lib/feature-flags.js`
Web gates	`apps/web/src/lib/planFeatureGates.ts`, `planLimits.ts`
Web hook	`apps/web/src/hooks/useEntitlements.ts`
Migrations	`0117`, `0119`, `0120`, `0145`, `0167`, `0175`

Part XI — API and Workflow Reference *(Internal Technical Reference)*

The Supplify API is an **Express 4** application (`apps/api/src/server.js`) exposing **554 HTTP routes** as of the latest `discover-routes.mjs` inventory (`docs/audits/route-inventory.json`, generated 2026-06-17). Routes use a consistent envelope:

```
{ "ok": true, "data": { … }, "error": null, "requestId": "…" }
```

Errors use `ok: false` with `error.name`, `error.message`, and optional `error.details`.

Global request pipeline

Stage	Purpose
Auth	Keycloak OIDC — cookie session (web) or Bearer (mobile)
Impersonation	Admin "view as tenant"; billing locks still apply
Billing	402 when subscription account locked (`billingAccess.js`)
CSRF	State-changing requests; `/api/public` bypassed
Tenant context	`req.tenantContext` — roles, permissions, tenant id/type
Plan gates	403 `FEATURE_NOT_AVAILABLE` / `LIMIT_EXCEEDED`

Health — `GET /health`, `GET /ready` (DB + migration readiness).

Route inventory by mount prefix

Counts from `route-inventory.json` (554 total). Regenerate: `node apps/api/scripts/discover-routes.mjs`.

Mount prefix	Routes	Primary domain
`/api/admin-dashboard`	50	Platform admin: tenants, plans, impersonation, feature flags
`/api/staff`	41	Staff portal: shifts, payroll, PTO, announcements
`/api/promotions`	32	Deals, coupons, restaurant/supplier promotions
`/api/consumer`	31	Consumer loyalty app (B2C)
`/api/supplier`	29	Supplier ops namespace (products, orders, settings)
`/api/public`	27	Unauthenticated: registration, invitations, reservations
`/api/orders`	25	Order CRUD, amendments, driver tracking, calendar
`/api/restaurant-inventory`	24	Stock, expiry, smart reorder, waste
`/api/fulfillment`	20	Board, exceptions, delivery routes
`/api/suppliers`	20	Supplier discovery, profile, relationships, branding
`/api/warehouses`	17	Warehouse CRUD, routing, pick/pack
`/api/org`	16	Multi-branch org: branches, users, settings
`/api/restaurants`	16	Restaurant profile, team, invitations
`/api/chat`	14	Conversations, support, admin
`/api/reports`	14	Restaurant & supplier analytics
`/api/reservations`	11	Table reservations (restaurant)
`/api/restaurant-org`	11	Org-level restaurant administration
`/api/quick-lists`	10	Quick lists & scheduled ordering
`/api/restaurant-finance`	10	Invoices, payments, finance dashboard
`/api/disputes`	9	Disputes & returns workflow
`/api/products`	9	Product catalog search & favorites
`/api/inventory`	8	Supplier inventory
`/api/quote-requests`	8	RFQ / quote workflow
`/api/billing`	7	Stripe checkout, payment methods, invoices
`/api/notifications`	7	In-app notifications
`/api/restaurant-pricing`	7	Contract pricing for restaurants
`/api/roles`	7	Tenant custom roles (`tenant-roles.routes.js`)
`/api/subscriptions`	7	Plans, entitlements, upgrades
`/api/drivers`	6	Driver roster
`/api/reviews`	6	Supplier reviews
`/api/files`	5	Presigned uploads
`/api/invoices`	5	Invoice PDF / payment
`/api/restaurant-onboarding`	5	Onboarding wizard
`/api/branches`	4	Branch locations
`/api/receiving`	4	Receiving & quality
`/api/audit`	3	Tenant activity log
`/api/prices`	3	Price lists
`/api/push`	3	Web push subscriptions
`/api/admin`	2	Legacy admin
`/api/credit-notes`	2	Credit notes
`/api/payments`	2	Payment webhooks
`/api/register`	2	Tenant registration
`/auth`	12	OIDC login, session, mobile refresh
`/health` , `/ready`	2	Probes
`/api/e2e`	1	E2E helpers (non-prod)

Mount map source: `apps/api/scripts/discover-routes.mjs` → `FILE_PREFIX_OVERRIDES` and `server.js` `app.use()` calls.

Authentication and tenant routes

Group	Key endpoints	Auth
Auth	`POST /auth/login`, `GET /auth/session`, `POST /auth/logout`, `POST /auth/mobile/refresh`	Public / session
Register	`POST /api/register/restaurant`, `POST /api/register/supplier`	Public
Subscriptions	`GET /api/subscriptions/plans`, `/current`, `/entitlements`	Tenant auth
Billing	`GET /api/billing/status`, `POST /api/billing/checkout`, payment methods	Tenant auth; always allowed when locked
Roles	`GET/POST/PATCH/DELETE /api/roles/*`	`advanced_roles` feature + permissions

Order status state machine

Enum values (PostgreSQL `order_status`)

Status	Introduced	Active use
`DRAFT`	`0001_init.sql`	Cart / unpublished orders
`PLACED`	`0001`	Restaurant submitted order
`ACKNOWLEDGED`	`0021_update_order_status_enum.sql`	Supplier confirmed (replaces `CONFIRMED`)
`PROCESSING`	`0021`	Supplier preparing / picking
`SHIPPED`	`0021`	Out for delivery / dispatched
`DELIVERED`	`0028_order_status_enhancements.sql`	Supplier marked delivered (awaiting receiving)
`COMPLETED`	`0001`	Legacy completion; supplier path may set inventory via `handleOrderDelivery`
`RECEIVED_PARTIAL`	`0028`	Restaurant received < ordered qty
`RECEIVED_FULL`	`0028`	Restaurant received all qty
`RECEIVED_WITH_DISPUTE`	`0110_order_status_received_with_dispute.sql`	Open dispute on received order
`INVOICED`	`0028`	Invoice issued post-receiving
`CANCELLED`	`0001`	Cancelled by restaurant or declined by supplier
`PENDING_APPROVAL`	`0069_approvals_budgets.sql`	**Legacy** — stuck orders migrated to `PLACED` (`0118`)

Removed legacy values — `CONFIRMED` → `ACKNOWLEDGED`, `FULFILLING` → `COMPLETED` (`0021`). Approvals product removed (`0114`); `PENDING_APPROVAL` no longer assigned.

Delivered-set (reviews, disputes eligibility): `COMPLETED`, `DELIVERED`, `RECEIVED_PARTIAL`, `RECEIVED_FULL`, `RECEIVED_WITH_DISPUTE`, `INVOICED` (`apps/api/src/lib/order-statuses.js`).

Lifecycle diagram

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Who performs transitions

Transition	Actor	Permission / role	Endpoint
→ `PLACED`	Restaurant	`ORDERS_CREATE`	`POST /api/orders`, quick-list checkout
→ `ACKNOWLEDGED` / `PROCESSING` / `SHIPPED`	Supplier	`ORDERS_EDIT`	`PATCH /api/orders/:id { status }`
→ `DELIVERED`	Supplier	`ORDERS_EDIT` or `handleOrderDelivery` on `COMPLETED`	`PATCH /api/orders/:id`
→ `COMPLETED`	Supplier	`ORDERS_EDIT`	`PATCH /api/orders/:id` → triggers `handleOrderDelivery` (sets `DELIVERED`)
→ `CANCELLED`	Restaurant	Own order only	`PATCH /api/orders/:id { status: "CANCELLED" }`
→ `CANCELLED` (decline)	Supplier	`ORDERS_MANAGE` + decline reason ≥3 chars	`PATCH /api/orders/:id`
→ `RECEIVED_*`	Restaurant	`RECEIVING_MANAGE`	`POST /api/receiving/receive`
→ `RECEIVED_WITH_DISPUTE`	System	On dispute create	`POST /api/disputes`
→ `INVOICED`	System	On receiving complete	Inside `POST /api/receiving/receive`
Driver sub-status	Supplier / driver	`FULFILLMENT_MANAGE`	`PATCH /api/orders/:id { delivery_status }`

Supplier status whitelist (`orders/update.js`) — Suppliers may only set: `ACKNOWLEDGED`, `PROCESSING`, `SHIPPED`, `DELIVERED`, `COMPLETED`, `CANCELLED`. Restaurants may only set `CANCELLED`.

Delivery route eligibility — `PLACED`, `PENDING_APPROVAL` (legacy), `ACKNOWLEDGED`, `PROCESSING`, `SHIPPED` for planned routes; dispatch on `PROCESSING` / `SHIPPED` (`delivery-route-order-statuses.js`).

Order workflow — key endpoints

Step	Method	Path	Notes
List / filter	`GET`	`/api/orders`	`status`, `supplier`, date range
Create	`POST`	`/api/orders`	Enforces `orders_per_day` meter
Manual create	`POST`	`/api/orders/manual`	Supplier-initiated
Detail	`GET`	`/api/orders/:id`	Includes amendments, dispute links
Update status	`PATCH`	`/api/orders/:id`	Role-gated transitions above
Remind supplier	`POST`	`/api/orders/:id/remind`	Notification
Packing slip	`GET`	`/api/orders/:id/packing-slip/pdf`	PDF export
Warehouse assign	`POST`	`/api/orders/:id/warehouses`	Multi-warehouse routing
Driver assign	`POST`	`/api/orders/:id/assign-driver`	Driver management feature
Delivery status	`PATCH`	`/api/orders/:id/delivery-status`	Driver lifecycle
Proof of delivery	`POST`	`/api/orders/:id/proof-of-delivery`	Photo / signature
GPS tracking	`GET`	`/api/orders/:id/tracking`	Live location
Calendar	`GET`	`/api/orders/calendar`	`order_calendar` feature

Receiving workflow

Feature gate — `receiving_quality` (`requireFeature` on router).

Receivable order statuses — `DELIVERED`, `COMPLETED` (`receiving.routes.js`).

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Endpoint	Method	Permission	Description
`/api/receiving/pending-orders`	`GET`	`RECEIVING_VIEW`	Restaurant: orders awaiting receive
`/api/receiving/pending-orders/supplier`	`GET`	`ORDERS_VIEW`	Supplier: counterpart view
`/api/receiving/receive`	`POST`	`RECEIVING_MANAGE`	Submit line-level quantities, quality, expiry
`/api/receiving/history`	`GET`	`RECEIVING_VIEW`	Past receiving reports

Receiving report statuses — `ACCEPTED` (full qty), `PARTIAL` (under-received). Line `quality_status` drives inventory updates (`ACCEPTED` only).

Side effects on receive — Inventory lots (`createLotFromReceivingLine`), loyalty earn, invoice generation, reorder forecast cache invalidation, optional review notification.

Disputes workflow

Feature gate — `disputes_returns` on all routes.

Dispute statuses — `open` → `under_review` → `resolved` | `rejected` | `cancelled`; `escalated` also resolvable.

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Endpoint	Method	Actor	Permission
`/api/disputes`	`POST`	Restaurant	`ORDERS_CREATE` or `RECEIVING_MANAGE`
`/api/disputes`	`GET`	Restaurant	`ORDERS_VIEW`
`/api/disputes/incoming`	`GET`	Supplier	`FULFILLMENT_VIEW`
`/api/disputes/:id`	`GET`	Both	`ORDERS_VIEW / FULFILLMENT_VIEW`
`/api/disputes/:id/attachments`	`POST`	Restaurant	`ORDERS_CREATE`
`/api/disputes/:id/cancel`	`POST`	Restaurant	`open` only
`/api/disputes/:id/review`	`POST`	Supplier	Moves to `under_review`
`/api/disputes/:id/reject`	`POST`	Supplier	`no_action` resolution
`/api/disputes/:id/resolve`	`POST`	Supplier	See resolution types below

Preconditions — Order must be in `DELIVERED_ORDER_STATUSES`. One active dispute per order (`open`, `under_review`, `escalated`). On create, order → `RECEIVED_WITH_DISPUTE` when prior status is `RECEIVED_*`, `DELIVERED`, or `COMPLETED`.

Resolution types (`resolveDispute`) — `credit_note` (creates `credit_note` row), `replacement` (spawns replacement `customer_order`), `refund` , `no_action` . On close, order status restored to `RECEIVED_PARTIAL` or `RECEIVED_FULL` from receiving aggregates.

Related — `GET/POST /api/credit-notes/*` ; replacement orders link `source_dispute_id` .

Order amendments workflow

Feature gate — `order_amendments` . Mounted at `/api/orders/:orderId/amendments` .

Mutable order statuses — `PLACED` , `PENDING_APPROVAL` , `ACKNOWLEDGED` , `PROCESSING` (`order-amendments.service.js`).

Endpoint	Method	Permission	Rules
`GET … /amendments`	`GET`	`ORDERS_VIEW`	List amendments + items
`POST … /amendments`	`POST`	`ORDERS_MANAGE`	`changeType` : quantity_change, item_substitution, item_removal, delivery_date_change, other
`POST … /:id/accept`	`POST`	`ORDERS_MANAGE`	Counterparty only; applies items
`POST … /:id/reject`	`POST`	`ORDERS_MANAGE`	Cannot reject own request
`POST … /:id/cancel`	`POST`	`ORDERS_MANAGE`	Requester only while `pending`

Fulfillment and logistics (related)

Prefix	Purpose
`/api/fulfillment/board`	Kanban-style order board
`/api/fulfillment/routes`	Delivery route planning & activation
`/api/fulfillment/exceptions`	Fulfillment exception log
`/api/warehouses/*`	Warehouse CRUD, pick lists, dispatch
`/api/drivers/*`	Driver roster

Warehouse inventory syncs on order status change via `syncWarehouseFulfillmentOnOrderStatus()` (`warehouseInventory.js`) for `ACKNOWLEDGED` / `PROCESSING` (picking) and `SHIPPED` / `COMPLETED` (dispatch).

Other high-traffic route groups

Restaurant operations

Prefix	Capabilities
`/api/restaurant-inventory`	Stock levels, expiry, waste, smart reorder / AI
`/api/quick-lists`	Lists, schedules, automated order placement
`/api/restaurant-finance`	Invoice list, payments, analytics
`/api/restaurant-pricing`	Contract / negotiated pricing
`/api/restaurant-org`	Central purchasing, org users
`/api/promotions` (restaurant)	Browse / redeem supplier deals

Supplier operations

Prefix	Capabilities
`/api/supplier`	Catalog, orders, settings (`supplier-ops.routes.js`)
`/api/suppliers`	Public profile, follow, search
`/api/inventory`	Supplier stock
`/api/promotions` (supplier)	Create/manage deals
`/api/supplier-growth`	Customer import, invites (via `supplier_growth` feature)

Platform admin

Prefix	Capabilities
`/api/admin-dashboard/tenants`	Tenant CRUD, limits, overrides
`/api/admin-dashboard/plans`	Plan catalog editing
`/api/admin-dashboard/subscriptions`	Subscription management
`/api/admin-dashboard/feature-flags`	Global + per-tenant feature overrides
`/api/admin-dashboard/impersonate`	Support impersonation

Communications

Prefix	Capabilities
`/api/chat/conversations`	Multi-party chat (`chats_per_day` / `open_conversations` limits)
`/api/notifications`	In-app notification feed
`/api/push`	Web push subscription

Error codes reference (workflow-related)

HTTP	`error.name`	When
402	Account locked	`billingAccessMiddleware` — overdue / Free Trial expired write
403	`FEATURE_NOT_AVAILABLE`	`requireFeature`
403	`LIMIT_EXCEEDED`	`requireWithinLimit` , `checkAndIncrementUsage`
403	`FORBIDDEN`	RBAC permission missing
409	`CONFLICT`	Duplicate receiving report, active dispute exists
400	`VALIDATION_ERROR`	Invalid status transition, Zod validation

Regenerating route inventory

```
cd apps/api
node scripts/discover-routes.mjs
```

Outputs:

- `docs/audits/route-inventory.json` — machine-readable 554 routes
- `docs/audits/DEV_API_ROUTE_TEST_MATRIX.md` — QA matrix

Source files (workflow index)

Workflow	Primary files
Order CRUD & status	`apps/api/src/routes/orders/update.js` , `orders.helpers.js` , `create.js`
Receiving	`apps/api/src/routes/receiving.routes.js`
Disputes	`apps/api/src/routes/disputes.routes.js` , `services/disputes.service.js`
Amendments	`apps/api/src/routes/order-amendments.routes.js` , `services/order-amendments.service.js`
Driver delivery	`apps/api/src/routes/orders-driver.routes.js` , `lib/driver-delivery.js`
Status constants	`apps/api/src/lib/order-statuses.js` , `lib/delivery-route-order-statuses.js`
Route discovery	`apps/api/scripts/discover-routes.mjs` , `apps/api/src/server.js`

Part XII — Demo Scripts

Audience: Sales, customer success, product, and engineering presenters.
Environment: Local or staging with `pnpm run seed:full` completed.
Evidence: `apps/api/scripts/seed-full.mjs` , `seed-demo-users.js` , `seed-plan-tier-demos.js` , `seed-billing.js` , `seed-demo-readiness-extras.js` , `docs/audits/SUPPLIFY_DEMO_READINESS_AUDIT.md` .

Before you present

One-command prep

```
pnpm local:infra          # Postgres, Keycloak (8180), MinIO, Redis
pnpm db:migrate
pnpm run seed:full        # WARNING: wipes all restaurants/suppliers
pnpm dev                  # API + web
```

If Keycloak was down during seed:

```
pnpm run seed:accounts && pnpm run seed:demo-users
```

Optional deterministic data: `SEED=1337 pnpm run seed:full`.

Primary demo accounts (Gold tier, active billing)

Account	Password	Tenant	Why use it
`admin@supplify.com`	`SupplifyAdmin1!`	Platform admin	Command center, deal approval, impersonation
`restaurant@supplify.com`	`SupplifyRestaurant1!`	Golden Fork Restaurant	**Gold**, active billing, coupon `DEMOFORK10`, expiring inventory tomorrow
`supplier@supplify.com`	`SupplifySupplier1!`	Fresh Foods Co.	**Gold**, active billing, linked to Golden Fork, rich catalog

Plan-tier matrix accounts (password `Supplify1!`)

Account	Plan	Slug	Notes
`restaurant-gold@supplify.com`	Gold	`plan-demo-restaurant-gold`	**Past-due grace** after `seed:billing` — shows billing banner; use for billing story only
`supplier-gold@supplify.com`	Gold	`plan-demo-supplier-gold`	Clean Gold supplier for tier comparisons
`supplier-free@supplify.com`	Free Trial	`plan-demo-supplier-free`	At **1/1 active deals** after seed (quota demo)
`restaurant-1@test.com` … `restaurant-10@test.com`	Prod-like	varies	Volume data; password `Supplify1!`

> **Presenter tip:** For a polished Gold demo without billing noise, lead with `restaurant@supplify.com` / `supplier@supplify.com`. Use `restaurant-gold@` only when demonstrating grace-period UX.

Data seeded by `seed:full` (talking points)

- ~10 prod-like restaurants, ~50 suppliers, ~2k products, ~1.5k orders, invoices, chats, quick lists, reservations, staff, disputes, approved deals.
- Extras (`seed-demo-readiness-extras.js`): inventory expiring tomorrow, coupon `DEMOFORK10`, Free-tier supplier at promotion limit.
- **Not seeded:** driver Keycloak logins, live GPS routes, multi-warehouse stock edge cases. See backups per step.

Avoid on live demos

Area	Reason
Supplier Settings → Delivery Zones / Contacts	UI not wired (`DELIVERY_ZONES_ENABLED = false` in `supplierSettingsShared.tsx`)
Restaurant finance → period statement opening balance	Hardcoded `0` (`restaurant-finance.routes.js:795`)
Dashboard 7d/30d/90d selector	Visual only; spend trend is fixed 30-day
Creating a **new** deal without pre-approving	New deals start `pending_approval` until admin approves

---

Step template (used below)

Each step lists: **Screen**, **User**, **Clicks**, **Narration**, **Business value**, **Expected result**, **Backup**, **Prep data**.

5-minute executive demo

Goal: Platform vision in one pass — marketplace, fulfillment, control plane.
Logins prepared: admin, restaurant@, supplier@ (three browser profiles or incognito tabs).

#	Screen	User	Clicks	Narration	Business value	Expected result	Backup	Prep
1	`/login`	Presenter	Sign in with Keycloak → `restaurant@supplify.com`	"Restaurants discover suppliers, build carts, and place B2B orders in one workspace."	Reduces phone/email ordering chaos	Lands on Command Center or Dashboard; sidebar shows OPERATIONS (Orders, Products, Cart)	Use `restaurant-gold@` + narrate billing banner	`seed:full`
2	`/app/products`	Restaurant	Open **Products** → filter/search → open one SKU	"Contract pricing and catalog search replace spreadsheets."	Price transparency, faster purchasing	Product list loads; "Your price" badge if contract price exists	Show **Suppliers** → one supplier detail	Fresh Foods products linked
3	`/app/cart`	Restaurant	**Cart** → review lines → **Place order** (do not need full checkout if time-tight)	"One click places the PO; supplier gets notified instantly."	Cuts order-to-ack time	Order created `PLACED`; redirect/toast success	Open existing order from **Orders**	Cart may already have items from prior seed orders
4	`/login` (new tab)	Supplier	`supplier@supplify.com` → **Orders** → open newest `PLACED` order → **Accept**	"Suppliers acknowledge, fulfill, and invoice without leaving the platform."	Supplier ops on one screen	Status → `ACKNOWLEDGED` or processing path visible	Show **Fulfillment** board with existing `PROCESSING` order	Pending orders in seed data
5	`/login` (new tab)	Admin	`admin@supplify.com` → `/app/admin` **Overview**	"We govern tenants, plans, growth, and compliance from a single command center."	SaaS operator control	KPI cards populate; Activity feed non-empty after seed	**Tenants** tab only if Overview empty	Admin Keycloak role

Close: "Supplify connects restaurant procurement to supplier fulfillment with plan-based monetization and platform oversight."

15-minute standard demo

Goal: End-to-end B2B order + money + chat.
Primary path: restaurant@ → supplier@ → admin@ (deals optional).

#	Screen	User	Clicks	Narration	Business value	Expected result	Backup	Prep
1	`/app/command-center` or `/app/dashboard`	Restaurant	Sign in → overview	"Command center surfaces spend, low stock, and pending orders."	Executive visibility for ops managers	Widgets load; pending order badge on sidebar	Skip to **Orders** if dashboard slow	Gold entitlements active
2	`/app/suppliers`	Restaurant	**Suppliers** → show Fresh Foods connected	"Restaurants follow suppliers; limits scale by plan."	Curated supplier network	Follow relationship visible	Use **Products** with supplier filter	`restaurant_supplier_follow` seeded
3	`/app/deals`	Restaurant	**Deals** → open active promotion	"Suppliers run campaigns; restaurants redeem with daily caps."	Promotional pull-through	Active deals list; redemption UI	Mention admin approval workflow	`seed-feature-demos` + approved deals
4	`/app/cart`	Restaurant	Add 2–3 SKUs → apply coupon `DEMOFORK10` if shown → **Place order**	"Deals and contract prices apply at checkout."	Margin protection + promos	Order `PLACED`; deal redemption recorded	Place without coupon	Coupon from `seed-demo-readiness-extras`
5	`/app/orders/:id`	Restaurant	Open new order → **Tracking** panel (if shipped)	"Restaurants see delivery ETA and driver location when enabled."	Delivery confidence	Tracking card or status timeline	Narrate GPS env flag if no live route	`GPS_TRACKING_ENABLED` default true
6	`/app/orders`	Supplier	Sign in supplier → filter **PLACED** → **Accept**	"Inbox replaces email POs."	Faster response SLA	Status updates; notification to restaurant	Show already-`ACKNOWLEDGED` order	60s polling on orders
7	`/app/fulfillment`	Supplier	**Fulfillment** → dispatch board → assign driver (or show existing assignment)	"Warehouse teams batch routes and assign drivers."	Last-mile efficiency	Board shows assignments; driver column	Narrate driver mobile app parity	Fulfillment feature on Gold
8	`/app/receiving`	Restaurant	**Receiving** → select delivered order → confirm quantities	"Receiving closes the loop and triggers invoicing."	Accurate goods-in	`RECEIVED_FULL` or partial path	Show pre-received order in list	Receiving orders in seed
9	`/app/invoices`	Restaurant	**Invoices** → open `ISSUED` invoice	"Finance sees AP in one ledger."	AP automation	Invoice lines match order	Supplier **Invoices** receivables view	~500 invoices seeded
10	`/app/chat`	Both	Restaurant **Chat** → open Fresh Foods thread → send message	"Contextual messaging beats WhatsApp chaos."	Fewer order errors	Message appears; typing/realtime if socket up	Refresh once if socket delayed	`seed:chats`
11	`/app/admin`	Admin	**Deals** tab → show approved vs pending	"Platform approves supplier promotions before they go live."	Brand/trust control	Filter pending/approved	**Plans** tab if no pending deals	Admin `ADMIN_GROWTH`

30-minute full demo

Goal: Standard demo plus inventory, reservations, growth, RBAC, and ops. Add **+15 min** after step 11 above.

#	Screen	User	Clicks	Narration	Business value	Expected result	Backup	Prep
12	`/app/restaurant-inventory`	Restaurant	**Inventory** → **Expiry** tab	"Expiry alerts prevent waste before service."	Food cost control	Item expiring tomorrow highlighted	Dashboard expiry summary widget	`seed-demo-readiness-extras`
13	`/app/quick-lists`	Restaurant	**Ordering Lists** → open list → **Order from list**	"Templates turn weekly buying into one click."	Purchaser productivity	Quick list → cart prefill	Scheduled list narrative only	`seed:quick-lists`
14	`/app/reservations`	Restaurant	**Reservations** → floor board	"FOH runs on the same platform as back-of-house buying."	Unified hospitality stack	Tables/slots visible	Public `/reserve/:slug` in second window	Reservations in prodlike seed
15	`/app/customer-growth`	Supplier	**Customer Growth** → metrics + import card	"Suppliers acquire restaurants via CSV, invites, referrals."	Supplier-led growth	KPI widgets; import history	Narrate `/register?ref=` flow	`supplier_growth` on Gold
16	`/app/promotions`	Supplier	**Deals** → create draft (optional)	"New deals enter compliance review."	Controlled promos	Status `pending_approval`	Show existing active deal instead	Don't submit if no admin follow-up
17	`/app/admin` → Usage	Admin	Search `plan-demo-supplier-free` → **Usage & quotas**	"Free tier hits promotion caps — upgrade path is visible."	Conversion funnel	`1/1` active deals	**Limits** override demo	`seed-demo-readiness-extras`
18	`/app/admin` → Operations	Admin	**Operations** → active deliveries / fulfillment issues	"Ops sees live logistics health."	NOC-style visibility	Panels load or empty state	**Health** tab cron status	`GET /api/admin-dashboard/operational/*`
19	`/app/settings` → Team	Restaurant	**Settings** → **Team** → show roles	"Granular RBAC: purchaser vs receiving vs accountant."	Least privilege	Role matrix visible	Mention `restaurant-gold-manager@` if `seed:tier-catalog` run	7 restaurant system roles
20	`/app/disputes`	Restaurant	**Disputes** → open active dispute	"Quality issues become structured workflows, not arguments."	Dispute resolution audit trail	Dispute detail with status	Supplier incoming disputes mirror	`seed-feature-demos`

Restaurant-only demo (12 minutes)

Login: `restaurant@supplify.com` / `SupplifyRestaurant1!`

#	Screen	User	Clicks	Narration	Business value	Expected result	Backup	Prep
1	`/app/dashboard`	Restaurant Owner	Sign in	"Your purchasing cockpit."	Visibility	Dashboard KPIs	Command center	Gold active
2	`/app/products`	Purchaser	Search → add to cart	"Browse all connected suppliers."	Assortment	Cart badge updates	**My Prices** for contracts	Catalog seeded
3	`/app/deals`	Purchaser	Redeem/view deal	"Save on promoted SKUs."	COGS	Deal applied at cart	Skip if empty	Approved deals
4	`/app/cart`	Purchaser	Place order	"PO in seconds."	Speed	`PLACED` order	Use quick list	—
5	`/app/orders`	Manager	Track status	"No more calling suppliers."	Accountability	Timeline updates	Open seeded `DELIVERED` order	—
6	`/app/receiving`	Receiving Staff	Receive shipment	"Scan-verify quantities."	Accuracy	Receiving complete	Photo upload if `receiving_quality`	—
7	`/app/invoices`	Accountant	Open invoice → record payment view	"AP ready for export."	Finance	Invoice `ISSUED` / `PAID`	Avoid period statement	—
8	`/app/chat`	Purchaser	Message supplier	"Clarify substitutions in-thread."	Fewer errors	Message sent	—	Chat thread exists
9	`/app/restaurant-inventory`	Manager	Expiry + par levels	"Stock ties to what you actually bought."	Waste reduction	Expiry row tomorrow	—	Extras seed
10	`/app/settings`	Owner	Plan & entitlements	"Upgrade when you outgrow limits."	Expansion revenue	Gold plan shown	Compare `restaurant-free@` in second tab	—

Supplier-only demo (12 minutes)

Login: `supplier@supplify.com` / `SupplifySupplier1!`

#	Screen	User	Clicks	Narration	Business value	Expected result	Backup	Prep
1	`/app/command-center`	Supplier Manager	Sign in	"Revenue, at-risk orders, fulfillment load."	Supplier exec view	KPIs render	Dashboard	—
2	`/app/products`	Catalog Manager	Open SKU → edit price	"Single catalog feeds all restaurants."	Catalog truth	Save succeeds	CSV import narrative	Products seeded
3	`/app/contract-pricing`	Catalog Manager	Show restaurant-specific price	"Negotiated rates per account."	Account management	Contract row for Golden Fork	—	—
4	`/app/orders`	Order Fulfillment	Accept/decline demo	"Structured decline reasons."	Quality feedback	Status change	Show declined order in seed	—
5	`/app/fulfillment`	Warehouse Manager	Dispatch board	"Pick, pack, route."	Throughput	Assignments visible	Routes tab	—
6	`/app/invoices`	Accountant	Receivables → record payment	"AR without QuickBooks export hell."	Cash application	Payment recorded	Credit note via dispute	—
7	`/app/promotions`	Promotions Manager	Active deal	"Growth through promotions."	Revenue lift	Active promotion	Locked card on Free tier account	—
8	`/app/customer-growth`	Manager	Invite link / CSV	"Acquire net-new restaurants."	Pipeline	Growth dashboard	—	Gold `supplier_growth`
9	`/app/restaurants`	Manager	Connected restaurants	"CRM for your buyer base."	Relationship mgmt	Golden Fork listed	—	Follow seeded
10	`/app/supplier-settings`	Owner	Profile, warehouses, team	"Configure org without IT."	Self-serve	Tabs load	**Do not** open Delivery Zones/Contacts	Warehouses API-backed

Operations / platform admin demo (15 minutes)

Login: `admin@supplify.com` / `SupplifyAdmin1!`

#	Screen	User	Clicks	Narration	Business value	Expected result	Backup	Prep
1	`/app/admin/overview`	Platform admin	Sign in → Overview	"Health of the marketplace."	Investor/ops metrics	KPI cards	Activity tab	`ADMIN_ACCESS`
2	`/app/admin/tenants`	Admin	Filter `ACTIVE` / `TRIALING`	"Every tenant at a glance."	Support efficiency	Paginated directory	Supplier/restaurant portals	Seeded tenants
3	`/app/admin/suppliers`	Admin	Open supplier row → impersonate (optional)	"Support sees exactly what they see."	Faster tickets	Impersonation banner on web	Narrate only if policy forbids impersonate	`POST /api/admin-dashboard/impersonate`
4	`/app/admin/plans`	Admin	Edit Free Trial days (7–90) → save	"Tune sandbox without deploy."	Product ops	PATCH succeeds	Revert after demo	Default 30 days
5	`/app/admin/subscriptions`	Admin	Find `restaurant-gold@` → show past due	"Grace before lockout."	Revenue protection	Past due + days left	Unlock action narrative	`seed-billing.js`
6	`/app/admin/limits`	Admin	Tenant override demo (narrate)	"Enterprise deals without new plan SKU."	Flexibility	Effective limit preview	Read-only if no permission	`ADMIN_PLANS`
7	`/app/admin/deals`	Admin	Approve pending promotion	"Compliance gate for public deals."	Trust & safety	Deal → active	Show already-approved	—
8	`/app/admin/operations`	Admin	Active deliveries, email logs	"Run the airline."	Incident response	Panels or empty states	Health tab	Operational APIs
9	`/app/admin/audit`	Admin	Audit log search	"Who changed what."	SOC narrative	Rows after impersonation/plan change	Tenant audit log on Gold restaurant	—
10	`/app/admin/growth-settings`	Admin	Referral discount fields	"Growth program knobs."	CAC/LTV tuning	GET/PATCH growth settings	—	`0169` migration

Admin demo (finance + governance focus, 10 minutes)

Subset for CFO/platform stakeholders — steps 1, 5, 6, 7, 9 from Operations demo, plus:

#	Screen	User	Clicks	Narration	Business value	Expected result	Backup	Prep
A	`/app/admin/finance`	Admin Finance	Financial overview	"MRR, churn, overdue."	Board reporting	Charts load	Conversion stats	`ADMIN_FINANCE`
B	`/app/admin/subscriptions`	Admin	Preview plan change	"Safe plan migrations."	Expansion/contraction	Preview modal	—	—
C	`/app/admin/feature-flags`	Admin Growth	Toggle feature flag (narrate risk)	"Kill switches for rollout."	Risk reduction	List loads; avoid mutating prod	Read-only walkthrough	—

Driver / logistics add-on (5 minutes)

Note: `seed:full` does **not** create driver Keycloak users. Prep manually or use fulfillment view as Warehouse Manager.

#	Screen	User	Clicks	Narration	Business value	Expected result	Backup	Prep
1	`/app/supplier-settings` → Drivers	Supplier Owner	Invite driver email (or use existing team member with Driver role)	"Drivers get a minimal UI — only their route."	Security isolation	Driver role assigned	**Fulfillment** board as Manager instead	`driver_management` feature
2	`/app/driver-deliveries`	Driver	Sign in as driver user → **My Deliveries**	"Mobile-first last mile."	Proof of delivery	Stop list + status buttons	Narrate mobile app	`DRIVER_DELIVERIES_*` perms
3	`/app/orders/:id`	Restaurant	Tracking map on in-transit order	"Buyers see ETA, not phone calls."	CX	Map or stale badge ≥5 min	Mention `GPS_STALE_AFTER_SECONDS=300`	Env `GPS_TRACKING_ENABLED`

Rehearsal checklist (day before)

- [ ] `seed:full` completes; Keycloak users exist
- [ ] `restaurant@` / `supplier@` login without `/login?expired=true`
- [ ] Admin Overview KPIs non-empty
- [ ] At least one `PLACED` order to accept live
- [ ] Coupon `DEMOFORK10` works on Golden Fork cart
- [ ] Chat message round-trip
- [ ] DevTools: no red errors on scripted path
- [ ] Mobile width: sidebar collapses (`Sidebar.mobile.test.tsx` behavior)

Related docs

- 02-complete-product-guide.md — domain reference
- 06-admin-onboarding.md — admin tab detail
- SUPPLIFY_DEMO_READINESS_AUDIT.md — known gaps
- 13-acceptance-criteria.md — pass/fail definitions

Part XIII — Acceptance Criteria

Audience: QA, product owners, implementation engineers, release managers.
Purpose: Pass/fail definitions for every major Supplify capability.
Evidence base: `apps/api/src/routes/` , `apps/web/src/pages/` , `apps/api/src/lib/permission-keys.js` , `apps/api/src/lib/feature-keys.js` , `tests/e2e/suites/` , `docs/audits/route-inventory.json` (554 routes).

Status legend: `Shipped` = production-intent in main branch; `Partial` = known gaps documented; `Planned` = not in codebase.

1. Authentication & session (OIDC)

Field	Criteria
Feature	Keycloak OIDC login, session cookies, logout
Preconditions	Keycloak reachable; `KEYCLOAK_*` env aligned; realm `Supplify`
Role	Any platform role
Plan	Any
Success path	`GET /auth/login` → Keycloak → `GET /auth/callback` → cookies set → `GET /auth/me` 200 with user + permissions
Alternatives	Bearer token (mobile); `POST /auth/refresh` on expired access token
Validation	Browser lands `/app/*` ; `access_token` httpOnly cookie; logout clears cookies + Keycloak end-session
Permissions	N/A (pre-auth)
API	`auth.routes.js` : login, callback, logout, me, refresh, session
UI	`/login` , `AuthGuard` , redirect `?expired=true`
DB	`app_user` upsert on first `/auth/me` via `upsertUser()`
Notifications	N/A
Error cases	Invalid state → redirect `/login?error=callback_failed` ; expired JWT → refresh or redirect expired
Security	OAuth state in session; CSRF on mutations; JWT verified via JWKS
Mobile	PKCE public client; Bearer auth skips CSRF (`csrf.test.js`)
Test coverage	`tests/e2e/suites/critical_e2e/auth.spec.ts` ; `apps/api/src/lib/mobile-auth.integration.test.js`
Status	Shipped

2. Tenant registration & activation

Field	Criteria
Feature	Self-serve org registration and account activation
Preconditions	Keycloak registration enabled; legal policies seeded
Role	`PENDING` → `RESTAURANT` or `SUPPLIER`
Plan	Free Trial default; `lock_reason = pending_activation` until activation
Success path	`/register/complete` → tenant + subscription + system roles → Free checkout or paid billing → lock cleared
Alternatives	Admin-created tenant; referral `?ref=` token
Validation	`GET /api/subscriptions/current` shows active; writes allowed (not 402)
Permissions	Owner role auto-assigned
API	`register.routes.js`, `billing.routes.js`
UI	`RegisterCompletePage`, `AccountActivationPage`
DB	`restaurant` / `supplier`, `subscription`, `tenant_roles`, `tenant_user_roles`
Notifications	Welcome email (SMTP configured)
Error cases	Duplicate slug; billing failure leaves lock; 402 on writes when locked
Security	CSRF; rate limits on public routes
Mobile	Registration via web; mobile uses existing accounts
Test coverage	`register-account` tests; e2e partial
Status	Shipped

3. RBAC — restaurant workspace roles

Field	Criteria
Feature	Seven system roles with permission keys
Preconditions	Tenant roles synced via `ensureTenantSystemRoles()`
Role	Owner, Manager, Purchaser, Receiving, Accountant, Viewer, FOH
Plan	`advanced_roles` for custom roles (Gold+)
Success path	User invited → role assigned → sidebar/API match matrix
Alternatives	Custom tenant role with permission subset
Validation	Purchaser: `ORDERS_CREATE` yes, `INVOICES_MANAGE` no; API returns 403 when denied
Permissions	52 keys in `permission-keys.js`; `_MANAGE` implies `_VIEW`
API	`requirePermission` on all mutating routes
UI	`RequirePermission`, `Sidebar` filtered by `navItemAllowed`
DB	`tenant_role_permissions`, `tenant_user_roles`
Notifications	Invite email on team add
Error cases	Last owner demotion edge case (documented gap)
Security	Server-side enforcement mandatory; UI mirrors only
Mobile	Same permission payload in `/auth/me`
Test coverage	`rbac-full-app.test.js`; `tests/e2e/suites/critical_e2e/rbac.spec.ts`
Status	Shipped

4. RBAC — supplier workspace roles

Field	Criteria
Feature	Nine supplier system roles including Driver isolation
Preconditions	Driver user linked in team/drivers
Role	Driver sees only `DRIVER_DELIVERIES_*`
Plan	`driver_management` , `fulfillment`
Success path	Driver login → `/app/driver-deliveries` only; status updates allowed enum
Alternatives	Warehouse Manager sees fulfillment board
Validation	Driver cannot access `/app/products` (403 or hidden nav)
Permissions	`driver-rbac.js` `DRIVER_ALLOWED_STATUS_UPDATES`
API	`fulfillment.routes.js` , `drivers.routes.js`
UI	Driver sidebar single item (`sidebarNavConfig.ts`)
DB	`tenant_user_roles` , driver assignment tables
Notifications	Assignment notifications to driver
Error cases	Invalid status transition rejected
Security	Driver scoped to assigned routes only
Mobile	Driver flows in `supplify-mobile`
Test coverage	`driver-rbac` tests; `drivers.routes.test.js`
Status	Shipped

5. Subscriptions & plan enforcement

Field	Criteria
Feature	Plan features and limits enforced at runtime
Preconditions	`subscription` + `subscription_plan` rows
Role	Tenant member
Plan	free, silver, gold, platinum per `plan-codes.js`
Success path	Action within limit → 200; feature on → UI visible
Alternatives	Admin limit override (increase-only); branch addons
Validation	`GET /api/subscriptions/usage/:meterType` ; 402 when billing locked
Permissions	`SUBSCRIPTIONS_VIEW` / `SUBSCRIPTIONS_MANAGE` for billing UI
API	`requireFeature()` , `checkPlanLimit()` , `billingAccessMiddleware`
UI	`FeatureLockedCard` , `UpgradeModal` , usage banners
DB	`subscription` , `plan_limit_override` , `tenant_limit_override`
Notifications	Trial ending emails (`trial-ending-soon.job`)
Error cases	Free sandbox expired → read-only; past due grace → banner
Security	Impersonation does not bypass billing lock
Mobile	Plan gates in mobile guards
Test coverage	`subscription-limits.spec.ts` ; `plan-enforcement` tests; `verify-tier-matrix`
Status	Shipped

6. Supplier catalog & products

Field	Criteria
Feature	Product CRUD, CSV import, image ZIP import
Preconditions	Supplier tenant; `supplier_products_skus` headroom
Role	Catalog Manager or Owner
Plan	Catalog always; storage `storage_mb` for images
Success path	Create/edit product → visible in `GET /api/products` for connected restaurants
Alternatives	CSV bulk; async ZIP import job (`0168`)
Validation	SKU unique per supplier; image thumb URL populated
Permissions	`CATALOG_VIEW` , `CATALOG_EDIT` , `CATALOG_MANAGE`
API	`products.routes.js`
UI	`ProductsPage` , `ProductImageImportDialog`
DB	`product` , `catalog` , `catalog_image_import_job`
Notifications	Import job completion (in-app)
Error cases	Limit exceeded → 402/403 with upgrade CTA; bad CSV row errors
Security	Supplier-scoped queries only
Mobile	Product browse parity
Test coverage	`products.routes.test.js` ; e2e `catalog.spec.ts`
Status	Shipped

7. Contract pricing

Field	Criteria
Feature	Per-restaurant negotiated prices
Preconditions	Restaurant–supplier relationship
Role	Supplier sets; Restaurant views
Plan	Effectively Gold workflows (contract pricing routes)
Success path	Supplier sets contract price → restaurant sees "Your price" on browse/cart
Alternatives	CSV contract import
Validation	`resolveProductPricesBatch` returns override
Permissions	`CATALOG_VIEW / CATALOG_EDIT`
API	`prices.routes.js` , `restaurant-pricing.routes.js`
UI	`/app/contract-pricing` , `/app/my-prices`
DB	`price` , `restaurant_pricing`
Notifications	N/A
Error cases	Price for unfollowed restaurant rejected
Security	Tenant isolation on both sides
Mobile	Price resolution on mobile catalog
Test coverage	`restaurant-pricing.routes.test.js`
Status	Shipped

8. Restaurant cart & order placement

Field	Criteria
Feature	Multi-supplier cart, checkout, order create
Preconditions	Followed supplier; items in stock
Role	Purchaser+ with `ORDERS_CREATE`
Plan	`orders_per_day` meter
Success path	Cart → place → `customer_order` status `PLACED`
Alternatives	Save draft `DRAFT`; quick list order
Validation	Supplier notification; order appears both sides
Permissions	`ORDERS_CREATE`
API	`POST /api/orders`
UI	`CartPage`
DB	`customer_order`, `order_item`
Notifications	`notifyTenantUsers` to supplier
Error cases	Daily limit → upgrade CTA; billing lock 402
Security	Restaurant can only order connected suppliers
Mobile	Cart/checkout parity
Test coverage	e2e `orders.spec.ts`; `orders.routes.test.js`
Status	Shipped

9. Supplier order inbox & decline

Field	Criteria
Feature	Accept, decline, process orders
Preconditions	Order `PLACED`
Role	Supplier Manager+
Plan	Core
Success path	Accept → `ACKNOWLEDGED` → fulfillment path
Alternatives	Decline with required `decline_reason` → `CANCELLED`
Validation	Restaurant sees status + reason
Permissions	`ORDERS_MANAGE`
API	`PATCH /api/orders/:id/status`
UI	`OrdersPage`, decline modal
DB	`cancel_reason`, `cancelled_by` columns
Notifications	Status change to restaurant
Error cases	Invalid transition rejected by `order-statuses.js`
Security	Supplier owns order via `supplier_id`
Mobile	Supplier order actions
Test coverage	Order status tests; e2e orders
Status	Shipped

10. Fulfillment, dispatch & routes

Field	Criteria
Feature	Fulfillment board, driver assignment, route planning
Preconditions	Order in fulfillable status; warehouses optional
Role	Warehouse Manager, Fulfillment Staff
Plan	`fulfillment` , `fulfillment_tools` , `warehouses`
Success path	Assign driver → route built → statuses through `SHIPPED` / `DELIVERED`
Alternatives	Manual status without driver
Validation	Board refreshes; assignment on order detail
Permissions	`FULFILLMENT_VIEW` , `FULFILLMENT_MANAGE`
API	`/api/fulfillment/*` , `routes/build-from-assignments`
UI	`FulfillmentPage` , `DriverDispatchBoard`
DB	fulfillment assignments, routes (`0127`)
Notifications	Driver assignment alerts
Error cases	Feature off → locked UI
Security	Supplier-scoped
Mobile	Driver delivery screen
Test coverage	`DriverDispatchBoard.test.tsx` ; fulfillment route tests
Status	Shipped

11. GPS tracking & delivery ETA

Field	Criteria
Feature	Driver location pings, restaurant tracking map, stale detection
Preconditions	`GPS_TRACKING_ENABLED=true` ; assignment in transit
Role	Driver posts; Restaurant views
Plan	Platform env gate (not plan-gated by design)
Success path	Driver location POST → restaurant order tracking shows map/ETA
Alternatives	Stale badge after `GPS_STALE_AFTER_SECONDS` (300)
Validation	`delivery-eta.service` payload; privacy flags for name/phone
Permissions	`DRIVER_DELIVERIES_MANAGE` ; restaurant `ORDERS_VIEW`
API	driver location endpoints; tracking on order
UI	`RestaurantOrderTrackingPanel` , maps components
DB	driver location history; retention job
Notifications	Optional push on delivery milestones
Error cases	GPS disabled globally → tracking hidden
Security	`GPS_RESTAURANT_SHOW_DRIVER_PHONE` default false
Mobile	Primary GPS capture surface
Test coverage	`delivery-eta.service.test.js` ; `RestaurantOrderTrackingPanel.test.tsx`
Status	Shipped

12. Receiving & quality

Field	Criteria
Feature	Goods-in against delivered orders
Preconditions	Order `DELIVERED` +
Role	Receiving Staff (`RECEIVING_MANAGE`)
Plan	`receiving_quality` for photos/scoring
Success path	Receive lines → `RECEIVED_FULL` or `RECEIVED_PARTIAL`
Alternatives	Partial receive
Validation	Inventory updated; invoice path unlocked
Permissions	`RECEIVING_VIEW` , `RECEIVING_MANAGE`
API	`receiving.routes.js`
UI	`ReceivingPage`
DB	`receiving_report` , `receiving_line_item`
Notifications	Supplier notified on receive
Error cases	Feature locked → `FeatureLockedCard`
Security	Restaurant-scoped
Mobile	Receiving partial parity
Test coverage	`receiving-delivered.spec.ts`
Status	Shipped

13. Disputes & credit notes

Field	Criteria
Feature	Open dispute, review, resolve, credit/replacement
Preconditions	Received order; `disputes_returns` feature
Role	Restaurant opens; Supplier resolves
Plan	`disputes_returns`
Success path	Dispute → `RECEIVED_WITH_DISPUTE` → resolve → credit note or replacement order
Alternatives	Cancel dispute
Validation	Credit note applies to invoice
Permissions	`ORDERS_*` / `RECEIVING_*` create; supplier `FULFILLMENT_VIEW` incoming
API	`disputes.routes.js` , `credit-notes.routes.js`
UI	`DisputesPage` , `DisputeDetailPage`
DB	`disputes` , `dispute_items` , `credit_note`
Notifications	Dispute opened/resolved
Error cases	Feature off hides nav
Security	Cross-tenant only via order relationship
Mobile	Limited
Test coverage	`disputes.routes.test.js`
Status	Shipped

14. Invoices & payments (AP/AR)

Field	Criteria
Feature	Invoice lifecycle, record payment, credit notes
Preconditions	Order received or manual invoice
Role	Accountant
Plan	`finance_invoices`
Success path	`DRAFT` → `ISSUED` → `PAID`
Alternatives	Partial payment; void
Validation	Both tenant invoice lists consistent
Permissions	`INVOICES_*`, `PAYMENTS_*`
API	`invoices.routes.js`, `payments.routes.js`
UI	`InvoicesPage`
DB	`invoice`, `payment`, `invoice_line`
Notifications	Overdue job emails
Error cases	Feature off → locked
Security	Tenant-scoped
Mobile	Invoice list parity
Test coverage	`invoices.routes.test.js`; `invoice-overdue.job.test.js`
Status	Shipped

15. Restaurant finance & statements

Field	Criteria
Feature	Per-supplier statements, aging, analytics
Preconditions	Invoices exist
Role	Accountant
Plan	`finance_invoices`, `reports` for analytics widgets
Success path	Statement shows charges/payments in selected period
Alternatives	Export
Validation	Closing balance within period correct
Permissions	`INVOICES_VIEW`
API	`restaurant-finance.routes.js`
UI	Finance tabs on invoices/dashboard
DB	invoice/payment aggregates
Notifications	N/A
Error cases	**Opening balance hardcoded 0** (`openingBalance: 0` TODO line 795)
Security	Restaurant-only data
Mobile	Partial
Test coverage	Sparse on statement rollup
Status	Partial

16. Quick lists & scheduled orders

Field	Criteria
Feature	Saved order templates; scheduled placement
Preconditions	`quick_lists` feature
Role	Purchaser
Plan	Limits: `quick_lists`, `quick_list_items`, `scheduled_quick_lists`
Success path	Create list → order from list → cron places scheduled
Alternatives	Free grace overflow once/day (`scheduled_order_grace_per_day`)
Validation	Order created from template; cron job registered
Permissions	`ORDERS_VIEW`
API	`quick-lists.routes.js`; `scheduled-orders.service.js`
UI	`QuickListsPage`
DB	`quick_list`, `quick_list_item`
Notifications	Scheduled order confirmation
Error cases	Limit banner at cap
Security	Restaurant-scoped
Mobile	Quick lists parity
Test coverage	`planLimits.test.ts`; e2e nightly
Status	Shipped

17. Smart reorder & AI assistant

Field	Criteria
Feature	Reorder suggestions; AI reorder assistant
Preconditions	Inventory history; `smart_reorder / ai_platform`
Role	Manager
Plan	`smart_reorder`; `ai_requests_per_day` on Gold+
Success path	Dashboard widget shows suggestions; AI chat returns recommendations
Alternatives	Manual quick list
Validation	`GET /api/restaurant-inventory/reorder-suggestions`
Permissions	`INVENTORY_VIEW`
API	reorder forecast job; AI routes
UI	Dashboard widget (no dedicated nav)
DB	`restaurant_inventory`, forecast queue
Notifications	N/A
Error cases	AI limit exceeded
Security	Tenant-scoped; no PII in prompts audit
Mobile	Partial
Test coverage	`ai-platform.test.js`; sparse E2E
Status	Partial (UI surfacing limited)

18. Restaurant & supplier inventory

Field	Criteria
Feature	On-hand stock, expiry, waste, supplier warehouse stock
Preconditions	`inventory_management`
Role	Manager / Warehouse Manager
Plan	SKU limits differ by tenant type
Success path	CRUD inventory → movements logged → expiry alerts
Alternatives	Waste tab (`waste_tracking`)
Validation	Par/low stock badges
Permissions	`INVENTORY_VIEW` , `INVENTORY_EDIT` , `INVENTORY_MANAGE`
API	`restaurant-inventory.routes.js` , `inventory.routes.js`
UI	`RestaurantInventoryPage` , supplier `InventoryPage`
DB	`restaurant_inventory` , `inventory` , `inventory_movement_log`
Notifications	Low stock alerts
Error cases	SKU cap on Free
Security	Branch-scoped where applicable
Mobile	Inventory views
Test coverage	`inventory-expiry.service.test.js`
Status	Shipped

19. Deals & promotions

Field	Criteria
Feature	Supplier promotions; admin approval; restaurant redemption
Preconditions	`promotions` (supplier); `supplier_deals` (restaurant)
Role	Promotions Manager; Admin approver
Plan	`promotions` limit; `deal_redemptions_per_day`
Success path	Create deal → admin approve → active → restaurant redeems at cart
Alternatives	Boost placement; coupon codes
Validation	Deal status transitions; redemption counter
Permissions	`PROMOTIONS_*` ; admin `ADMIN_GROWTH`
API	`promotions.routes.js` , `deal-promotions.service.js`
UI	`PromotionsPage` , `DealsPage` , admin Deals tab
DB	`deal_promotion` , redemption tables
Notifications	Approval/rejection
Error cases	Pending deals invisible to restaurants; locked shows `FeatureLockedCard`
Security	Supplier cannot approve own deals
Mobile	Deals browse
Test coverage	`promotions-deals-gates.spec.ts` ; `PromotionsPage.locked.test.tsx`
Status	Shipped

20. Chat & realtime messaging

Field	Criteria
Feature	Restaurant–supplier chat with files, typing, order links
Preconditions	`chat` feature; Socket.IO + Redis adapter in prod
Role	Users with `CHAT_VIEW` / `CHAT_SEND`
Plan	`chats_per_day`, `open_conversations`
Success path	Open conversation → send message → realtime delivery
Alternatives	Admin support chat
Validation	Message persisted; read receipts per plan tier
Permissions	`CHAT_*`
API	`chat.routes.js`
UI	`ChatPage`, `useChatRealtime`
DB	`conversation`, `message`
Notifications	In-app + optional push
Error cases	Daily chat limit; socket disconnect retry
Security	Conversation membership enforced
Mobile	Chat parity
Test coverage	`useChatRealtime.test.ts`; chat route tests
Status	Shipped

21. Reservations (FOH)

Field	Criteria
Feature	Floor plan, bookings, waitlist, public guest portal
Preconditions	Branch hours configured
Role	FOH Staff or Manager
Plan	No dedicated feature key (core module)
Success path	Create reservation → guest manages via token URL
Alternatives	Waitlist auto-promo (`waitlist_auto_promo`)
Validation	Table assignment; availability rules
Permissions	`RESERVATIONS_*`
API	`reservations.routes.js`, `public.routes.js`
UI	`ReservationsPage`, `/reserve/*`
DB	`reservation`, `reservation_table`, waitlist
Notifications	Guest SMS/email/WhatsApp
Error cases	Double-booking prevented
Security	Public tokens unguessable
Mobile	Public booking responsive
Test coverage	`reservations.routes.test.js`; `ReservationsPage.test.tsx`
Status	Shipped

22. Staff directory & staff portal

Field	Criteria
Feature	Roster, shifts, staff portal self-service
Preconditions	Staff records; portal account provisioned
Role	`STAFF_PORTAL` isolated from `/app`
Plan	Team limits `users`
Success path	Manager provisions portal → staff logs `/staff/login` → dashboard
Alternatives	PTO / shift swap
Validation	Staff gets 403 on main app APIs
Permissions	`STAFF_*`
API	`staff.routes.js`, `staff-portal-auth.js`
UI	`StaffPage`, `/staff/dashboard`
DB	staff tables; `staff_portal` link
Notifications	Shift reminders
Error cases	Disabled Keycloak user cannot login
Security	`assertStaffPortalRouteAccess`
Mobile	Staff portal web responsive
Test coverage	`staff.routes.test.js`
Status	Shipped

23. Consumer B2C ordering

Field	Criteria
Feature	Public storefront, guest checkout, consumer loyalty
Preconditions	Menu configured; branch hours
Role	Guest / light consumer account
Plan	Restaurant enables consumer modules
Success path	`/order/:slug` → menu → checkout → track
Alternatives	Takeaway vs delivery zones on branch
Validation	Order in `consumer_orders`; receipt token works
Permissions	Restaurant admin `consumer-menu`, `consumer-orders`
API	`apps/api/src/routes/consumer/*`
UI	`apps/web/src/pages/consumer/*`
DB	migrations `0161` – `0164`
Notifications	Guest order status
Error cases	Outside hours; zone not serviceable
Security	Public routes rate-limited
Mobile	Responsive PWA
Test coverage	`consumer-ordering.spec.ts` smoke
Status	Shipped

24. Supplier customer growth program

Field	Criteria
Feature	CSV import, invites, referrals, sponsorship
Preconditions	`supplier_growth` on Gold+
Role	Manager with `GROWTH_VIEW`, `CUSTOMERS_IMPORT`
Plan	Sponsorship limits per year
Success path	Import prospects → invite → connection or registration
Alternatives	Referral link `/register?ref=`
Validation	Growth metrics API
Permissions	`GROWTH_VIEW`, `CUSTOMERS_IMPORT`
API	growth program routes; `0169` tables
UI	`CustomerGrowthPage`
DB	import batches, referrals, sponsorship
Notifications	Invite emails
Error cases	Feature off hides nav
Security	Import data tenant-scoped
Mobile	N/A
Test coverage	`supplier-growth-program.test.js`
Status	Shipped

25. Quote requests (RFQ)

Field	Criteria
Feature	Multi-supplier RFQ, compare responses, add to cart
Preconditions	Connected suppliers
Role	Restaurant creates; Supplier responds
Plan	Core ordering
Success path	RFQ sent → supplier quotes lines → restaurant compares
Alternatives	Add winning lines to cart (manual checkout)
Validation	Notifications `quote_request_received`
Permissions	`ORDERS_CREATE`
API	`quote-requests.service.js`
UI	`/app/quote-requests/*`
DB	`0153` schema
Notifications	Email/in-app on quote events
Error cases	Quoted price informational only at order create
Security	Suppliers see only their RFQ lines
Mobile	Limited
Test coverage	Service tests
Status	Shipped (quote price not auto-applied at checkout)

26. Reports & analytics

Field	Criteria
Feature	KPI dashboards, usage, waste, supplier revenue
Preconditions	`reports` feature
Role	Manager+ with report permissions
Plan	Tier strings: basic → advanced forecasting
Success path	`/app/reports` loads charts for date range
Alternatives	Dashboard widgets subset
Validation	`GET /api/reports/*` returns data
Permissions	`ORDERS_VIEW` / `INVOICES_VIEW` / analytics anyOf
API	`reports.routes.js`
UI	`ReportsPage`
DB	analytics indexes `0071`
Notifications	N/A
Error cases	Feature off hides nav
Security	Tenant-scoped aggregates only
Mobile	Reports limited
Test coverage	Report route tests
Status	Shipped

27. Admin platform command center

Field	Criteria
Feature	Tenant admin, plans, limits, impersonation, ops health
Preconditions	`role: ADMIN` + granular `adminPermissions`
Role	Platform admin
Plan	N/A
Success path	`/app/admin` tabs load with lazy queries + `skip` when inactive
Alternatives	Supplier/restaurant scoped admin portals
Validation	Overview KPIs; audit on impersonation
Permissions	`ADMIN_*` keys
API	`/api/admin-dashboard/*` (47+ routes)
UI	`AdminDashboardPage` + tab components
DB	all tenants
Notifications	N/A
Error cases	Tab hidden without permission
Security	Impersonation signed cookie; audit logged
Mobile	Admin usable but desktop-first
Test coverage	`admin-rbac.spec.ts` ; `admin-impersonation.spec.ts`
Status	Shipped

28. Warehouses & multi-branch

Field	Criteria
Feature	Supplier warehouses; restaurant branches; org billing inheritance
Preconditions	`multi_branch`, `warehouses` features
Role	Owner / Warehouse Manager
Plan	Branch/warehouse limits per tier
Success path	Create warehouse → stock → fulfill from location
Alternatives	Branch switcher cookie `active_tenant`
Validation	Entitlements resolve at org parent
Permissions	`WAREHOUSES_*`, branch APIs
API	`warehouses.routes.js`, `branches.routes.js`
UI	`BranchSwitcher`, warehouse tabs
DB	`warehouse`, `branch`
Notifications	N/A
Error cases	Limit exceeded on create
Security	Supplier-only warehouse mutations
Mobile	Branch context
Test coverage	`warehouses.routes.test.js`; branches audit
Status	Shipped

29. PWA & web push notifications

Field	Criteria
Feature	Installable PWA, service worker, push subscriptions
Preconditions	HTTPS; VAPID keys; `push_notifications` feature
Role	Any tenant user
Plan	`push_notifications`
Success path	Register SW → opt in → `POST /api/push/subscribe`
Alternatives	In-app notifications only
Validation	`manifest.webmanifest` valid; push received on event
Permissions	Settings notification toggles
API	`push.routes.js`
UI	`usePushNotifications`, onboarding opt-in
DB	push subscription rows
Notifications	Web Push + bell
Error cases	Browser denies permission; SW unsupported
Security	VAPID; subscription bound to user
Mobile	Native push in mobile app separate
Test coverage	`pwaManifest.test.ts`; `registerServiceWorker.test.ts`
Status	Shipped

30. Tenant audit log

Field	Criteria
Feature	Immutable activity log per tenant
Preconditions	`tenant_audit_log` on Gold+
Role	Owner / settings viewers
Plan	Gold+
Success path	Settings → Activity shows entries; export CSV
Alternatives	Admin platform audit separate
Validation	`GET /api/audit/logs` paginated
Permissions	`SETTINGS_VIEW`
API	`tenant-audit.routes.js`
UI	Activity tab in settings
DB	`tenant_audit_log`
Notifications	N/A
Error cases	Feature off hides tab
Security	Tenant-scoped; no cross-tenant leak
Mobile	N/A
Test coverage	Audit route tests
Status	Shipped

Cross-feature release gate

Before marking a release **accepted**:

1. `pnpm typecheck` — pass
2. `pnpm --filter @supplify/api test:run` — pass (~1008 tests)
3. `pnpm --filter @supplify/web test:run` — pass
4. `pnpm e2e:playwright` critical suite — pass on staging
5. `pnpm verify:tier-matrix` — pass against staging DB
6. Manual smoke: auth, place order, accept, invoice, admin overview

Part XIV — Troubleshooting Guide

Audience: Developers, DevOps, support engineers, demo presenters.
Scope: Common failures across auth, infrastructure, API errors, data, GPS, PWA, and mobile.
Method: Code-traced symptoms → diagnosis → safe fix. No destructive production actions without explicit approval.

Quick health URLs (local defaults):

Service	URL	Check
Web	`http://localhost:5173`	SPA loads
API	`http://localhost:3000/api/health`	200 JSON
Keycloak	`http://localhost:8180`	Admin console
Postgres	`localhost:5433` (docker)	`pnpm db:migrate`
Redis	`localhost:6379`	Socket.IO + cache
MinIO	`localhost:9000`	File uploads

1. Cannot log in / stuck on login page

Symptoms

- Redirect loop to `/login`
- `/login?expired=true` after idle
- `/login?error=callback_failed`
- Keycloak form shows then returns to login with no app session

Likely causes

Cause	Evidence
Keycloak down or wrong URL	`auth.routes.js` login catch → `error=callback_failed`
`KEYCLOAK_BASE_URL` mismatch	API `.env` vs Docker port (8180 vs 8080)
OAuth `state` mismatch	Session store not persisting (`oauthState`)
Cookie not set (Secure on HTTP)	E2E hint in `tests/e2e/auth.setup.ts`
`WEB_ORIGIN` / `OAUTH_CALLBACK_BASE_URL` wrong	Cookies set on wrong domain
User missing realm role	No `restaurant` / `supplier` / `admin` role in Keycloak
Demo user missing	Keycloak realm imported without users

Diagnose

```
## Keycloak up?
curl -s -o /dev/null -w "%{http_code}" http://localhost:8180/realms/Supplify

## API auth probe
curl -i http://localhost:3000/auth/session

## Recreate demo users
pnpm run seed:demo-users
```

Browser: DevTools → Application → Cookies on API origin — expect `access_token`, `refresh_token` after login.

Logs:

- API: `Login error`, `Error saving session` — `apps/api/src/routes/auth.routes.js`
- Keycloak container: `docker compose logs keycloak`

Files

- `apps/api/src/routes/auth.routes.js`
- `apps/api/src/lib/auth.js` (JWKS, token exchange)
- `apps/api/src/lib/rbac.js` (`setAuthCookies`)
- `apps/api/scripts/seed-demo-users.js`
- `apps/web/src/components/AuthGuard.tsx`
- `apps/web/src/services/api/base.ts` (`redirectToLoginForAuthError`)

Safe fix

1. Align env: `KEYCLOAK_BASE_URL=http://localhost:8180`, `WEB_ORIGIN=http://localhost:5173`, `OAUTH_CALLBACK_BASE_URL` = API public URL.
2. `pnpm run seed:demo-users` (passwords: `SupplifyAdmin1!`, etc.).
3. Clear site cookies; retry incognito.
4. Local HTTP: ensure `COOKIE_SECURE=false` in dev.
5. If session store fails: verify Postgres session table / `SESSION_SECRET` set.

Escalation

- Production: verify Keycloak realm client redirect URIs include exact API callback URL.
- Railway: check `KEYCLOAK_USE_OPTIMIZED` and memory (`docs/infra/KEYCLOAK_RAILWAY_MEMORY_FIX.md`).

---

2. Keycloak admin / seed account failures

Symptoms

- `Keycloak admin token failed: 401`
- `seed:full` warns "Keycloak accounts failed"
- Users exist in DB but cannot sign in

Likely causes

- Keycloak not started before seed
- Wrong `KEYCLOAK_ADMIN_USERNAME` / `KEYCLOAK_ADMIN_PASSWORD` (default `admin` / `admin`)
- Realm name mismatch (`KEYCLOAK_REALM=Supplify`)
- `SKIP_KEYCLOAK=true` left set

Diagnose

```
pnpm run seed:accounts      # prod-like emails
pnpm run seed:demo-users    # admin@, restaurant@, supplier@
```

Files

- `apps/api/scripts/seed-full.mjs` (lines 77–97)
- `apps/api/scripts/seed-accounts-for-prodlike.js`
- `docker-compose.yml` Keycloak service

Safe fix

1. `docker compose up -d keycloak` — wait for healthy.
2. Re-run account scripts (idempotent).
3. First login creates `app_user` — email must match `restaurant.contact_email` or `supplier.contact_email`.

Escalation

- Import realm JSON from `deploy/keycloak/` if realm corrupted.

---

3. HTTP 401 Unauthorized on API calls

Symptoms

- API returns `{ error: "Unauthorized" }` or 401
- RTK Query errors; empty dashboards
- Mobile: token rejected

Likely causes

Code path	Cause
Missing/expired JWT	`requireAuth` in `rbac.js`
Invalid issuer/JWKS	Keycloak realm URL changed
Staff portal on wrong route	`assertStaffPortalRouteAccess`
Bearer token expired (mobile)	No refresh cookie

Diagnose

```
curl -b cookies.txt http://localhost:3000/api/auth/me
curl -H "Authorization: Bearer $TOKEN" http://localhost:3000/api/auth/me
```

Logs: `JWT verify failed`, `User not found for sub`.

Files

- `apps/api/src/lib/rbac.js` (`requireAuth`, `verifyAccessToken`)
- `apps/api/src/lib/auth.js`
- `apps/web/src/services/api/base.ts`

Safe fix

1. Log out and log in (`GET /auth/logout`).
2. `POST /auth/refresh` if refresh cookie valid.
3. Verify `app_user.keycloak_sub` matches token `sub` (re-login upserts).
4. Staff users: use `/staff/login` only.

Escalation

- Clock skew between API and Keycloak (rare) — sync NTP.

---

4. HTTP 403 Forbidden

Symptoms

- Action visible but API returns 403
- "You don't have permission" toasts

Likely causes

- Missing `requirePermission` key for role
- Plan feature off (`requireFeature`)
- Billing lock (`billingAccessMiddleware`) — sometimes 402, not 403
- Driver accessing non-delivery routes

- Admin tab without `adminPermissions`
- CSRF header missing on web mutations

Diagnose

- Compare role in **Settings** → **Team** with `docs/architecture/rbac-permission-matrix.md`.
- `GET /api/auth/me` → `tenantPermissions` array.
- `GET /api/subscriptions/current` → features/limits.

Files

- `apps/api/src/middlewares/billingAccess.js`
- `apps/api/src/lib/plan-enforcement.js`
- `apps/api/src/middlewares/csrf.js`
- `apps/web/src/components/RequirePermission.tsx`

Safe fix

1. Assign correct tenant role (Owner for full access).
2. Upgrade plan or admin unlock subscription.
3. Web: ensure `X-CSRF-Token` header sent (`api/base.ts`).
4. Impersonation: permissions follow impersonated tenant, not admin superpowers on billing writes.

Escalation

- Permission cache stale: Redis key `perm:{userId}:{tenantId}:{tenantType}` TTL 180s — wait or restart API.

---

5. HTTP 402 Payment Required / billing lock

Symptoms

- Writes fail; read-only mode banners
- "Activate your account" / "Trial expired"

Likely causes

- `lock_reason`: `pending_activation`, `free_sandbox_expired`, `SUSPENDED`, past due beyond grace
- `seed-billing` demo: `supplier-silver@` locked, `restaurant-gold@` past due

Diagnose

```
## As tenant owner
GET /api/billing/status
GET /api/subscriptions/current
```

Files

- `apps/api/src/middlewares/billingAccess.js`
- `apps/api/scripts/seed-billing.js`

Safe fix

1. **Demo:** use `restaurant@supplify.com` / `supplier@supplify.com` (active Gold).
2. Complete activation checkout (`/app/settings` → billing).
3. Admin: `POST /api/admin-dashboard/subscriptions/:id/unlock` or extend trial.

Escalation

- Stripe/payment provider misconfig — check `billing.routes.js` logs.

---

6. HTTP 429 Too Many Requests

Symptoms

- `Too many requests from this IP`
- Public endpoints fail under load test

Likely causes

- `express-rate-limit` on API (`server.js`)
- Public routes: 60/min prod, 200/min dev

- Driver location rate limit (`driver-location.service.js`)

Diagnose

- Response headers `RateLimit-*`
- Redis keys `rl:*` if Redis store enabled

Files

- `apps/api/src/server.js` (rate limit config)
- `apps/api/src/config/env.js` `RATE_LIMIT_MAX`

Safe fix

1. Dev: increase `RATE_LIMIT_MAX` temporarily.
2. Back off retries in client.
3. Do not disable rate limits in production without review.

Escalation

- DDoS / bot traffic — WAF/nginx layer.

---

7. HTTP 500 / 502 / 503

Symptoms

- Generic error toasts
- nginx 502 Bad Gateway
- API process crash loop

Likely causes

Status	Cause
500	Unhandled exception; DB query error
502	API down behind proxy; upstream timeout
503	Health check failing; DB pool exhausted

Diagnose

```
docker compose logs api --tail 100
curl http://localhost:3000/api/health
```

Logs: `request-timing` middleware; uncaught stack in API stdout.

Files

- `apps/api/src/server.js`
- `apps/api/src/lib/db.js` (pool)
- `deploy/nginx/*`

Safe fix

1. Restart API container.
2. Verify `DATABASE_URL` connectivity.
3. Run pending migrations (`pnpm db:migrate`).
4. Check disk/memory (Keycloak OOM — `KEYCLOAK_RAILWAY_MEMORY_FIX.md`).

Escalation

- Postgres connection limit — reduce pool size or scale DB.

---

8. Redis connection / cache failures

Symptoms

- Socket.IO chat not realtime (falls back or disconnects)
- Permission cache misses causing slow auth
- Rate limiter errors on startup
- Logs: `ECONNREFUSED` Redis, `MaxRetriesPerRequestError`

Likely causes

- `REDIS_URL` unset (dev may run without Redis — degraded mode)
- Railway: using `REDIS_PUBLIC_URL` for internal traffic (egress/fees) — wrong URL
- Redis down

Diagnose

```
redis-cli -u $REDIS_URL ping
```

Files

- `apps/api/src/config/resolve-redis-url.js`
- `apps/api/src/lib/cache.js`
- Socket.IO Redis adapter setup in server

Safe fix

1. Local: `docker compose up -d redis`; set `REDIS_URL=redis://localhost:6379`.
2. Railway: use private `REDIS_URL`, not public proxy (`isLikelyPublicRedisUrl`).
3. Temporary: API may boot without Redis — expect no cross-instance sockets.

Escalation

- Redis memory eviction clearing sessions — increase plan or TTL review.

---

9. Database migration failures

Symptoms

- API crash on startup: missing table/column
- `migrate` container: `WARN: partial SQL migrations`
- `role "api_user" does not exist`
- `cannot drop type order_status`

Likely causes

- Partial failed migration
- Skipped migration file
- Wrong Postgres port (`5432` vs `5433` docker)
- 175 migrations not all applied

Diagnose

```
SELECT version FROM schema_migrations ORDER BY version DESC LIMIT 10;
```

```
pnpm db:migrate
docker compose logs migrate
```

Files

- `apps/api/db/migrations/*.sql`
- `apps/api/scripts/run-migration.js`
- `apps/api/src/lib/migrator.js` (runtime backfill)
- `docs/guides/database-migrations.md`

Safe fix

1. **Dev only:** `pnpm db:reset` or new Docker volume.
2. Fix failing SQL; re-run `run-migration.js`.
3. `api_user` grants: already commented in `0019` / `0020` / `0039` — use `postgres` user locally.
4. `0021` enum issue: reset DB per migration guide.

Escalation

- Production: never `db:reset` — forward-fix migration with idempotent `IF NOT EXISTS`.

---

10. `seed:full` / demo data problems

Symptoms

- Empty admin tenant lists
- Login works but no orders/products
- Keycloak users missing
- Wrong plan tier data

Likely causes

- Seed aborted mid-way
- `ALLOW_PRODLIKE_SEED` not set (handled by `seed:full`)
- Keycloak step failed but DB seeded

Diagnose

Re-run full seed (**wipes commercial data**):

```
pnpm run seed:full
```

Files

- `apps/api/scripts/seed-full.mjs`
- Individual scripts listed in seed-full output

Safe fix

1. Full re-seed on local only.
2. Partial: `pnpm run seed:demo-tenants`, `seed:plan-tiers`, `seed:demo-readiness`.
3. Keycloak only: `pnpm run seed:accounts && pnpm run seed:demo-users`.

Escalation

- Staging with real data: never run `seed:full` — use targeted scripts.

---

11. GPS / delivery tracking not showing

Symptoms

- Restaurant order tracking empty
- Map never loads
- "Last seen" stale immediately
- Driver location POST fails

Likely causes

- `GPS_TRACKING_ENABLED=false`
- No driver assignment / order not in transit
- `MAP_PROVIDER` missing API key (`GOOGLE_MAPS_API_KEY`, `MAPBOX_ACCESS_TOKEN`)
- Stale threshold: `GPS_STALE_AFTER_SECONDS=300`
- Privacy: `GPS_ALLOW_RESTAURANT_LIVE_TRACKING=false`

Diagnose

- Env in `apps/api/src/config/env.js` lines 244–252
- Order status must be in transit family
- Network tab: driver location POST responses

Files

- `docs/features/drivers-and-gps-tracking.md`
- `apps/api/src/services/driver-location.service.js`
- `apps/api/src/services/delivery-eta.service.js`
- `apps/web/src/components/orders/RestaurantOrderTrackingPanel.tsx`

Safe fix

1. Enable `GPS_TRACKING_ENABLED=true`.
2. Complete fulfillment path: assign driver → mark shipped/in transit.
3. Add map API key to `apps/web/.env`.
4. Demo without live GPS: narrate ETA text-only; show fulfillment board.

Escalation

- Mobile app not sending background location — check mobile permissions docs.

---

12. PWA / service worker / push notifications

Symptoms

- Install prompt never appears
- Push opt-in fails
- `serviceWorker registration failed`
- Notifications never arrive

Likely causes

- Not HTTPS (required except localhost)
- Browser blocked notifications
- Missing VAPID keys on API
- `push_notifications` plan feature off
- User denied permission

Diagnose

- `apps/web/static/manifest.webmanifest` — tested by `pwaManifest.test.ts`
- `GET /api/push/vapid-public-key`
- Browser Application → Service Workers

Files

- `apps/web/src/lib/registerServiceWorker.ts`
- `apps/web/src/hooks/usePushNotifications.ts`
- `apps/api/src/routes/push.routes.js`

Safe fix

1. Use HTTPS or localhost.
2. Reset notification permission in browser settings.
3. Configure VAPID env vars on API.
4. Gold plan tenant for push feature gate.

Escalation

- iOS Safari PWA limitations — document platform constraints.

---

13. Socket.IO / chat realtime

Symptoms

- Messages appear only after refresh
- Typing indicator stuck
- Console WebSocket errors

Likely causes

- Redis adapter unavailable (single instance still works locally)
- Wrong socket base URL (`socketBaseUrl.ts`)
- CORS origin mismatch
- Auth cookie not sent cross-origin

Diagnose

- `apps/web/src/lib/socketBaseUrl.test.ts`
- Network WS connection to API origin

Files

- `apps/web/src/hooks/useChatRealtime.ts`
- API Socket.IO setup in `server.js`

Safe fix

1. Ensure web proxies API in dev Vite config.

2. Start Redis for multi-instance.
3. Same-site cookies: align API and web domains.

Escalation

- Production: sticky sessions or Redis adapter mandatory.

---

14. File upload / MinIO / S3 errors

Symptoms

- Product images fail
- Chat attachments 500
- `STORAGE_DRIVER` misconfiguration

Likely causes

- MinIO not running
- `S3_ENDPOINT`, keys wrong
- `storage_mb` plan limit exceeded

Diagnose

```
docker compose ps minio
curl http://localhost:9000/minio/health/live
```

Files

- `apps/api/src/config/env.js` `STORAGE_DRIVER`
- `apps/api/src/routes/files.routes.js`

Safe fix

1. `docker compose up -d minio`
2. `STORAGE_DRIVER=s3` with local MinIO credentials from `docker/.env`.

Escalation

- Production S3 bucket policy / IAM.

---

15. CSRF errors on POST/PATCH/DELETE

Symptoms

- 403 with CSRF message
- Mutations work in Postman but not browser

Likely causes

- Missing `X-CSRF-Token: 1` header
- Cookie session without CSRF setup
- Mobile Bearer incorrectly blocked (should skip — see `csrf.test.js`)

Files

- `apps/api/src/middlewares/csrf.js`
- `apps/web/src/services/api/base.ts`

Safe fix

1. Use generated API client (sets header).
2. Mobile: use `Authorization: Bearer` only.
3. Public routes `/api/public/*` exempt.

Escalation

- Custom integrators must document CSRF header requirement.

---

16. CORS / cookie / third-party login issues

Symptoms

- API calls blocked by CORS
- Cookies not sent on cross-subdomain setup

Likely causes

- `WEB_ORIGIN` not in CORS allowlist
- `COOKIE_DOMAIN` wrong for subdomain split
- `SameSite=None` without `Secure`

Files

- `apps/api/src/server.js` CORS config
- `apps/api/src/lib/rbac.js` cookie options

Safe fix

1. Set `WEB_ORIGIN` exactly (no trailing slash mismatch).
2. Production: first-party API+web domain pattern per `callbackOrigin()` docs.

Escalation

- Split domains require `COOKIE_DOMAIN=.example.com` + HTTPS.

---

17. Impersonation issues (admin)

Symptoms

- Impersonation banner but wrong tenant
- 403 while impersonating
- Cannot exit impersonation

Likely causes

- `impersonation_token` cookie invalid/expired
- Admin missing `ADMIN_TENANTS`
- Billing lock still enforced (by design)

Files

- `apps/api/src/lib/impersonation.js`
- `apps/web/src/hooks/useImpersonation.ts`

Safe fix

1. `POST /api/admin-dashboard/impersonate/stop`
2. Clear cookies; re-login admin.

Escalation

- Audit log review for impersonation events.

---

18. Mobile app auth / parity

Symptoms

- Mobile login fails; web works
- Plan gates differ mobile vs web

Likely causes

- Keycloak mobile client not configured (`docs/mobile/KEYCLOAK_MOBILE_CLIENT.md`)
- `EXPO_PUBLIC_API_URL` points to localhost on physical device
- Types out of sync with `apps/web/src/types/index.ts`

Safe fix

1. Configure Keycloak public client with PKCE + redirect `supplify://auth/callback` .
2. Use LAN IP or public API URL on device.
3. Run `cd supplify-mobile && npm run typecheck` .

Escalation

- See `docs/mobile/MOBILE_FEATURE_PARITY.md`.

---

19. Cron / background jobs not running

Symptoms

- Scheduled quick lists never place
- Trial expiry not locking
- Invoices not marking overdue

Likely causes

- `CRONS_ENABLED=false`
- `DELIVERY_ROLLOVER_ENABLED=false` (rollover no-op by design)
- API single instance crashed

Files

- `apps/api/src/lib/register-cron-jobs.js`
- `docs/operations/cron-jobs.md`

Safe fix

1. Enable crons in env for non-dev.
2. Manual: `node apps/api/scripts/run-delivery-rollover.mjs --force`

Escalation

- Move jobs to dedicated worker if API scales horizontally.

---

20. Typecheck / build / test failures (local dev)

Symptoms

- `pnpm typecheck` fails
- Vitest failures after pull

Safe fix

1. `pnpm install`
2. `pnpm db:migrate`
3. `pnpm --filter @supplify/api test:run`
4. `pnpm --filter @supplify/web test:run`

Escalation

- Compare with CI logs; check Node 18+.

Log locations summary

Component	Where
API	stdout / Railway logs / `docker compose logs api`
Web	Vite dev console; nginx access in prod
Keycloak	`docker compose logs keycloak`
Postgres	`docker compose logs postgres`
Migrations	`docker compose logs migrate`
E2E failures	`tests/e2e/test-results/`

Escalation matrix

Severity	Condition	Action
P0	Production auth down	Status page; rollback API; verify Keycloak
P1	Orders cannot be placed	Check billing lock, DB, API 500s
P2	Chat/realtime degraded	Redis + Socket.IO
P3	Reports slow	Analytics indexes; read replicas
P4	Demo script gap	Use backups in 12-demo-script.md

Part XV — Security Review *(Internal Technical Reference)*

Date: 2026-06-17
Scope: Security posture of Supplify as evidenced by repository code, configuration, and onboarding documentation.
Method: Static analysis and documentation cross-check only — **no destructive testing, no penetration testing, no production probing.**

Reviewer context: This document assesses what the codebase *implements* and where *gaps* remain. It is not a SOC 2 or ISO certification.

Executive summary

Supplify's security architecture is **appropriately layered for a B2B SaaS MVP targeting production**: OIDC via Keycloak, server-side JWT validation, mandatory API permission checks, CSRF on cookie-based web mutations, rate limiting, helmet headers, tenant-scoped data access, billing lock enforcement even under admin impersonation, and audited admin impersonation.

Residual risk clusters around: operational secrets handling, Redis/session hardening in multi-node deploys, incomplete features that previously misled users (now mitigated), GPS privacy configuration discipline, and documentation of public endpoints. No **Critical** code defect was identified in this static pass; highest practical risks are **High** configuration and process items.

Findings summary

Severity	Count	Themes
Critical	0	—
High	4	Secrets, Redis/session, admin surface, GPS privacy
Medium	6	Rate limits, CSRF scope, file uploads, audit completeness
Low	5	Verbose errors, demo passwords, lint debt
Informational	6	Positive controls, doc hygiene

Critical

None identified in static review.

High

H-1: Production secrets must be uniquely generated and rotated

Field	Detail
Finding	Default demo credentials (`SupplifyAdmin1!` , Keycloak `admin/admin`) and example `.env` values are suitable for local dev only.
Evidence	`apps/api/scripts/seed-demo-users.js` lines 21–45; `docker/.env` patterns; `docs/guides/setup.md`
Risk	Credential stuffing if defaults reach production or staging URLs.
Recommendation	Enforce unique `SESSION_SECRET` , Keycloak client secrets, DB passwords; disable demo seed scripts in prod; use secret manager (Railway variables).
Status	Process — not auto-enforced in code

H-2: Session and token storage depend on correct cookie flags

Field	Detail
Finding	Auth relies on httpOnly cookies (`access_token` , `refresh_token`). Misconfigured `COOKIE_SECURE` , `COOKIE_SAME_SITE` , or `COOKIE_DOMAIN` enables session theft or breaks auth.
Evidence	`apps/api/src/lib/rbac.js` `setAuthCookies()` ; `tests/e2e/auth.setup.ts` Secure-cookie HTTP warning
Risk	Session hijack on HTTP deployments; cross-site request risk if `SameSite` too permissive
Recommendation	HTTPS everywhere in prod; `COOKIE_SECURE=true` ; document `OAUTH_CALLBACK_BASE_URL` first-party pattern (`09-authentication-rbac.md`)
Status	Config-dependent

H-3: Admin impersonation is powerful — audit and permission gating required

Field	Detail
Finding	Admins can impersonate tenants via signed `impersonation_token` cookie; billing writes remain blocked but read access is tenant-equivalent.
Evidence	`apps/api/src/lib/impersonation.js` ; `tests/api/admin-impersonation.spec.ts` ; route matrix marks impersonate `UNSAFE` in `DEV_API_ROUTE_TEST_MATRIX.md`
Risk	Insider abuse; support account compromise
Recommendation	Restrict `ADMIN_TENANTS` ; log all impersonation to `audit-logs` ; MFA on admin Keycloak realm; time-boxed impersonation tokens
Status	Partially mitigated (audit exists; MFA is org process)

H-4: GPS tracking exposes driver location to restaurants — privacy env discipline

Field	Detail
Finding	Live driver GPS is globally gated (`GPS_TRACKING_ENABLED`) with optional name/phone exposure to restaurants.
Evidence	`apps/api/src/config/env.js` 244–252; `docs/features/drivers-and-gps-tracking.md` ; `GPS_RESTAURANT_SHOW_DRIVER_PHONE` default `false`
Risk	Workforce surveillance liability; phone leak if env mis-set
Recommendation	Keep phone hidden default; document DPA/worker consent; per-tenant opt-out roadmap
Status	Mitigated by defaults; ops vigilance required

Medium

M-1: CSRF protection skipped for Bearer (mobile) — correct but increases API token sensitivity

Field	Detail
Evidence	`apps/api/src/middlewares/csrf.test.js` — Bearer bypasses CSRF
Risk	Stolen mobile access token = full API access until expiry
Recommendation	Short access TTL; refresh rotation; remote wipe; biometric on mobile

M-2: Rate limiting may be insufficient for authenticated abuse

Field	Detail
Evidence	`server.js` global limiter; public 60/min prod; auth endpoints shared limit
Risk	Authenticated credential abuse, enumeration
Recommendation	Per-user rate limits on sensitive routes (`supplier-ops.routes.js` has local limiter pattern)

M-3: File upload attack surface (size, type, storage quota)

Field	Detail
Evidence	`files.routes.js` , MinIO/S3 driver, `storage_mb` plan meter
Risk	Malware hosting; DoS via large uploads
Recommendation	Content-type validation, virus scan hook, signed URLs with short TTL

M-4: Public and guest endpoints expand attack surface

Field	Detail
Evidence	`/api/public/*` , `/reserve/*` , consumer `/order/*` — CSRF exempt on public API
Risk	Scraping, reservation spam, checkout abuse
Recommendation	CAPTCHA on high-abuse endpoints; tighten public rate limits; monitor `email-logs` admin tab

M-5: Redis optional in dev — production must require Redis for Socket.IO scale

Field	Detail
Evidence	`resolve-redis-url.js` ; cache permission keys
Risk	Inconsistent security state across API instances; stale permissions up to 180s
Recommendation	Mandate Redis in prod; invalidate perm cache on role change (already implemented)

M-6: Tenant audit log not universal

Field	Detail
Evidence	`tenant_audit_log` Gold+; role change audit described as thin in demo audit
Risk	Forensics gap for Silver tenants
Recommendation	Platform audit for all subscription events; expand tenant audit coverage

Low

L-1: Error responses may leak implementation details in dev

Field	Detail
Evidence	`logger.error` with stack in `auth.routes.js` login catch
Risk	Verbose errors if `NODE_ENV` mis-set
Recommendation	Sanitize production error middleware

L-2: Demo seed scripts wipe commercial data

Field	Detail
Evidence	`seed-full.mjs` warning banner
Risk	Accidental run against shared staging
Recommendation	`NODE_ENV=production` guard in seed scripts

L-3: Supplier Settings unwired tabs (mitigated)

Field	Detail
Evidence	`DELIVERY_ZONES_ENABLED=false` ; fake toasts removed per `SUPPLIFY_DEMO_READINESS_AUDIT.md`
Risk	Was user-trust issue; now honest messaging
Recommendation	Hide tabs until wired

L-4: ESLint `exhaustive-deps` warnings (46)

Field	Detail
Evidence	Demo readiness audit §6
Risk	Stale closure bugs — indirect security (wrong tenant data displayed)
Recommendation	Triage per file

L-5: `sql-migrator.js` treats `42P07` as success

Field	Detail
Evidence	`docs/audits/supplify-quick-performance-ui-db-security-audit.md`
Risk	Partial migrations marked applied
Recommendation	Strict migration CI on fresh DB

Informational (positive controls)

I-1: Server-side RBAC is mandatory

`requirePermission` on routes; comprehensive tests in `rbac-full-app.test.js` and e2e `rbac.spec.ts` .

I-2: JWT validation uses remote JWKS with issuer normalization

`apps/api/src/lib/auth.js` — industry standard.

I-3: Staff portal isolation

`STAFF_PORTAL` users blocked from main `/app` APIs via `assertStaffPortalRouteAccess` .

I-4: Driver role minimized

Only `DRIVER_DELIVERIES_VIEW` and `DRIVER_DELIVERIES_MANAGE` ; `driver-rbac.js` status enum enforcement.

I-5: Billing lock cannot be bypassed by impersonation

`billingAccessMiddleware` tests; documented in billing regression audit.

I-6: Security headers via Helmet

`apps/api/src/server.js` helmet middleware configured.

I-7: Permission cache invalidation on role changes

Redis `perm:*` keys; TTL 180s documented in `09-authentication-rbac.md`.

I-8: Route inventory and test matrix for 554 API routes

`docs/audits/route-inventory.json`, `DEV_API_ROUTE_TEST_MATRIX.md` — visibility for untested unsafe routes.

Documentation security assessment

Doc area	Assessment
09-authentication-rbac.md	Accurate OIDC flow, cookie table, permission list — suitable for engineers; does not expose secrets
12-demo-script.md	Contains demo passwords intentionally — **mark as INTERNAL**; do not publish to public web
14-troubleshooting.md	Safe fixes only; no exploit instructions
Onboarding guides	No production credentials committed in tracked files (verify `.env` gitignored)
Gap	No dedicated `SECURITY.md` responsible disclosure policy in onboarding set — recommend root `SECURITY.md`

Threat model sketch (documentation level)

Primary assets: Tenant commercial data (orders, invoices, PII), credentials, payment methods.
Primary controls: OIDC, RBAC, tenant_id scoping, TLS, CSRF (web), rate limits.

Compliance-oriented notes (non-exhaustive)

Topic	Status in codebase
GDPR data export/delete	Partial — admin tools; verify DPA requirements per deployment
PCI	Card data via payment provider — confirm SAQ scope with Stripe/integration
Audit trail	Platform admin audit + tenant audit (Gold)
Data residency	Not enforced in code — deployment choice

Recommended next security work (priority order)

1. Add production guard to destructive seed scripts.
2. Publish `SECURITY.md` with disclosure contact.
3. MFA for admin Keycloak realm in production.
4. Per-user rate limiting on auth and file upload routes.
5. Hide or wire Supplier Settings Delivery Zones/Contacts tabs.
6. Automated dependency scanning in CI (if not already — verify `.github/workflows`).

Assessment conclusion

Supplify's **documented and implemented security model is coherent for enterprise B2B demos and controlled production rollout**, provided operators follow environment hardening (HTTPS, secrets, Redis, Keycloak tuning). The largest gaps are **operational** (secret hygiene, admin power, GPS privacy config) rather than missing authentication entirely.

This review did not include dynamic scanning, fuzzing, or social engineering.

Part XVI — Implementation Status

Date: 2026-06-17
Branch baseline: `ab5695e` (per `docs/onboarding/_artifacts/bootstrap-metrics.md`)
Purpose: What actually works, what is partial, what is missing tests, inconsistencies, dead code, and deployment risks.

Not marketing. If a feature is UI-only, it says so.

TL;DR verdict

Dimension	Grade	One-line truth
Core B2B order flow	**A**	Restaurant cart → supplier accept → fulfill → receive → invoice works end-to-end
Admin platform	**A-**	14 lazy-loaded tabs; impersonation + plans + deals production-usable
Monetization / tiers	**A-**	FE/BE keys aligned; Free Trial = Gold features + Free limits is confusing by design
Hospitality add-ons	**B+**	Reservations, staff, consumer B2C shipped; less demo polish than core B2B
Logistics / GPS	**B**	Fulfillment board strong; driver accounts not in `seed:full`; GPS env-dependent
Test coverage	**B+**	~1008 API + web unit tests; E2E only 16 Playwright files; 554 routes mostly API-unit
Production readiness	**B**	Railway docs exist; Keycloak memory/OOM history; cron in-process; lint gate still noisy

Demo readiness: Yes, with scripted path (`12-demo-script.md`). **Free roam: No** — known UI-only settings tabs and finance gaps.

Metrics (code-verified)

Metric	Value	Source
API routes	554	`docs/audits/route-inventory.json`
SQL migrations	179	`apps/api/db/migrations/`
API test files	213	bootstrap-metrics
Web test files	309	bootstrap-metrics
Playwright e2e specs	16	`tests/e2e/suites/`
Frontend routes	~80	`apps/web/src/App.tsx`
Permission keys	52	`permission-keys.js`
Restaurant feature keys	26	`feature-keys.js`
Supplier feature keys	24	`feature-keys.js`

What is fully working

These domains have **UI + API + RBAC + plan gates + meaningful tests**:

Domain	Evidence	Confidence
OIDC auth & session refresh	`auth.routes.js` , e2e auth	High
Tenant RBAC (16 system roles)	`role-matrix.js` , `rbac-full-app.test.js`	High
Plan features & limits	`plan-enforcement.js` , `verify-tier-matrix`	High
Supplier catalog CRUD + CSV + image ZIP	`products.routes.js` , migration `0168`	High
Restaurant browse, cart, place order	e2e `orders.spec.ts`	High
Supplier accept/decline orders	`order-decline.md` , order status enum	High
Fulfillment board & dispatch	`FulfillmentPage` , fulfillment routes	High
Receiving	`receiving.routes.js` , API spec	High
Disputes & credit notes	`disputes.service.js`	High
Invoices & payments (core)	`invoices.routes.js`	High
Chat + Socket.IO	`chat.routes.js` , `useChatRealtime`	Medium-High (Redis-dependent)
Deals/promotions + admin approval	`deal-promotions.service.js`	High
Admin command center	`admin-dashboard/*` , lazy tabs	High
Reservations FOH + public booking	`reservations.routes.js`	High
Consumer B2C storefront	migrations `0161` – `0164` , smoke e2e	Medium-High
Supplier growth program	migration `0169` , tests	High
Supplier ops wave 2	run sheet, pick lists, collections, POD, quote lock, accounting export	High
Quote requests RFQ	migration `0153`	Medium-High
Warehouses & branches	`warehouses.routes.js` , branches audit	High
Tenant audit log (Gold)	`tenant-audit.routes.js`	High

Partial — UI exists, backend incomplete, or behavior wrong

Feature	What works	What does not	Evidence
Supplier Settings → Contacts	Same	UI-only; honest toast after audit fix	`SupplierSettingsPage.tsx`
Restaurant finance statements	Period charges/payments/closing	`openingBalance` **hardcoded** `0`	`restaurant-finance.routes.js:795`
Dashboard period selector	UI toggles 7d/30d/90d	**Does not refilter** spend trend (fixed 30d)	`DashboardPage.tsx` ; demo audit
Smart reorder	API + forecast job	No dedicated nav; dashboard widget only	`reorder-forecast.job.js`
Delivery rollover cron	Manual script + per-assignment API	**Hourly cron no-op** unless `DELIVERY_ROLLOVER_ENABLED=true`	`env.js` , `cron-jobs.md`
Credit notes nav	Via disputes/invoices	No top-level restaurant nav	feature audit
Restaurant `restaurant-gold@` **demo**	Gold entitlements	**Past-due billing** seeded intentionally	`seed-billing.js`
Supplier `supplier-silver@` **demo**	Silver entitlements	**Locked account** seeded	`seed-billing.js`
Deal redemption metering	Limit check at redemption	Increment path flagged for manual QA	demo audit §4.8
Role change audit	Tenant audit on some events	Thin coverage for all RBAC mutations	demo audit §8
Last-owner guard	Prevents demotion in org context	Org-less owner can self-demote edge case	demo audit §8

Missing or weak test coverage

Area	Unit/API tests	E2E	Gap severity
Auth login flow	Partial	`auth.spec.ts`	Low
Fulfillment + GPS live	Component tests	None full path	**Medium**
Driver deliveries E2E	`driver-rbac` unit	None	**Medium**
Restaurant finance statements	Sparse	None	**Medium**
Consumer B2C checkout	Some API	Smoke only	Medium
Admin mutations (plan change)	API tests	Skipped in route matrix (`SKIP_MUTATION`)	Medium
554 API routes	~213 test files	Live route matrix skips unsafe	**High breadth gap**
Mobile app	Sibling repo	Not in this metrics	External
PWA push end-to-end	Manifest unit	None	Low
Impersonation	`admin-impersonation.spec.ts`	Manual	Low

Honest statement: Unit test count is **impressive** (~1300+ tests combined), but **E2E breadth is narrow** (16 files). Most routes are validated only by inventory/matrix classification, not automated HTTP tests.

Permission / plan inconsistencies

Issue	Detail	Severity
Free Trial feature parity	Free plan uses **Gold feature JSON** with **Free limits** — looks like Gold in UI, hits limits quickly	Intentional; document in sales
GPS tracking	**Not plan-gated** — env flag only	Product decision; surprises tier buyers
Reservations	**No** `feature-keys` **entry** — always on	May not match packaging docs
`promotions` limit	Supplier-only; restaurant uses `supplier_deals` + `deal_redemptions_per_day`	Correct in code; easy doc confusion
Enterprise tier	**Removed** from catalog (`0066`); maps to platinum in code only	Legacy data may exist
Bronze display name	Mapped to Silver in UI (`formatPlanDisplayName`)	Cosmetic
Contract pricing feature key	Placeholder gating	Low
Frontend-only permission checks	UX hiding; **API is source of truth**	OK if API always called

Verification tools that pass: `pnpm verify:tier-matrix` , `plan-catalog-audit` tests, `adminLimitLabels.test.ts` .

Dead code, deprecated, and removed features

Item	Status	Evidence
Approvals & budgets	**Removed**	migration `0114` ; `REMOVED_FEATURE_KEYS`
Enterprise plan selectable	**Deactivated**	`0066_remove_enterprise_tier.sql`
Supplier command-center broken link	**Fixed**	was `/app/supplier/command-center` → `/app/command-center`
Fake success toasts (supplier settings)	**Fixed**	honest messaging
`api_user` DB role grants	**Commented out**	migrations `0019` , `0020` , `0039`
Uncommitted admin refactor	Was in audit — verify committed on your branch	demo audit §10

Possible dead UI: Supplier Settings tabs behind `false` flags — not dead, deliberately disabled.

Seed data honesty (`seed:full`)

Seeded well	Not seeded / weak
10 prod-like restaurants, 50 suppliers	Driver Keycloak logins
~2k products, ~1.5k orders	Live GPS route history
Invoices, chats, quick lists	Multi-warehouse stock edge cases
Tier demos Free–Platinum	Smart-reorder demand history depth
Disputes, approved deals	Receiving line items for every order
Coupon `DEMOFORK10` , expiry item, near-limit Free supplier	Near-limit restaurant examples
~70 Keycloak accounts	`seed:tier-catalog` team roles (`*-manager@`) **not** in default full seed

Best demo logins: `restaurant@supplify.com` + `supplier@supplify.com` (Gold, **active billing**, richest extras).
Avoid for smooth demo: `restaurant-gold@` (past due), `supplier-silver@` (locked).

Deployment risks

Risk	Why it hurts	Mitigation
Keycloak OOM on Railway	Historical 7–10 GB spikes	`KEYCLOAK_RAILWAY_MEMORY_FIX.md` , JVM caps
In-process crons	Duplicate runs if API scaled horizontally	`CRONS_ENABLED` + single replica or external scheduler
Redis optional	Socket.IO / perm cache inconsistent multi-node	Mandate Redis prod
Migration partial failure	Compose migrate continues on error (`WARN`)	CI fresh-DB migration gate
Public Redis URL on Railway	Egress fees / wrong host	`resolve-redis-url.js`
Session store in Postgres	Load on auth	Acceptable at current scale; watch pool
Lint max 0 warnings	46 warnings remain — CI may fail	demo audit §6
Destructive seed on staging	`seed:full` wipes tenants	Process guard
GPS / Maps API keys	Tracking UI blank in prod	Env checklist
SMTP not configured	Silent email failures	Mailpit dev; Resend prod

Mobile parity status

Area	Web	Mobile (`supplify-mobile`)
Auth PKCE	Cookie	Bearer
Orders	Full	Parity expected
Driver GPS	Web view	Primary capture
Admin	Full	Limited/none
Consumer B2C	Full	Partial

Rule: `.cursor/rules/mobile-parity.mdc` — changes must be evaluated. Drift documented in `docs/mobile/MOBILE_FEATURE_PARITY.md` .

CI / quality gates (honest)

Gate	Typical status	Notes
`pnpm typecheck`	Pass	per demo audit
API `vitest run`	Pass (~1008 tests)	fixed stale mocks Jun 2026
Web `vitest run`	Pass	+ locked deals regression
`pnpm lint`	**May fail**	46 warnings (22 exhaustive-deps)
Playwright e2e	Requires infra	auth, orders, rbac, catalog, subscription-limits
`verify:tier-matrix`	Needs live DB	not always in CI

Feature status by persona

Restaurant — 85% complete

Working: orders, cart, suppliers, receiving, invoices, inventory, deals, chat, quick lists, disputes, reservations, consumer modules, reports (tiered).
Weak: finance statement opening balance, smart reorder surfacing, dashboard period filter.

Supplier — 82% complete

Working: catalog, orders, fulfillment, invoices, promotions, growth, contract pricing, warehouses, **run sheet, pick lists, collections reminders, accounting export, warehouse zones UI, quote price lock, POD media**.
Weak: settings contacts tab, driver seed login, deal approval dependency for new promos.

Admin — 88% complete

Working: overview, tenants, plans, subscriptions, limits, deals, finance, ops health, audit, impersonation.
Weak: some mutation paths manual QA only; password reset marked unsafe in route matrix.

Driver — 75% complete

Working: role isolation, delivery board, status enum, proof of delivery APIs.
Weak: no default demo account; GPS depends on ops env; E2E gap.

Public / guest — 80% complete

Working: reservations, consumer order flow, public supplier catalog (no anon prices), quote flows.
Weak: abuse protection (rate limits only).

Recommended engineering priorities

1. **Wire or hide** Supplier Settings Contacts (product decision).
2. **Fix** restaurant finance `openingBalance` calculation.
3. **Add** driver demo to `seed:full` (Keycloak + one assignment).
4. **E2E:** fulfillment → driver status → restaurant tracking path.
5. **Dashboard:** wire period selector or remove control.
6. **Production:** migration CI on empty DB; Redis required flag.
7. **Lint:** burn down 46 warnings or adjust gate policy explicitly.

How this doc stays honest

- Claims cite files, migrations, or audit docs — not roadmap slides.
- "Working" means code path exists and was verified in audits/tests — not that every customer edge case is handled.
- Partial features are **not** labeled Shipped in 13-acceptance-criteria.md.

Related artifacts

Document	Use
SUPPLIFY_DEMO_READINESS_AUDIT.md	Jun 2026 demo pass
full-app-feature-audit.md	Feature matrix
DEV_API_ROUTE_TEST_MATRIX.md	Route test classification
12-demo-script.md	What to show anyway

Part XVII — Glossary

Audience: Sales, support, onboarding specialists, developers, and customers who need a shared vocabulary for Supplify.

Source of truth: Application code (`apps/api` , `apps/web`), migrations, and companion docs in `docs/onboarding/` .

Terms are grouped by theme. Where a concept has both a **plan entitlement** (subscription feature) and an **RBAC permission**, both are noted — they are independent gates (see 09-authentication-rbac.md and 10-subscriptions-and-plans.md).

Platform & identity

Tenant

A **tenant** is a billable organization row in the database: either a `restaurant` or a `supplier` . Every tenant has its own subscription, team, branding, and data isolation. API handlers resolve the active tenant via `tenant-resolve.js` (`getRestaurantIdForRequest` , `getSupplierIdForRequest`). Platform admins (`ADMIN` role) are not tenants.

Workspace

A **workspace** is the authenticated product experience for one tenant. Users see one restaurant **or** one supplier workspace at a time. Table `user_workspace_membership` enforces **at most one** active restaurant or supplier account per user (migration `0104_user_workspace_membership.sql`). The account creator is **main admin** (`is_main_admin`) with the **Owner** system role.

Organization (org)

Restaurant and supplier tenants can have an **organization** parent (`restaurant_organizations` , `supplier_organizations`) with child **branches**. Org-level billing rolls up to the root tenant via `resolveOrgBillingTenantId` . Multi-branch features require plan key `multi_branch` (Gold+ on paid tiers; enabled on Free Trial via Gold feature parity).

Branch

A **branch** is a physical or logical site under an org: `restaurant_branch` or `supplier_branch` . Branches scope orders, inventory, quick lists, and delivery coordinates. Plan limit `branches` caps active locations (Free/Silver: 1; Gold: 3; Platinum: unlimited). Branch invites use `/invite/branch` .

Platform role (`app_user.role`)

Keycloak-linked persona stored on `app_user` : `PENDING` (registration incomplete), `RESTAURANT` , `SUPPLIER` , `ADMIN` , or `STAFF_PORTAL` . This is **not** the same as tenant RBAC roles (Owner, Manager, etc.).

Main admin

The user who created the tenant (`is_main_admin = true`). Cannot be removed without transfer. Always mapped to **Owner** unless explicitly reassigned within guard rules in `rbac-guards.js` .

Pending activation

Subscription state `lock_reason = pending_activation` after registration. `billingAccessMiddleware` blocks writes until the user completes `/app/activate` (free or paid checkout). Distinct from Free Trial **sandbox expiry**.

Free Trial / sandbox expiry

Free plan workspaces get `subscription.free_sandbox_expires_at` (default 7 days from `platform_setting.free_sandbox_days`). After expiry, account is locked: reads mostly allowed, writes return **402 Payment Required**.

Access control

RBAC (role-based access control)

Tenant-scoped RBAC maps users to roles (`tenant_user_roles`) and roles to permission keys (`tenant_role_permissions`). Canonical permission constants live in `apps/api/src/lib/permission-keys.js` (52 keys). Backend enforcement is mandatory via `requirePermission` ; the React app mirrors checks for UX only.

Permission

A granular capability key such as `ORDERS_CREATE` , `RECEIVING_MANAGE` , or `INVOICES_VIEW` . `hasPermission` treats domain `_MANAGE` as satisfying `_VIEW` / `_EDIT` checks. Permissions are cached in Redis (`perm:{userId}:{tenantId}:{tenantType}` , TTL ~180s).

System role

Predefined role template synced per tenant from `role-matrix.js` : e.g. Restaurant **Owner**, **Purchaser**, **Receiving Staff**; Supplier **Warehouse Manager**, **Driver**. Owner has `permissions: 'ALL'` .

Custom role

Tenant-defined role created under **Settings → Team → Roles** when plan feature `advanced_roles` is enabled (Gold+). Names cannot collide with reserved system role names. Assigner cannot grant permissions they do not hold.

Admin permission

Platform-scoped keys for `ADMIN` users: `ADMIN_ACCESS` , `ADMIN_TENANTS` , `ADMIN_PLANS` , `ADMIN_SUPPORT` , `ADMIN_FINANCE` , `ADMIN_GROWTH` . Stored in legacy `role` / `role_permission` tables. Tab visibility in `/app/admin` follows `resolveAdminDashboardPermission()` .

Entitlement

Runtime subscription payload from `GET /api/subscriptions/entitlements` : effective **plan**, **features**, **limits**, **usage**, overrides, and addons. Frontend hook: `useEntitlements()` . Entitlements answer: *did this tenant pay for the module?*

Feature (plan feature key)

Boolean or tier string in `subscription_plan.features` JSON, e.g. `finance_invoices` , `smart_reorder` , `driver_management` . Enforced by `requireFeature()` → **403 FEATURE_NOT_AVAILABLE**. Canonical keys: `feature-keys.js` .

Limit (plan limit key)

Numeric cap in plan JSON, e.g. `orders_per_day` , `warehouses` , `branches` . Value `-1` or `null` means unlimited. Enforced by `requireWithinLimit()` or inline `checkLimit()` . Resolution order: plan → plan override → tenant override → location addons.

Feature gate vs permission gate

Both must pass for many actions:

Layer	Question	Example
Entitlement	Tenant on right plan?	`disputes_returns` for dispute API
RBAC	User allowed?	`RECEIVING_MANAGE` to post receiving
Billing lock	Account writable?	Not locked / not expired Free Trial

Impersonation

Admin **view-as** tenant workflow. Cookie `impersonation_token` (JWT signed with `IMPERSONATION_SECRET`). Requires `ADMIN_SUPPORT` or `SUPER_ADMIN`. Effective permissions come from **view-as role**, not blanket Owner bypass. Cleared on login/logout. Documented in `impersonation.js`.

Staff portal

Separate operational surface for restaurant **scheduling staff** (`STAFF_PORTAL` app role, Keycloak `staff_portal`). Not tenant Team RBAC. Allowlisted API paths only; home `/staff/dashboard`.

Commerce & orders

Customer order

B2B order from restaurant to supplier (`customer_order` + line items). Status lifecycle includes placement, supplier acceptance/decline, fulfillment, delivery, invoicing. Restaurants need `ORDERS_CREATE`; suppliers need `ORDERS_VIEW` / `ORDERS_MANAGE` to decline or manage.

Quick list / ordering list

Saved reorder template (`quick_list`, `quick_list_item`). UI label **Ordering Lists**; route `/app/quick-lists`. Plan feature `quick_lists`; limits `quick_lists`, `quick_list_items`, `scheduled_quick_lists`. Can scope to `branch_id`.

Scheduled quick list

Quick list with `is_scheduled = true` for automated or calendar-driven reorder. Free Trial has hidden limit `scheduled_order_grace_per_day` (one daily order overflow).

Smart reorder

Restaurant feature `smart_reorder` (Gold+): cadence detection, at-risk SKUs, dashboard widgets. Uses `reorder-cadence` service and optional `ai_platform` LLM assistant (`ai_requests_per_day` limit on Gold/Platinum).

Reorder cadence

Computed ordering rhythm per restaurant SKU/branch from historical order and inventory signals. API: `POST /api/restaurant-inventory/reorder-cadence/recompute`; supplier at-risk view: `GET /api/supplier/reorder-cadence/at-risk`. Requires inventory + smart reorder entitlements.

Order amendment

Post-placement change request (`order_amendments` tables). Plan feature `order_amendments` (all tiers including Free Trial). Restaurant accepts/rejects; does not silently mutate lines without workflow.

Substitution

Supplier-proposed replacement product when original is unavailable. Creates `order_fulfillment_issue` with status `substitution_suggested` and may spawn pending **amendment** for mapped products. Order lines are **not** auto-changed. API under `/api/supplier/orders/:orderId/fulfillment-issues/substitution`.

Shortage

Supplier-reported inability to fulfill ordered quantity. Creates fulfillment issue `shortage_reported`; may open contextual chat (`ORDER_REFERENCE` message type).

Fulfillment

Supplier-side pick/pack/dispatch workflow. Plan features `fulfillment` and/or `fulfillment_tools` (supplier only; off for restaurants). Permissions `FULFILLMENT_VIEW`, `FULFILLMENT_MANAGE`. UI: `/app/fulfillment` board, routes, warehouse assignment.

Fulfillment issue

Structured shortage/substitution record (`order_fulfillment_issue`, migration `0134_order_fulfillment_issues.sql`). Statuses: `shortage_reported`, `substitution_suggested`, `waiting_restaurant_approval`, `accepted`, `rejected`.

Decline reason

Supplier rejection of an order with coded/text reason before fulfillment starts. Requires `ORDERS_MANAGE`.

Contract pricing

Negotiated price list between supplier and restaurant; overrides catalog default on eligible lines.

Supplier follow

Restaurant relationship `restaurant_supplier_follow` / `supplier_follow`. Limit `suppliers_per_restaurant` by plan.

Deal / promotion

Supplier commercial offer. Restaurant redemption: `supplier_deals`, `supplier_deals_redeem`; supplier promos: `promotions` with limit `promotions`. Admin may approve certain deal types.

Quote request

Restaurant-initiated RFQ-style flow to supplier outside standard catalog checkout (see product guide).

B2C consumer order

Public storefront order at `/order/:restaurantSlug` — separate from B2B `customer_order` procurement.

Logistics & receiving

Warehouse

Supplier ship-from location (`warehouse` table). Plan feature `warehouses`; limit `warehouses` (0 on Free = feature effectively off; Silver: 1; Gold: 3; Platinum: ∞). Permissions `WAREHOUSES_VIEW`, `WAREHOUSES_EDIT`, `WAREHOUSES_MANAGE`. Multi-warehouse routing: `multi_warehouse` (Gold+).

Receiving

Restaurant confirmation of goods delivered against an order. Plan feature `receiving_quality` (photos, quality scoring tiers). Permissions `RECEIVING_VIEW`, `RECEIVING_MANAGE`. API: `receiving.routes.js`. Distinct from supplier warehouse **receiving** in inventory context.

Receiving session

Structured receive flow tying order lines to quantities received, optional quality photos/scores, and optional lot creation.

Proof of delivery (POD)

Driver-captured evidence (notes, optional photo) on `delivered` transition. Driver permissions `DRIVER_DELIVERIES_MANAGE`.

Delivery status

Driver assignment lifecycle: `assigned` → `picked_up` → `out_for_delivery` → `delivered` | `failed` | `rescheduled`. API: `PATCH /api/orders/:id/delivery-status`.

Driver route

Ordered sequence of delivery stops (`fulfillment` routes API). Driver can build route from assignments: `POST /api/fulfillment/routes/build-from-assignments`.

GPS tracking / ETA

Live driver location shared during `out_for_delivery` when supplier plan and restaurant **delivery coordinates** are set. Restaurant map privacy: driver-focused surfaces per product rules.

Delivery coordinates

Latitude/longitude on restaurant or branch (`PATCH /api/restaurants/me/delivery-location`). Required for accurate ETA — street address alone is insufficient.

Inventory & quality

Restaurant inventory

On-hand stock per restaurant SKU (`restaurant_inventory`). Plan feature `inventory_management`; limit `restaurant_inventory_skus`.

Inventory lot

Batch-level record (`restaurant_inventory_lot`) with `quantity`, `expiry_date`. Status at read time: `safe`, `expiring_soon`, `expired` (default threshold 7 days). Platinum tier string includes `lot_expiry_tracking`. Migration `0133_restaurant_inventory_lots.sql`.

Supplier inventory

Warehouse-scoped available quantity (`inventory` / `available_qty` on supplier side). Drives fulfillment shortage detection.

Waste tracking

Restaurant feature `waste_tracking` for recording spoilage/shrink with analytics tiers on paid plans.

Stock status

Computed label (in stock, low, out) from thresholds — aggregate quantity; expiry handled at lot level.

Dispute

Post-receiving disagreement (quality, quantity). Plan `disputes_returns`. May escalate to returns workflow.

Quality score

Optional numeric/structured score on receiving when plan tier enables `receiving_quality` quality scoring (Gold+).

Finance

Invoice

Bill document tied to fulfilled/delivered orders. Plan feature `finance_invoices`. Permissions `INVOICES_VIEW`, `INVOICES_CREATE`, `INVOICES_EDIT`, `INVOICES_MANAGE`.

Receivable

Supplier-side outstanding amount owed by restaurants (`GET /api/supplier/invoices/receivables*`). Requires `INVOICES_VIEW` + `finance_invoices` feature.

Payable (restaurant)

Restaurant-side obligation to pay supplier invoices; recorded payments reduce open balance.

Aging

Buckets of open receivables/payables by days outstanding (e.g. current, 30, 60, 90+). Shown in finance dashboards when `finance_invoices` tier includes analytics.

Payment recording

Manual or stub-gateway payment applied to invoice (`PAYMENTS_VIEW`, `PAYMENTS_MANAGE`). Accountant role typical owner.

Account statement

Period summary of orders, invoices, and payments between restaurant and supplier pair.

Billing checkout

`POST /api/billing/checkout` — activates plan, clears `pending_activation`, or upgrades tier. Stub card `4242424242424242` when `BILLING_GATEWAY=stub`.

Plan code

Canonical subscription tier: `free`, `silver`, `gold`, `platinum` (`plan-codes.js`). Legacy `enterprise` deactivated; `bronze` aliases to `silver`.

Pending downgrade

`subscription.pending_plan_id` + `pending_effective_at` — lower tier applies on next billing cycle read.

Limit override / feature override

Admin or growth tools to raise caps (`tenant_limit_override`, `plan_limit_override`) or toggle features (`feature-flags.js`) without changing base plan row.

Growth & discovery

Supplier growth

Supplier feature `supplier_growth` — referrals, customer import, command-center analytics. Free Trial includes via migration `0175`.

Mini-store / public catalog

Unauthenticated supplier catalog at `/supplier/:idOrSlug` for discovery and quote flows.

Supplier review

Restaurant rating of supplier; feature `supplier_reviews`.

Referral token

`referralToken` on `POST /api/register/complete` from `/register?ref=…` linking new tenant to growth program.

Customer import

Supplier bulk import of restaurant contacts (`CUSTOMERS_IMPORT` permission).

Reservations & FOH

Reservation

Table booking for restaurant FOH. Permissions `RESERVATIONS_*` . Public guest flow at `/reserve` .

Waitlist

Queue when no tables available; feature `waitlist_auto_promo` for auto-promotion rules (Gold+).

FOH Staff

Restaurant system role with reservations permissions only — no order create.

Chat & notifications

B2B chat

Tenant messaging with plan feature `chat` (tiered: multi_supplier, group_chat_files, real_time_media). Limits `chats_per_day` , `open_conversations` .

ORDER_REFERENCE message

Chat message type linking thread to specific order — used in substitution/shortage flows.

Push notification

Mobile/web push when `push_notifications` enabled; requires device registration.

Notification preference

Per-user/category opt-in stored in notification settings (e.g. `notify_inventory_expiring`).

Admin & platform

Super Admin

Platform role with all `ADMIN_*` permissions.

Support Admin

`ADMIN_ACCESS` , `ADMIN_TENANTS` , `ADMIN_SUPPORT` — tenant directory, impersonation, password reset.

Finance Admin

Billing overview, revenue metrics — `ADMIN_FINANCE` .

Growth Admin

Feature flags, deal approvals, experiments — `ADMIN_GROWTH` .

Audit log

Platform and tenant activity records. Tenant feature `tenant_audit_log` (Gold+). Admin audit at `/app/admin` → Audit.

Feature flag

Runtime toggle layered on plan JSON via `resolveFeatureEnabled()` — may enable beta features without plan migration.

Conversion event

Monetization telemetry when user hits `BLOCKED_FEATURE` or `BLOCKED_LIMIT` — feeds admin conversion stats.

Technical

PWA (Progressive Web App)

Web app installable on mobile/desktop with service worker caching (`apps/web` Vite PWA plugin). Driver and field workflows are **PWA-friendly** (offline-limited; mutations require network). Not a separate native app — see `supplify-mobile` for React Native parity.

OIDC / Keycloak

Identity provider for login. Authorization code flow via `/auth/login` → Keycloak → `/auth/callback`. Tokens in HttpOnly cookies (`access_token` , `refresh_token`).

CSRF token

`X-CSRF-Token` header on state-changing API calls when using cookie auth.

RTK Query

Redux Toolkit data layer in `apps/web` for API hooks (`useGetEntitlementsQuery` , etc.).

Tenant context

Request attachment from `resolveTenantContext` : `tenantId` , `tenantType` , `permissions` , active branch.

Active tenant cookie

`active_tenant` — branch/workspace switcher state for multi-branch users.

Redis cache

Shared cache for permissions, entitlements, subscription (recommended production: `REDIS_URL`).

Migration

Sequential SQL file in `apps/api/migrations/` — schema source of truth alongside runtime code.

Route inventory

Machine-readable API catalog `docs/audits/route-inventory.json` (554 routes as of 2026-06-17).

Acronyms

Acronym	Meaning
B2B	Business-to-business (restaurant ↔ supplier)
B2C	Business-to-consumer (public menu orders)
ETA	Estimated time of arrival (delivery)
FOH	Front of house (reservations, guest-facing)
GPS	Global positioning system (driver location)
KPI	Key performance indicator (dashboards/reports)
LLM	Large language model (AI reorder assistant)
MOQ	Minimum order quantity (supplier policy)
OIDC	OpenID Connect (auth protocol)
POD	Proof of delivery
PTO	Paid time off (staff portal)
PWA	Progressive Web App
RBAC	Role-based access control
RFQ	Request for quote
SKU	Stock keeping unit (product identifier)
SLA	Service level agreement (support tier)
VAT	Value-added tax identifier

Related docs

- 09-authentication-rbac.md — permissions and roles in depth
- 10-subscriptions-and-plans.md — feature and limit matrices
- 02-complete-product-guide.md — feature-to-route mapping

- 11-api-and-workflow-reference.md — API workflows

Part XVIII — Frequently Asked Questions

Audience: Sales, customer support, onboarding specialists, and developers answering real customer questions.

Grounding: Answers reflect current plan matrices (`10-subscriptions-and-plans.md`), RBAC (`09-authentication-rbac.md` , `role-matrix.js`), and registration/billing flows verified against the repository.

Sales & pricing

What plans does Supplify offer?

Four self-serve tiers for **both** restaurant and supplier workspaces: **Free Trial** (`free`), **Silver** ($49/mo), **Gold** ($149/mo), and **Platinum** ($349/mo). Yearly pricing is available at roughly 10× monthly. Legacy **Enterprise** was removed; `enterprise` codes normalize to `platinum` for comparisons only.

What is included in Free Trial?

Free Trial uses **Gold feature gates** with **Free limit caps** — prospects can explore nearly the full product surface for ~7 days (configurable `free_sandbox_days`), not a crippled demo. After sandbox expiry, the account becomes read-only for most GETs; writes return **402** until upgrade.

Which plan should a single-location restaurant start on?

Silver if they need paid support and modest volume (20 orders/day, 5 suppliers, 1 branch). **Gold** if they need multi-branch (3 branches), smart reorder, advanced roles, API keys, or higher daily order volume (100/day). **Platinum** removes most numeric caps and adds advanced reporting/AI forecast tiers.

Which plan should a regional distributor start on?

Silver enables one warehouse and basic fulfillment. **Gold** adds multi-warehouse (3 warehouses, 3 branches), driver management, and warehouse pick/pack fulfillment tools. **Platinum** adds routing suite and unlimited warehouses/branches/SKUs.

Can a customer mix restaurant and supplier accounts on one login?

No. `user_workspace_membership` allows **one** active restaurant **or** supplier workspace per email. Same-organization branch invites are allowed; a second unrelated tenant on the same email is rejected at invite accept.

Do restaurants and suppliers need separate subscriptions?

Yes. Each tenant row has its own `subscription` . A company that is both buyer and seller must register two workspaces (two emails or sequential accounts per policy).

What happens when they hit a limit mid-month?

API returns **403 LIMIT_EXCEEDED** with upgrade payload (`recommendPlan()` suggests next tier). Daily meters (`orders_per_day` , `chats_per_day` , `ai_requests_per_day`) reset at UTC day boundary. Admins can apply **tenant limit overrides** (increase-only) without changing plan code.

Is custom branding available on Silver?

No. `custom_branding` is off on Silver for both tenant types. Gold enables logo/colors; Platinum adds white-label domain tier string.

Can we quote API access on Silver?

No. `api_integrations` is disabled on Silver. Gold grants `api_key_access` ; Platinum grants `full_api_webhooks` .

Onboarding & activation

Why can't the customer save settings or place orders after signup?

New tenants have `lock_reason = pending_activation` . They must complete `/app/activate` — either activate Free or complete paid checkout. Until then, `billingAccessMiddleware` blocks writes (**402**).

What is the difference between pending activation and Free Trial expiry?

State	Trigger	Effect
Pending activation	Registration complete, no activation checkout	No writes until `/app/activate`
Free sandbox expired	`free_sandbox_expires_at` passed	Read-mostly; writes and sensitive exports blocked

What data is required to register a supplier vs restaurant?

Both need account type, **business name**, legal acceptance. Phone optional. Supplier registration creates default catalog and warehouse scaffold. Restaurant registration creates org and default branch.

How long does onboarding take?

Self-serve minimum path: register → activate → profile → first catalog/order — **under 30 minutes** with prepared data. Full production rollout (team, branches, warehouses, integrations) typically **1–2 weeks** depending on catalog size and training.

Can admins create tenants without self-service signup?

Suppliers: yes via `POST /api/suppliers` (admin API). Subscription starts pending activation; owner must still be linked via invite. **Restaurants:** primarily self-service `/register/complete`; confirm latest admin API in `06-admin-onboarding.md`.

What stub card works in demo/staging billing?

`4242424242424242` when `BILLING_GATEWAY=stub`.

RBAC & team access

What is the difference between a permission and a plan feature?

Plan feature = tenant paid for module (`finance_invoices`). **Permission** = user may act (`INVOICES_VIEW`). Both must pass. Example: Accountant role has invoice permissions, but Free Trial still needs `finance_invoices` enabled (it is, via Gold parity).

Can a purchaser receive goods?

Not by default. Purchaser has `ORDERS_CREATE` but not `RECEIVING_MANAGE`. Assign **Receiving Staff** or **Restaurant Manager** for receiving.

Can a supplier driver see the full supplier portal?

No. Driver role only has `DRIVER_DELIVERIES_VIEW` and `DRIVER_DELIVERIES_MANAGE`. Sidebar shows **My Deliveries** (`/app/driver-deliveries`) only.

Who can invite team members?

Restaurant: **Owner** (all permissions); Manager cannot `STAFF_INVITE` per role matrix. Supplier: **Owner** and roles with `STAFF_INVITE` / `STAFF_MANAGE` — Manager lacks staff manage. Custom roles possible with `advanced_roles` (Gold+).

Can we create custom roles on Silver?

No. `advanced_roles` is off on Silver for both tenant types. System roles only until Gold upgrade.

What is the Viewer role for?

Read-only audit/training accounts. All `*_VIEW` keys for that tenant type; **no** create/edit/manage. Useful for executives or external accountants who should not mutate data.

Does Owner bypass permissions in the API?

Yes for tenant Owner role in `requirePermission`. **Impersonation does not** automatically grant Owner — admin view-as uses selected role permissions.

What is Staff Portal vs Team member?

Staff Portal (`STAFF_PORTAL`) = scheduling/PTO for hourly staff at `/staff` — separate from procurement RBAC. **Team member** = `tenant_user_roles` with permissions like Purchaser or Receiving Staff.

Restaurants — operations

How many suppliers can a Free Trial restaurant follow?

One (`suppliers_per_restaurant` limit on Free). Gold allows 30; Platinum unlimited.

Why doesn't ETA show on tracking?

Common causes: (1) restaurant has not set **delivery coordinates** (lat/long required — address text alone is insufficient); (2) driver has not set status to `out_for_delivery`; (3) supplier lacks `driver_management` (Gold+).

Can receiving staff create orders?

No unless given a role with `ORDERS_CREATE`. **Receiving Staff** is view orders + receive only.

What plan is needed for disputes?

`disputes_returns` is enabled on **all tiers** including Free Trial (Gold feature parity). User still needs appropriate permissions (often Manager or Receiving for restaurant side).

What plan is needed for smart reorder suggestions?

`smart_reorder` — **off on Silver**; full on Gold (`full_90day_trends`); AI forecast on Platinum. Gold also enables `ai_platform` with `ai_requests_per_day` limit (20 on Gold, 100 on Platinum).

Are quick lists available to suppliers?

No. `quick_lists` is not enabled on supplier plan JSON — restaurant-only ordering lists feature.

Can Silver restaurants use multiple branches?

No. `multi_branch` is off on Silver (limit still 1 branch). Gold enables multi-branch up to 3 branches; Platinum unlimited.

Suppliers — operations

How many warehouses on Free Trial?

Limit `warehouses: 0` on Free — warehouse feature keys exist via parity but count cap is zero on Free supplier limits table. **Silver** includes 1 warehouse.

Who can decline orders?

Users with `ORDERS_MANAGE` — typically **Owner**, **Supplier Manager**, **Promotions Manager**. Fulfillment Staff can edit fulfillment progress but not decline at manage level.

What roles should we assign for warehouse pickers?

Warehouse Manager or **Order Fulfillment Staff** — both have `FULFILLMENT_*` ; Warehouse Manager also has `WAREHOUSES_EDIT` .

How do substitutions work?

Supplier reports substitution from order detail → fulfillment issue + chat notification → pending **amendment** if product mapped → restaurant accepts/rejects. Lines do **not** auto-change.

Can Catalog Manager see receivables?

No. Receivables API requires `INVOICES_VIEW` . **Catalog Manager** has catalog/inventory edit only — finance APIs return **403**.

Is driver management on Silver?

No. `driver_management` requires **Gold+**. Without it, delivery board features for assigning drivers are gated.

Finance & billing

Which roles can record payments?

Accountant on either side (`PAYMENTS_MANAGE`). Restaurant Manager and Supplier Manager have invoice **view** only, not payment manage per matrix.

When are invoices created?

Typically from delivered/fulfilled orders via supplier finance workflow (see product guide). Requires `finance_invoices` — enabled Silver+ with `record_payments` tier; Free Trial has feature via parity.

What does “aging” show?

Open receivable buckets by days outstanding on supplier finance dashboards — available when finance feature tier includes analytics (Gold `expense_analytics` / Platinum `advanced_finance_dashboard`).

Can accountants change subscription plan?

Restaurant/supplier **Accountant** has `SUBSCRIPTIONS_VIEW` only — **not** `SUBSCRIPTIONS_MANAGE` . Owner handles plan changes.

Why does export return 402 on expired Free Trial?

`billingAccessMiddleware` treats sensitive GETs (`/api/reports/*` , `*/export` , invoice PDF) as blocked when account locked — even though ordinary reads work.

Support & admin

How does support impersonate a customer?

Admin with `ADMIN_SUPPORT` → tenant row → impersonate → `impersonation_token` cookie. Session respects view-as role. All impersonation should be audit-logged. Stop via impersonate stop endpoint or logout.

Can support upgrade a tenant without payment?

Admins with `ADMIN_PLANS` can change plan in admin dashboard / subscription APIs. Use for comps and escalations; document reason in audit.

Why does admin see 402 while impersonating?

By design. Impersonating admins **do not** bypass billing lock — they experience what the tenant experiences for monetization enforcement.

Where are audit logs?

Platform: `/app/admin` → Audit (`ADMIN_ACCESS`). Tenant activity log: Settings when `tenant_audit_log` enabled (Gold+).

How do we reset a user password?

Admin **Users** tab (`ADMIN_SUPPORT`) or Keycloak admin console — not tenant Owner action for another user's Keycloak password.

Developers & technical

Where is the permission list defined?

`apps/api/src/lib/permission-keys.js` — 52 keys. Role defaults in `role-matrix.js` . Tests: `tenant-role-matrix.test.js` .

Where are plan features defined?

DB seeds in migrations `0117` , `0119` , `0120` , `0145` ; runtime keys in `feature-keys.js` ; Free → Gold override in `free-trial-plan-features.js` .

How do I gate a new API route?

1. `requireAuth` + `requireRole` + `resolveTenantContext`
2. `requirePermission('DOMAIN_ACTION')`
3. `requireFeature('feature_key')` if module is plan-gated
4. `requireWithinLimit('limit_key')` if creating countable resource
5. Ensure route listed in route inventory audit

How does frontend check entitlements?

`useEntitlements()` + helpers in `planFeatureGates.ts` / `planLimits.ts` . Check `planFeatures` **and** `features` for Free Trial parity.

Is there a mobile app?

supplify-mobile (sibling repo) for native parity. Web is PWA-capable; drivers often use mobile browser or PWA.

What auth cookies exist?

`access_token` , `refresh_token` , optional `impersonation_token` , `active_tenant` , session cookie for OAuth state.

How long are permissions cached?

~180 seconds Redis (`perm:…`). Invalidate on role assignment via `invalidateUserPermissionCache()` .

Where is tenant ID resolved?

`apps/api/src/lib/tenant-resolve.js` — impersonation → active branch cookie → workspace membership → primary contact fallback. **Do not** resolve only by contact email.

Troubleshooting quick reference

Symptom	Likely cause	Fix
402 on POST	Pending activation or expired Free Trial	`/app/activate` or upgrade
403 FEATURE_NOT_AVAILABLE	Plan lacks feature	Upgrade tier or admin override
403 LIMIT_EXCEEDED	Plan cap hit	Upgrade or admin limit override
403 permission	Role lacks key	Change role or custom role (Gold+)
Empty driver board	Not linked in `drivers` table or wrong role	Supplier admin links driver user
Invite accept fails	Email mismatch or second workspace	Use invited email; one workspace rule
Sidebar missing Finance	Feature off or `can()` false	Check entitlements + permissions
CSRF error on POST	Missing `X-CSRF-Token`	Frontend base query must send header

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Related docs

- 17-glossary.md — term definitions
- 19-onboarding-checklists.md — printable checklists
- 03-supplier-onboarding.md — supplier steps
- 04-restaurant-onboarding.md — restaurant steps
- 06-admin-onboarding.md — platform admin
- 10-subscriptions-and-plans.md — full matrices

Part XIX — Onboarding Checklists

Audience: Onboarding specialists, customer success, implementation partners.

Usage: Print each section (page break before each checklist heading). Check boxes during prep calls, live sessions, go-live, and hypercare. Cross-reference step detail in persona guides (`03` – `06`).

Checklist 1 — Supplier prep (before live session)

Owner: Onboarding specialist · **Duration:** 30–45 min prep

Customer contacts

- ☐ Primary owner email matches future Keycloak login
- ☐ Finance contact identified (Accountant role candidate)
- ☐ Warehouse/ops contact identified (Warehouse Manager / Fulfillment Staff)
- ☐ Driver contacts list (if Gold+ and `driver_management`)

Business data to collect

- ☐ Legal business name, VAT/tax ID, phone, address
- ☐ Public slug for mini-store (`/supplier/:slug`)
- ☐ Logo file (PNG/SVG, < 2 MB)
- ☐ MOQ, payment terms, return policy text
- ☐ Business hours and holiday blackout dates
- ☐ Warehouse name(s) and ship-from address(es)

Catalog data

- ☐ Product CSV or spreadsheet (SKU, name, unit, price, category)
- ☐ Product images (per SKU or ZIP import plan)
- ☐ Contract pricing list (if applicable)

Plan & access

- ☐ Target plan confirmed (Silver / Gold / Platinum)
- ☐ Warehouse count within plan limit (`warehouses` , `multi_warehouse`)
- ☐ SKU count within `supplier_products_skus` limit
- ☐ Customer understands one workspace per email rule

Environment

- ☐ Demo or production URL shared
- ☐ Stub billing card noted if sandbox (`4242424242424242`)
- ☐ Browser: Chrome/Edge current version

Checklist 2 — Supplier live onboarding session

Owner: Onboarding specialist + supplier owner · **Duration:** 90–120 min

Account & activation

- ☐ Owner registers at `/login` → Register
- ☐ Completes `/register/complete` with accountType **Supplier**
- ☐ Activates at `/app/activate` (free or paid)
- ☐ `GET /api/billing/status` → not locked

Profile & policies

- ☐ Settings → Profile: name, logo, contact, slug saved
- ☐ Settings → Business: MOQ, hours, terms saved
- ☐ Public page `/supplier/{slug}` loads

Warehouses & fulfillment

- [ ] Settings → Warehouses: at least one active warehouse
- [ ] Fulfillment settings saved (`PATCH /api/suppliers/me/fulfillment`)
- [ ] `/app/fulfillment` board accessible (plan `fulfillment` / `fulfillment_tools`)

Catalog

- [ ] At least 10 live products (or agreed pilot set)
- [ ] Categories and units correct
- [ ] Test product visible to test restaurant account

Team (if Gold+ `advanced_roles`)

- [ ] Invites sent: Manager, Fulfillment, Driver (as needed)
- [ ] Driver invitee sees only `/app/driver-deliveries` after accept

Smoke test

- [ ] Test restaurant places order
- [ ] Supplier sees order on fulfillment board
- [ ] Supplier can accept/decline or progress status
- [ ] Chat message on order thread works (`chat` feature)

Wrap-up

- [ ] Owner knows Settings → Team for future invites
- [ ] Support contact and plan limits documented
- [ ] Next session date for finance/drivers (if deferred)

Checklist 3 — Restaurant prep (before live session)

Owner: Onboarding specialist · **Duration:** 30–45 min prep

Customer contacts

- [ ] Owner/purchasing manager email for login
- [ ] Receiving manager contact (Receiving Staff role)
- [ ] FOH lead (if using reservations)
- [ ] Accountant contact (if using finance module)

Operational data

- [ ] Restaurant legal name, address, phone
- [ ] Logo and branding assets
- [ ] Branch list (names, addresses) if multi-site
- [ ] **GPS delivery coordinates** per site (lat/long — not address-only)
- [ ] List of current suppliers to follow (within plan `suppliers_per_restaurant`)

Plan & access

- [ ] Target plan: Silver / Gold / Platinum
- [ ] Branch count within `branches` limit
- [ ] Expected daily order volume vs `orders_per_day`
- [ ] Inventory SKU count vs `restaurant_inventory_skus` if using inventory

Supplier linkage

- [ ] Pilot supplier(s) live on platform with catalog
- [ ] Or: plan for restaurant to discover/follow suppliers in session

Environment

- [ ] URL and login instructions sent
- [ ] Test user credentials for specialist (if co-browsing)

Checklist 4 — Restaurant live onboarding session

Owner: Onboarding specialist + restaurant owner · **Duration:** 90–120 min

Account & activation

- [ ] Register → `/register/complete` accountType **Restaurant**
- [ ] Activate at `/app/activate`
- [ ] Sidebar shows Orders, Suppliers, Settings

Profile & locations

- [ ] Settings/Onboarding → Profile complete
- [ ] Delivery coordinates saved (`PATCH … /delivery-location`)
- [ ] Branches created if multi-branch entitled (`multi_branch` , Gold+)

Suppliers & catalog

- [ ] Follow at least one supplier (`/app/suppliers`)
- [ ] Browse products `/app/products`
- [ ] Confirm contract pricing if applicable

First order

- [ ] Build cart and place order
- [ ] Order visible in `/app/orders`
- [ ] Supplier acknowledges (coordinate with supplier or test tenant)

Receiving (if in scope)

- [ ] Receiving Staff or Manager walks receive flow
- [ ] Optional quality photo if `receiving_quality` tier allows
- [ ] Optional inventory lot/expiry capture

Team

- [ ] Invites: Purchaser, Receiving Staff, Accountant as needed
- [ ] Role sidebar matches least privilege

Optional modules (plan permitting)

- [ ] Quick list created (`/app/quick-lists`)
- [ ] Finance: invoice list loads (`finance_invoices` + `INVOICES_VIEW`)
- [ ] Reservations smoke test (`/app/reservations`) if FOH

Wrap-up

- [ ] Plan limits explained (orders/day, suppliers, branches)
- [ ] Disputes/receiving escalation path documented

Checklist 5 — Driver onboarding

Owner: Supplier admin + driver · **Duration:** 30–45 min

Prerequisites (supplier admin)

- [ ] Supplier on Gold+ (`driver_management`) or equivalent entitlement
- [ ] Driver user invited with **Driver** system role
- [ ] User linked to `drivers` row (admin fulfillment setup)
- [ ] At least one order assigned to driver for training

Driver session

- [ ] Login at `/login` — only **My Deliveries** in sidebar
- [ ] Board loads: `GET /api/supplier/deliveries/board`
- [ ] Driver understands statuses: assigned → out_for_delivery → delivered
- [ ] Practice **I'm on the way** (`out_for_delivery`)
- [ ] Practice **Delivered** with notes/POD if required
- [ ] Practice **Problem** (failed) and **Reschedule** paths
- [ ] If 2+ stops: **Build my route** demonstrated
- [ ] GPS permission granted on mobile browser/PWA
- [ ] Restaurant confirms ETA/tracking on test order

Safety & policy

- [ ] Driver knows not to share login
- [ ] Privacy: restaurant address visible; limited financial data
- [ ] Who to call at supplier dispatch for reassignment

Checklist 6 — Platform admin onboarding

Owner: Internal ops / support lead · **Duration:** 2–3 hours

Access

- [ ] Admin user exists (`role: ADMIN` in `app_user`)
- [ ] Admin permissions assigned (SUPER_ADMIN or scoped roles)

- [ ] Login → `/app/admin` loads overview

Portal navigation

- [ ] Platform portal vs Supplier admin vs Restaurant admin understood
- [ ] Tab gating matches permission map (`ADMIN_TENANTS` , `ADMIN_PLANS` , etc.)

Core workflows practiced

- [ ] Search tenants (`/app/admin/tenants`)
- [ ] Filter by subscription status (TRIALING, ACTIVE, SUSPENDED, …)
- [ ] View subscription row and change plan (test tenant)
- [ ] Apply feature flag or limit override on test tenant (Growth/Plans)
- [ ] Impersonate tenant → verify **no** Owner bypass without view-as Owner
- [ ] Stop impersonation
- [ ] Review audit log entry for impersonation/plan change

Support tools

- [ ] User search (`ADMIN_SUPPORT`)
- [ ] Password reset procedure documented
- [ ] Health tab: `/api/admin-dashboard/health` or equivalent loads
- [ ] Conversion stats after blocked feature test

Governance

- [ ] Impersonation policy acknowledged (customer consent, logging)
- [ ] DPA / data access policy reviewed (`DATA_PROCESSING_ADDENDUM.md`)
- [ ] Escalation path to engineering documented

Checklist 7 — Go-live (production cutover)

Owner: Implementation lead + customer exec sponsor · **Duration:** 1 day window

Pre cutover (T-1)

- [ ] Production URLs and SSL verified
- [ ] Keycloak production realm configured
- [ ] `REDIS_URL` , database, storage buckets production-ready
- [ ] Billing gateway mode confirmed (stub vs live)
- [ ] Plan/subscription correct on production tenant rows
- [ ] Data migration complete (catalog, users, branches) if applicable
- [ ] Rollback plan documented

Cutover (T-0)

- [ ] DNS / bookmark update communicated to users
- [ ] All users complete activation (no `pending_activation`)
- [ ] Owner and backup Owner confirmed
- [ ] Critical roles invited and accepted (purchasing, receiving, fulfillment)
- [ ] First production order placed and fulfilled end-to-end
- [ ] Invoice/payment path verified if finance in scope
- [ ] Monitoring: error rate, 402/403 spikes, health endpoints green

Post cutover (T+0)

- [ ] War room channel open for 4 business hours
- [ ] Known issues log started
- [ ] Customer sign-off email template sent

Checklist 8 — First week hypercare

Owner: Customer success · **Duration:** 5 business days

Daily

- [ ] Review support tickets tagged for new tenant
- [ ] Check admin activity for 402/403 conversion events
- [ ] Confirm order volume within plan limits

Day 1

- [ ] Owner can log in; no activation lock
- [ ] At least one order cycle completed

- [ ] Team invites accepted or nudged

Day 3

- [ ] Receiving or fulfillment workflow used in production
- [ ] Chat or notifications working if in scope
- [ ] Address any RBAC misconfigurations (wrong role assignments)

Day 5

- [ ] Usage vs entitlements review (branches, SKUs, orders/day)
- [ ] Upgrade conversation if consistently near limits
- [ ] Schedule Day 30 check-in
- [ ] Customer satisfaction pulse (email/call)

Exit criteria

- [ ] No P1 open issues
- [ ] Primary workflows adopted without manual workarounds
- [ ] Documentation links sent (persona guide + FAQ)

Checklist 9 — First month success review

Owner: Account manager + customer sponsor · **Duration:** 60 min meeting

Adoption metrics

- [ ] Orders per week trend
- [ ] Active users vs invited users
- [ ] Feature adoption: quick lists, inventory, finance, drivers
- [ ] Support ticket themes categorized

Plan fit

- [ ] Limit headroom: `orders_per_day`, `branches`, `warehouses`, SKUs
- [ ] Feature gaps vs next tier documented
- [ ] ROI narrative draft (time saved, error reduction)

RBAC hygiene

- [ ] No shared Owner credentials
- [ ] Viewer/Accountant roles used appropriately
- [ ] Custom roles documented if `advanced_roles`

Roadmap

- [ ] Phase 2 modules agreed (API, multi-branch expansion, AI reorder)
- [ ] Training gaps scheduled
- [ ] Reference/customer story consent if applicable

Checklist 10 — Technical deployment (new environment)

Owner: DevOps / platform engineer · **Duration:** 4–8 hours

Infrastructure

- [ ] PostgreSQL provisioned; migrations applied (`pnpm run migrate` or CI)
- [ ] Redis provisioned (`REDIS_URL` internal URL in production)
- [ ] Object storage for uploads configured
- [ ] Keycloak realm + clients (`web`, `api`) with correct redirect URIs
- [ ] Environment variables set per `docs/operations/railway.md` or host equivalent

API (`apps/api`)

- [ ] `OAUTH_CALLBACK_BASE_URL` matches public API origin
- [ ] `WEB_ORIGIN` matches SPA origin
- [ ] `COOKIE_SECURE`, `COOKIE_DOMAIN`, `COOKIE_SAME_SITE` correct
- [ ] `IMPERSONATION_SECRET` set (production strength)
- [ ] `BILLING_GATEWAY` configured
- [ ] `/health` and `/ready` return 200

Web (`apps/web`)

- [ ] Build with correct `VITE_API_URL`
- [ ] PWA assets served over HTTPS

- [ ] CSRF flow verified against API

Auth smoke

- [ ] Register → callback → `/auth/me` returns user
- [ ] Refresh token rotation works
- [ ] Logout clears cookies

Seeds (non-prod only)

- [ ] `seed:demo-users` if demo environment
- [ ] Plan catalog rows exist (`subscription_plan`)

Checklist 11 — Production validation (post-deploy)

Owner: QA / engineer · **Duration:** 2–4 hours

Automated

- [ ] API test suite green (`apps/api` CI)
- [ ] Web typecheck/build green
- [ ] Route inventory spot-check against `docs/audits/route-inventory.json`

Auth & security

- [ ] OIDC login/logout full cycle
- [ ] CSRF rejected without token (401/403)
- [ ] Staff portal allowlist enforced
- [ ] Impersonation audit row written

Monetization

- [ ] Free tenant: entitlements show Gold features + Free limits
- [ ] `requireFeature` returns 403 on disabled feature (Silver `smart_reorder` test)
- [ ] `requireWithinLimit` returns 403 at cap
- [ ] Expired Free Trial: write 402, read mostly OK

RBAC

- [ ] Purchaser cannot `RECEIVING_MANAGE` (403)
- [ ] Driver cannot access `/api/suppliers/me` catalog mutations
- [ ] Viewer cannot POST orders

Critical paths

- [ ] Restaurant place order → supplier fulfill → driver deliver → restaurant receive
- [ ] Invoice create/view (finance feature + permissions)
- [ ] Admin tenant search + read-only impersonation browse

Performance

- [ ] Entitlements cache hit acceptable (< 500 ms p95 on warm)
- [ ] Permission cache invalidates on role change within TTL

Checklist 12 — Demo environment prep (sales / POC)

Owner: Sales engineer · **Duration:** 1–2 hours

Tenant setup

- [ ] Demo supplier tenant seeded with catalog (50+ SKUs ideal)
- [ ] Demo restaurant tenant follows demo supplier
- [ ] Both tenants activated (not `pending_activation`)
- [ ] Plans set to Gold or Platinum for full demo story (or explain Free Trial parity)

Personas

- [ ] `owner@` supplier and restaurant credentials documented
- [ ] `driver@` linked driver with assigned delivery
- [ ] `viewer@` read-only optional
- [ ] Admin `admin@` for impersonation demo

Scenario data

- [ ] 3+ orders in various statuses (placed, in fulfillment, delivered)

- [ ] One open dispute or amendment for narrative
- [ ] One quick list with scheduled order if showing automation
- [ ] Receiving session with quality photo example

Demo script assets

- [ ] Slug URLs bookmarked: `/supplier/{slug}`, `/app/orders`, `/app/fulfillment`
- [ ] Upgrade modal trigger prepared (e.g. hit branch limit on Silver test user)
- [ ] FAQ one-pager link: `18-frequently-asked-questions.md`

Reset procedure

- [ ] Document how to re-seed or reset demo DB
- [ ] No real PII in demo tenants
- [ ] Billing stub only — no live cards

Related docs

- 03-supplier-onboarding.md
- 04-restaurant-onboarding.md
- 05-driver-onboarding.md
- 06-admin-onboarding.md
- 07-technical-architecture.md
- 18-frequently-asked-questions.md

Part XX — Source Evidence Index

Purpose: Central traceability table linking onboarding documentation claims to repository evidence. Use for audits, demo certification, and dispute resolution ("where is this implemented?").

Verification status legend:

Status	Meaning
Verified	Path/symbol exists; behavior confirmed in code or automated test
Partial	Implemented with documented gaps or UI/API asymmetry
Doc-only	Described in docs; verify before customer commitment
Generated	Machine output; regenerate to refresh

Last reviewed: 2026-06-17

Documentation section	Claim	Repository path	Symbol / route	Verification status	N
01 Executive	Restaurant–supplier marketplace product scope	`docs/product/overview.md`	—	Verified	C
01 Executive	554 API routes in inventory	`docs/audits/route-inventory.json`	`count: 554`	Generated	F
01 Executive	Four plan tiers free/silver/gold/platinum	`apps/api/src/lib/plan-codes.js`	`PLAN_CODES`	Verified	
01 Executive	Tenant types RESTAURANT/SUPPLIER/ADMIN	`docs/architecture/rbac-overview.md`	—	Verified	M
01 Executive	Driver role limited deliveries surface	`apps/api/src/lib/role-matrix.js`	`Driver` role	Verified	
01 Executive	Staff portal separate from Team RBAC	`apps/api/src/lib/staff-portal-auth.js`	`STAFF_PORTAL_APP_ROLE`	Verified	F
02 Product guide	Order amendments API	`apps/api/src/routes/order-amendments.routes.js`	`/api/orders/:id/amendments`	Verified	C
02 Product guide	Receiving workflow	`apps/api/src/routes/receiving.routes.js`	`/api/receiving/*`	Verified	
02 Product guide	Disputes API	`apps/api/src/routes/disputes.routes.js`	`/api/disputes/*`	Verified	
02 Product guide	Fulfillment board	`apps/web/src/pages/FulfillmentPage.tsx`	`/app/fulfillment`	Verified	S
02 Product guide	Public supplier mini-store	`apps/web/src/App.tsx`	`/supplier/:idOrSlug`	Verified	P
02 Product guide	B2C consumer ordering	`apps/web/src/App.tsx`	`/order/:restaurantSlug`	Verified	P
02 Product guide	Reservation guest portal	`apps/web/src/App.tsx`	`/reserve`	Verified	U
03 Supplier	OAuth register flow	`apps/api/src/routes/auth.routes.js`	`GET /auth/register`	Verified	K
03 Supplier	Registration complete creates supplier	`apps/api/src/routes/register.routes.js`	`POST /api/register/complete`	Verified	
03 Supplier	Pending activation lock	`apps/api/src/middlewares/billingAccess.js`	`billingAccessMiddleware`	Verified	4
03 Supplier	Activation page	`apps/web/src/pages/ActivatePage.tsx`	`/app/activate`	Verified	A
03 Supplier	Supplier settings hub	`apps/web/src/pages/SupplierSettingsPage.tsx`	`/app/settings`	Verified	S
03 Supplier	Supplier profile API	`apps/api/src/routes/suppliers/`	`GET/PATCH /api/suppliers/me`	Verified	T
03 Supplier	Warehouse CRUD	`apps/api/src/routes/warehouses.routes.js`	`/api/suppliers/me/warehouses`	Verified	
03 Supplier	Fulfillment settings patch	`apps/api/src/routes/suppliers/`	`PATCH /api/suppliers/me/fulfillment`	Verified	F
03 Supplier	Team invites	`apps/api/src/routes/invites.routes.js`	`POST /api/invites/accept`	Verified	V
03 Supplier	Product catalog CRUD	`apps/api/src/routes/products/`	`/api/products/*`	Verified	
03 Supplier	CSV product import	`apps/api/src/routes/products/import.routes.js`	`POST /api/products/import`	Verified	
03 Supplier	Promotions	`apps/api/src/routes/promotions/supplier.js`	`/api/supplier/promotions`	Verified	
03 Supplier	Command center	`apps/web/src/pages/SupplierCommandCenterPage.tsx`	`/app/command-center`	Verified	
03 Supplier	Receivables API	`apps/api/src/routes/supplier-ops.routes.js`	`GET /api/supplier/invoices/receivables`	Verified	
03 Supplier	Customer import	`apps/api/src/routes/supplier-growth.routes.js`	`CUSTOMERS_IMPORT`	Verified	C
04 Restaurant	Restaurant registration	`apps/api/src/routes/register.routes.js`	`accountType: RESTAURANT`	Verified	C
04 Restaurant	Restaurant settings/onboarding UI	`apps/web/src/components/restaurant/onboarding/`	`/app/onboarding`	Verified	T
04 Restaurant	Delivery location coordinates	`apps/api/src/routes/restaurants/`	`PATCH … /delivery-location`	Verified	E
04 Restaurant	Branch CRUD	`apps/api/src/routes/restaurant-branches.routes.js`	`/api/restaurants/branches/*`	Verified	
04 Restaurant	Supplier discovery	`apps/api/src/routes/suppliers/`	`GET /api/suppliers`	Verified	M
04 Restaurant	Supplier follow limit	`apps/api/src/lib/limit-resolution.js`	`suppliers_per_restaurant`	Verified	F
04 Restaurant	Place order	`apps/api/src/routes/orders.routes.js`	`POST /api/orders`	Verified	
04 Restaurant	Quick lists	`apps/api/src/routes/quick-lists.routes.js`	`/api/quick-lists`	Verified	
04 Restaurant	Restaurant inventory	`apps/api/src/routes/restaurant-inventory.routes.js`	`/api/restaurant-inventory`	Verified	S
04 Restaurant	Waste tracking	`apps/api/src/routes/restaurant-inventory.routes.js`	waste analytics routes	Verified	
04 Restaurant	Reservations	`apps/api/src/routes/reservations.routes.js`	`/api/reservations`	Verified	
04 Restaurant	Finance invoices (restaurant)	`apps/api/src/routes/restaurant-finance.routes.js`	`/api/restaurant-finance`	Verified	
05 Driver	Driver home route	`apps/web/src/App.tsx`	`/app/driver-deliveries`	Verified	M

Documentation section	Claim	Repository path	Symbol / route	Verification status	N
05 Driver	Deliveries board API	`apps/api/src/routes/orders-driver.routes.js`	`GET /api/supplier/deliveries/board`	Verified	
05 Driver	Delivery status update	`apps/api/src/routes/orders-driver.routes.js`	`PATCH /api/orders/:id/delivery-status`	Verified	S
05 Driver	Build route from assignments	`apps/api/src/routes/fulfillment-routes.routes.js`	`POST … /routes/build-from-assignments`	Verified	I
05 Driver	Active route fetch	`apps/api/src/routes/fulfillment-routes.routes.js`	`GET /api/fulfillment/routes/active`	Verified	A
05 Driver	Route stop reorder	`apps/api/src/routes/fulfillment-routes.routes.js`	`POST … /stops/reorder`	Verified	D
05 Driver	Driver management plan gate	`apps/api/src/lib/feature-keys.js`	`driver_management`	Verified	C
05 Driver	PWA-friendly driver UI	`apps/web/vite.config.ts`	PWA plugin	Verified	I
06 Admin	Admin dashboard landing	`apps/web/src/pages/AdminDashboardPage.tsx`	`/app/admin`	Verified	T
06 Admin	Admin overview API	`apps/api/src/routes/admin-dashboard/overview.js`	`GET /api/admin-dashboard/overview`	Verified	
06 Admin	Tenant directory suppliers	`apps/api/src/routes/admin-dashboard/tenants.js`	`GET … /tenants/suppliers`	Verified	
06 Admin	Tenant directory restaurants	`apps/api/src/routes/admin-dashboard/tenants.js`	`GET … /tenants/restaurants`	Verified	R
06 Admin	Impersonate start	`apps/api/src/routes/admin-dashboard/audit.js`	`POST /api/admin-dashboard/impersonate`	Verified	U
06 Admin	Impersonate stop	`apps/api/src/routes/admin-dashboard/audit.js`	`POST … /impersonate/stop`	Verified	C
06 Admin	Admin create supplier	`apps/api/src/routes/suppliers/admin.js`	`POST /api/suppliers`	Verified	A
06 Admin	Subscription admin list	`apps/api/src/routes/admin-dashboard/subscriptions.js`	`/api/admin-dashboard/subscriptions`	Verified	
06 Admin	Feature flags admin	`apps/api/src/routes/admin-dashboard/features.js`	feature override routes	Verified	
06 Admin	Audit logs	`apps/api/src/routes/admin-dashboard/audit.js`	`GET /api/admin-dashboard/audit-logs`	Verified	R
06 Admin	Admin nav config	`apps/web/src/lib/adminNavConfig.ts`	portal tabs	Verified	S
07 Architecture	Express server entry	`apps/api/src/server.js`	`app.listen`	Verified	M
07 Architecture	Middleware order billing after impersonation	`apps/api/src/server.js`	mount sequence	Verified	S
07 Architecture	PostgreSQL session store	`apps/api/src/lib/session-store.js`	connect-pg-simple	Verified	C
07 Architecture	Redis permission cache	`apps/api/src/lib/permissions.js`	`perm:*` key	Verified	T
07 Architecture	RTK Query API client	`apps/web/src/store/api.ts`	`baseQuery`	Verified	C
07 Architecture	Vite SPA build	`apps/web/vite.config.ts`	—	Verified	
07 Architecture	Railway deployment doc	`docs/operations/railway.md`	env vars table	Doc-only	C
08 Database	Tenant roles tables	`apps/api/migrations/`	`tenant_roles`, `tenant_user_roles`	Verified	R
08 Database	Subscription tables	`apps/api/migrations/`	`subscription`, `subscription_plan`	Verified	R
08 Database	Inventory lots	`apps/api/migrations/0133_restaurant_inventory_lots.sql`	`restaurant_inventory_lot`	Verified	E
08 Database	Fulfillment issues	`apps/api/migrations/0134_order_fulfillment_issues.sql`	`order_fulfillment_issue`	Verified	S
08 Database	Reorder cadence	`apps/api/migrations/0135_reorder_cadence_and_quick_list_branch.sql`	cadence tables	Verified	S
08 Database	Workspace membership one-per-user	`apps/api/migrations/0104_user_workspace_membership.sql`	unique constraint	Verified	I
08 Database	Free sandbox expiry column	`apps/api/migrations/0113_free_sandbox_expiry.sql`	`free_sandbox_expires_at`	Verified	T
09 Auth RBAC	52 permission keys	`apps/api/src/lib/permission-keys.js`	`PERMISSION_KEYS`	Verified	P
09 Auth RBAC	7 restaurant system roles	`apps/api/src/lib/role-matrix.js`	`RESTAURANT_SYSTEM_ROLES`	Verified	L
09 Auth RBAC	9 supplier system roles	`apps/api/src/lib/role-matrix.js`	`SUPPLIER_SYSTEM_ROLES`	Verified	I
09 Auth RBAC	Role matrix unit tests	`apps/api/src/lib/tenant-role-matrix.test.js`	test cases	Verified	C
09 Auth RBAC	OIDC login	`apps/api/src/routes/auth.routes.js`	`GET /auth/login`	Verified	C
09 Auth RBAC	Auth me payload	`apps/api/src/routes/auth.routes.js`	`GET /auth/me`	Verified	p
09 Auth RBAC	HttpOnly auth cookies	`apps/api/src/lib/rbac.js`	`setAuthCookies`	Verified	a
09 Auth RBAC	requirePermission Owner bypass	`apps/api/src/lib/rbac.js`	`requirePermission`	Verified	~
09 Auth RBAC	Impersonation JWT	`apps/api/src/lib/impersonation.js`	`createImpersonationToken`	Verified	H
09 Auth RBAC	Impersonation context middleware	`apps/api/src/middlewares/impersonationContext.js`	`req.impersonationContext`	Verified	E
09 Auth RBAC	Admin dashboard permission map	`apps/api/src/lib/route-permissions.js`	`resolveAdminDashboardPermission`	Verified	T

Documentation section	Claim	Repository path	Symbol / route	Verification status	N
09 Auth RBAC	Staff portal path gate	`apps/api/src/lib/staff-portal-auth.js`	`assertStaffPortalRouteAccess`	Verified	4
09 Auth RBAC	Frontend usePermissions	`apps/web/src/hooks/usePermissions.ts`	`can()`, `canAny()`	Verified	U
09 Auth RBAC	AuthGuard registration gate	`apps/web/src/components/AuthGuard.tsx`	`needsSetup`	Verified	
09 Auth RBAC	Custom roles API	`apps/api/src/routes/tenant-roles.routes.js`	`POST /api/roles`	Verified	S
10 Plans	Restaurant 26 feature keys	`apps/api/src/lib/feature-keys.js`	`RESTAURANT_FEATURE_KEYS`	Verified	A
10 Plans	Supplier 24 feature keys	`apps/api/src/lib/feature-keys.js`	`SUPPLIER_FEATURE_KEYS`	Verified	A
10 Plans	Free uses Gold features runtime	`apps/api/src/lib/subscription/free-trial-plan-features.js`	`resolveEffectivePlanFeatures`	Verified	
10 Plans	Silver migration limits	`apps/api/migrations/0117_silver_tier_limits_features.sql`	plan rows	Verified	S
10 Plans	Gold migration limits	`apps/api/migrations/0119_gold_tier_limits_features.sql`	plan rows	Verified	S
10 Plans	Platinum migration limits	`apps/api/migrations/0120_platinum_tier_limits_features.sql`	plan rows	Verified	S
10 Plans	Free catalog sync	`apps/api/migrations/0145_plan_catalog_audit_sync.sql`	free limits	Verified	A
10 Plans	Supplier growth free parity	`apps/api/migrations/0175_free_trial_supplier_growth_parity.sql`	`supplier_growth`	Verified	F
10 Plans	AI requests limit	`apps/api/migrations/0167_ai_platform_and_usage.sql`	`ai_requests_per_day`	Verified	L
10 Plans	requireFeature 403 payload	`apps/api/src/lib/subscription/entitlements.js`	`buildFeatureNotAvailablePayload`	Verified	U
10 Plans	requireWithinLimit	`apps/api/src/lib/subscription/entitlements.js`	`requireWithinLimit`	Verified	L
10 Plans	Entitlements GET	`apps/api/src/routes/subscriptions.routes.js`	`GET /api/subscriptions/entitlements`	Verified	A
10 Plans	Entitlements cache 300s	`apps/api/src/lib/subscription/entitlements.js`	cache TTL	Verified	U
10 Plans	billingAccess 402	`apps/api/src/middlewares/billingAccess.js`	`buildAccountLockedError`	Verified	S
10 Plans	Admin bypass unless impersonating	`apps/api/src/middlewares/billingAccess.js`	ADMIN check	Verified	I
10 Plans	recommendPlan upsell	`apps/api/src/lib/subscription/plans.js`	`recommendPlan`	Verified	C
10 Plans	useEntitlements hook	`apps/web/src/hooks/useEntitlements.ts`	`useGetEntitlementsQuery`	Verified	S
10 Plans	planFeatureGates helpers	`apps/web/src/lib/planFeatureGates.ts`	`canUseFulfillment` etc.	Verified	A
10 Plans	evaluatePlanFeatureValue	`apps/api/src/lib/subscription/entitlements.js`	tier string = on	Verified	M
10 Plans	Bronze → silver alias	`apps/api/migrations/0116_rename_bronze_to_silver.sql`	`LEGACY_PLAN_CODE_ALIASES`	Verified	L
10 Plans	Enterprise removed	`apps/api/migrations/0066_remove_enterprise_tier.sql`	deactivated row	Verified	M
11 API	554 routes inventory	`docs/audits/route-inventory.json`	`routes[]`	Generated	2
11 API	Route discover script	`apps/api/scripts/discover-routes.mjs`	CLI	Verified	R
11 API	Standard response envelope	`apps/api/src/lib/response.js`	`ok/data/error`	Verified	C
11 API	CSRF middleware	`apps/api/src/middlewares/csrf.js`	`csrfProtection`	Verified	P
11 API	Health endpoints	`apps/api/src/server.js`	`GET /health`, `/ready`	Verified	S
11 API	Orders router mutation guard	`apps/api/src/lib/route-permissions.js`	`ordersRouterMutationGuard`	Verified	C
17 Glossary	Tenant resolve algorithm	`apps/api/src/lib/tenant-resolve.js`	`getEffectiveTenant`	Verified	I
17 Glossary	Substitution creates amendment	`docs/features/inventory-expiry-and-reorder.md`	fulfillment-issues	Verified	M
17 Glossary	Reorder cadence recompute	`apps/api/src/services/reorder-cadence.service.js`	service	Verified	T
17 Glossary	Inventory lot expiry statuses	`apps/api/src/routes/restaurant-inventory.routes.js`	`/expiry`, `/expiry/summary`	Verified	F
18 FAQ	Purchaser no receiving	`apps/api/src/lib/role-matrix.js`	Purchaser permissions	Verified	M
18 FAQ	Catalog manager no receivables	`docs/archive/audits/rbac-roles-permissions-audit.md`	checklist	Verified	4
18 FAQ	One workspace per user	`apps/api/src/lib/workspace-membership.js`	`assertUserCanJoinWorkspace`	Verified	4
18 FAQ	Stub card 4242…	`docs/onboarding/03-supplier-onboarding.md`	`BILLING_GATEWAY=stub`	Doc-only	E
19 Checklists	Demo user seed	`package.json` / scripts	`seed:demo-users`	Verified	
19 Checklists	Migration command	`apps/api/package.json`	`migrate` script	Verified	C
Features	Fulfillment shortage API	`apps/api/src/routes/supplier-ops.routes.js`	… `/fulfillment-issues/shortage`	Verified	I
Features	Fulfillment substitution API	`apps/api/src/routes/supplier-ops.routes.js`	… `/substitution`	Verified	A
Features	Smart reorder at-risk supplier	`apps/api/src/routes/supplier-ops.routes.js`	`GET /api/supplier/reorder-cadence/at-risk`	Verified	C
Features	Admin impersonation doc	`docs/features/admin-impersonation.md`	—	Verified	V
Features	Access control two layers	`docs/architecture/access-control.md`	requireFeature + requirePermission	Verified	A
Features	Tenant roles feature doc	`docs/features/tenant-roles.md`	custom roles	Verified	
Features	Inventory expiry doc	`docs/features/inventory-expiry-and-reorder.md`	lot model	Verified	M

Documentation section	Claim	Repository path	Symbol / route	Verification status	N
Features	Branch invitations	`docs/features/branch-invitations.md`	`/invite/branch`	Verified	E
Architecture	RBAC permission matrix doc	`docs/architecture/rbac-permission-matrix.md`	—	Verified	H
Architecture	Security baseline	`docs/architecture/security-baseline.md`	CSRF, cookies	Verified	S
Sales	Admin impersonation sales doc	`docs/sales/06_admin_and_operations.md`	impersonate	Verified	S
Web	Sidebar plan gating	`apps/web/src/components/Sidebar.tsx`	`canUse*` helpers	Verified	H
Web	Limit exceeded banner	`apps/web/src/components/LimitExceededBanner.tsx`	component	Verified	M
Web	Monetization Redux slice	`apps/web/src/store/monetizationSlice.ts`	upgrade modal	Verified	4
Web	Branch context switcher	`apps/web/src/contexts/BranchContext.tsx`	`active_tenant` cookie	Verified	M
Web	Multi-branch gate	`apps/web/src/lib/planLimits.ts`	`multiBranchEnabled`	Verified	P
API	Reports feature gate	`apps/api/src/routes/reports.routes.js`	`requireFeature('reports')`	Verified	S
API	Disputes feature gate	`apps/api/src/routes/disputes.routes.js`	`disputes_returns`	Verified	P
API	Chat routes	`apps/api/src/routes/chat.routes.js`	`/api/chat`	Verified	D
API	Billing checkout	`apps/api/src/routes/billing.routes.js`	`POST /api/billing/checkout`	Verified	C
API	Billing status	`apps/api/src/routes/billing.routes.js`	`GET /api/billing/status`	Verified	L
API	Register status	`apps/api/src/routes/register.routes.js`	`GET /api/register/status`	Verified	
API	GPS driver location	`apps/api/src/routes/orders-driver.routes.js`	`recordDriverLocation`	Verified	
API	Notifications	`apps/api/src/routes/notifications.routes.js`	`/api/notifications`	Verified	C
API	Tenant audit log	`apps/api/src/routes/tenant-audit.routes.js`	tenant audit	Verified	
API	Public staff magic link	`apps/api/src/routes/public/staff.routes.js`	`POST /api/public/staff/request-link`	Verified	P
API	Org billing entitlements child	`apps/api/src/lib/org-billing-entitlements.test.js`	`getEntitlements` child	Verified	E
API	Free sandbox expiry job	`apps/api/src/jobs/free-sandbox-expiry.job.js`	cron job	Verified	S
Tests	Driver access tests	`apps/api/src/routes/orders-driver-access.routes.test.js`	—	Verified	P
Tests	Reorder cadence tests	`apps/api/src/services/reorder-cadence.service.test.js`	—	Verified	C
Tests	Staff portal access tests	`apps/api/src/lib/staff-portal-access.test.js`	—	Verified	A
Tests	Warehouse routes tests	`apps/api/src/routes/warehouses.routes.test.js`	mocked gates	Verified	L
Mobile	Parity checklist	`docs/mobile/MOBILE_PARITY_CHECKLIST.md`	—	Doc-only	S
Mobile	Feature parity doc	`docs/mobile/MOBILE_FEATURE_PARITY.md`	—	Doc-only	C
Legal	DPA impersonation clause	`apps/web/static/legal/DATA_PROCESSING_ADDENDUM.md`	support access	Verified	C
Audits	API route test matrix	`docs/audits/DEV_API_ROUTE_TEST_MATRIX.md`	—	Generated	C
Audits	Demo readiness audit	`docs/audits/SUPPLIFY_DEMO_READINESS_AUDIT.md`	—	Partial	P
Bootstrap	Onboarding metrics artifact	`docs/onboarding/_artifacts/bootstrap-metrics.md`	—	Generated	D

How to verify a row

1. **Routes** — Open `route-inventory.json` or grep `apps/api/src/routes` for path string.
2. **Permissions** — Cross-check `role-matrix.js` and `tenant-role-matrix.test.js`.
3. **Plan gates** — Read `subscription_plan` seeds in migrations and `requireFeature` mount in route file.
4. **UI** — Grep `apps/web/src` for route path in `App.tsx` or page component.
5. **Regenerate inventory** — From repo root: `node apps/api/scripts/discover-routes.mjs`.

Related docs

- README.md — reading order and doc index
- 11-api-and-workflow-reference.md — API pipeline
- docs/audits/route-inventory.json — machine route list

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